Understanding Society: India and Beyond

Social Science Textbook for Grade 9 | Part 2

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NCERT

राष्ट्रीय शैक्षिक अनुसंधान और प्रशिक्षण परिषद्
NATIONAL COUNCIL OF EDUCATIONAL RESEARCH AND TRAINING

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The whole world is one family.

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Chapter 1

Oceans and Life

The Big Questions

1. *What is the importance of ocean relief?*
2. *How does ocean water movement impact life and environment?*
3. *Why are maritime zones important for a country?*
4. *How can we use ocean resources in a sustainable way?*
5. *How do early warning systems help protect people from tsunamis and cyclones?*

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1. Ocean water is always in motion.

Waves, tides, and currents shape our climate, support marine life, and impact travel.

2. The gravitational pull creates tides.

The daily rise and fall of sea level is crucial for navigation and generating tidal energy.

3. The ocean floor is a hidden landscape.

Like land, it has features like mountains, plains, mid-oceanic ridges, trenches, etc.

4. The 'Blue Economy': sustaining ocean resources.

Oceans provide food, energy, and jobs, which must be used sustainably.

5. Early warning systems protect coastal life.

Seismic sensors and satellites detect tsunamis and cyclones, allowing for timely evacuations.

6. Maritime zones create 'rules of the sea'.

International laws like UNCLOS define a country's rights from its coast to the high seas.

Fig. 1.1. The dynamic ocean system

In earlier grades, you learnt about continents and oceans. In Indian culture, the ocean—commonly called *Samudra*—is considered powerful and sacred, and features prominently in stories and rituals.

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Oceans cover most of our planet and conceal many secrets beneath their waters. Have you ever wondered what the land under the oceans looks like, why ocean water is always moving, or what wealth and resources lie beneath the sea? How do areas like maritime zones help countries use the oceans, and how can we protect ourselves from ocean hazards, such as cyclones and tsunamis?

In this chapter, we will explore these human–ocean connections and discover how oceans shape both our daily lives and the world around us.

Ocean Relief

As we explore human–ocean connections, let us begin with the ocean floor and its features. Just as land has mountains, valleys, and plains, the ocean floor, too, has several landforms, together known as **submarine relief** or **ocean relief.** These include underwater mountains, valleys, plateaus, and trenches. The Mariana Trench in the Pacific Ocean is the deepest place on Earth, while the Mid-Ocean Ridge is the world's longest mountain range, hidden beneath the ocean. Some underwater mountains like Mauna Kea (Hawai'i) are even taller than Mount Everest.

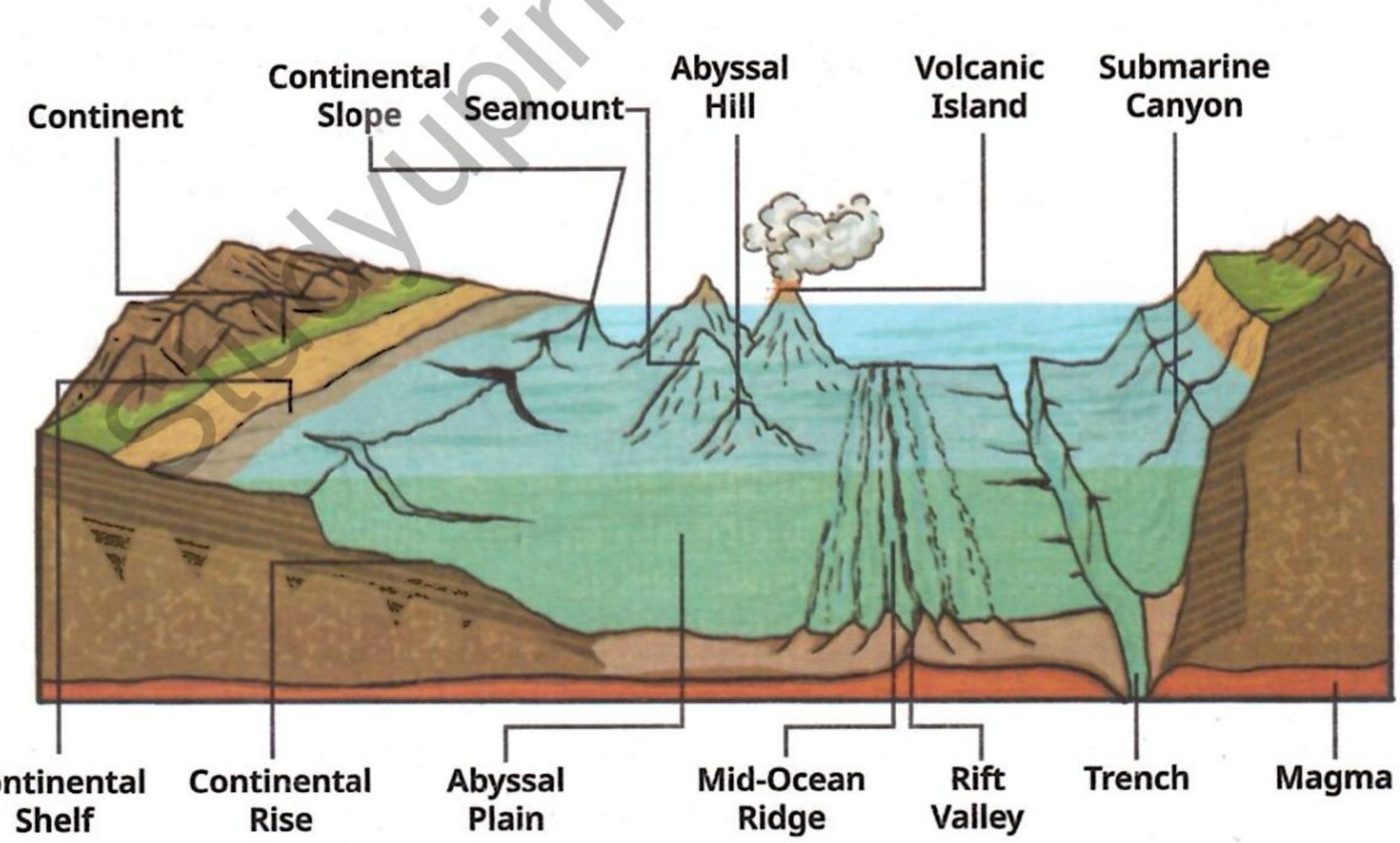

Fig. 1.2. Submarine relief

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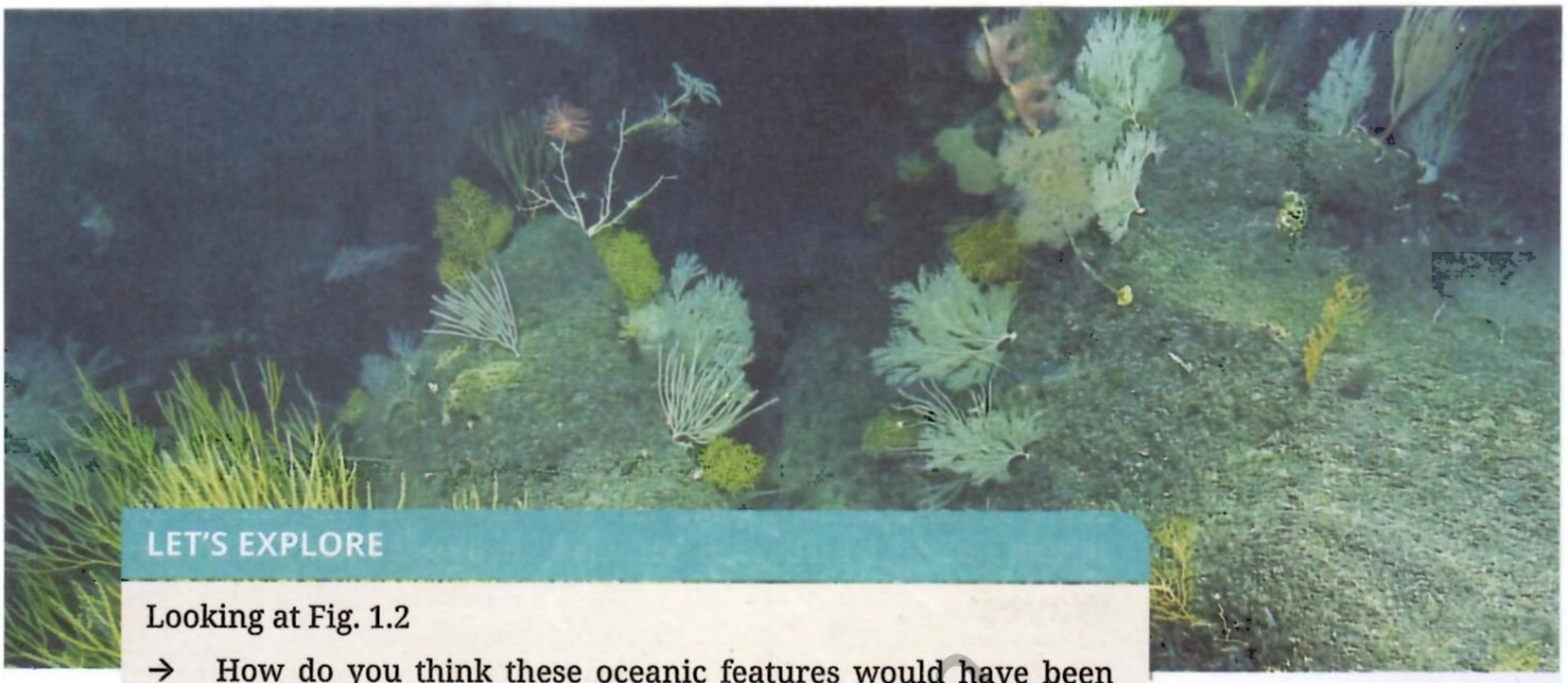

LET'S EXPLORE

Looking at Fig. 1.2

- → How do you think these oceanic features would have been formed?
- → Which part of the ocean floor appears deep and shallow?
- → How do you think these underwater features would have an effect on marine life and ocean movements?

Important submarine relief features

Continental Shelf

The submerged, shallow edge of a continent; extends under the ocean. Rich in marine life and resources like oil and gas.

Continental Slope

The steep slope marking the true edge of a continent.

Abyssal Plain

The vast, flat, and deep ocean floor.

Mid-Ocean Ridge

An immense underwater mountain range formed by volcanic activity.

Ocean Trenches

The deepest parts of the ocean.

Seamounts and Guyots

Seamounts are often volcanic, with some flattened by wave action over millennia to form **guyots**.

Seamount:
An underwater mountain, usually of volcanic origin, that rises from the ocean floor but does not reach the sea surface.

Guyot:
A flat-topped underwater mountain, usually a former seamount whose summit was eroded and later submerged.

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Movement of Ocean Water—Waves, Tides, and Currents

The water in the oceans is constantly in motion. Waves, tides, and currents are not merely movements of ocean water; they help shape the Earth's surface, support marine life, aid fishing and travel, and even influence climate. Let us learn more about these oceanic movements.

Waves

When the surface of the ocean rises and falls alternately, it is called a **wave.** Waves are primarily caused by winds blowing across the water. They carry energy across the oceans over long distances. Observe Fig. 1.3 and note the crests (high peaks) and troughs (lower valleys).

Wavelength: The horizontal distance between two consecutive wave crests or troughs.

Waves are often called coastal architects as they shape the shoreline through erosion and sediment transport. Observe the various landforms formed by waves in Fig. 1.4 and note how wave action creates erosional features, such as sea cracks, caves, arches, sea stacks, and stumps. Also notice the original headland shown in the figure.

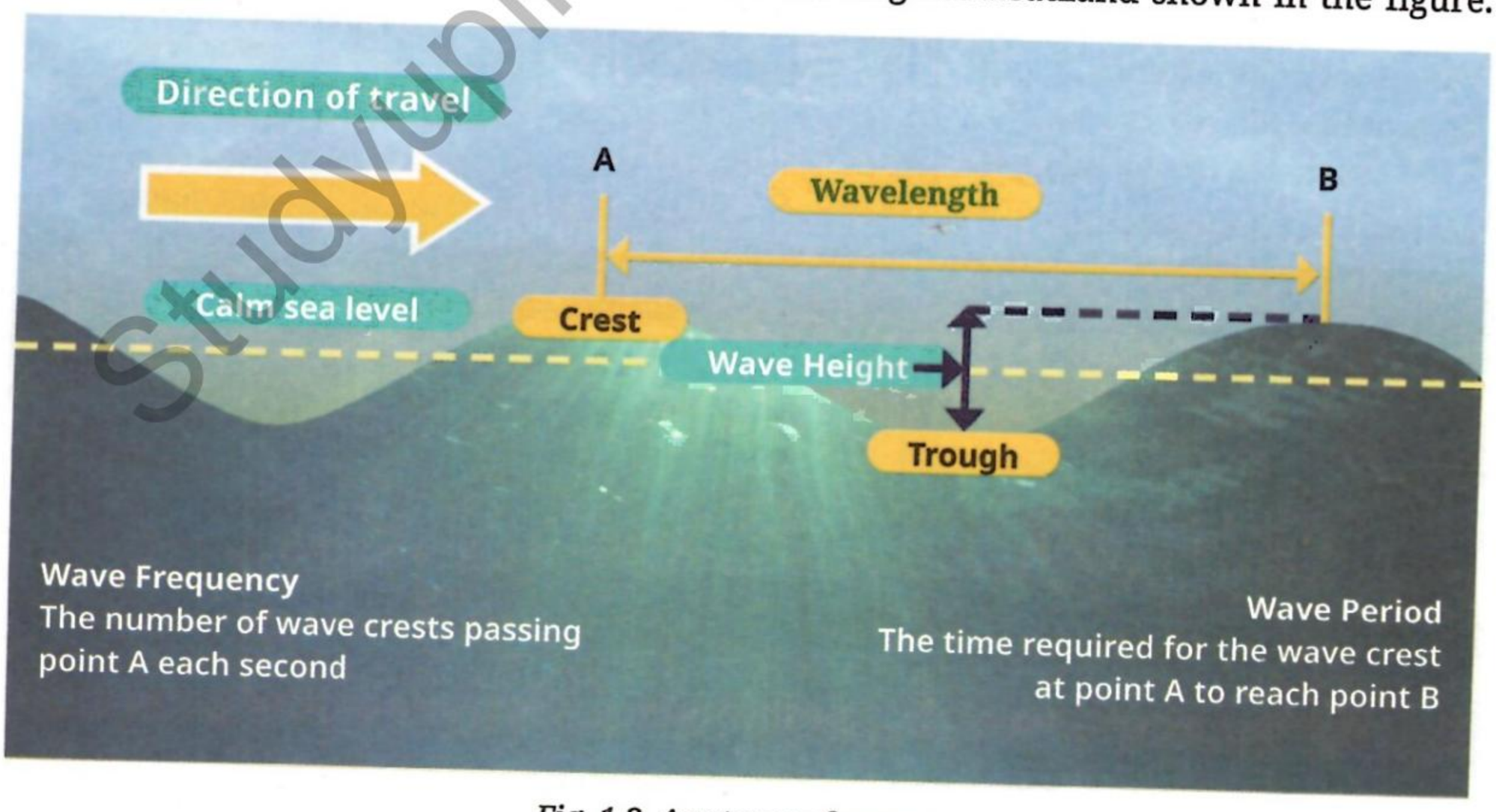

Fig. 1.3. Anatomy of waves

In addition to erosional landforms, waves also form depositional features such as beaches.

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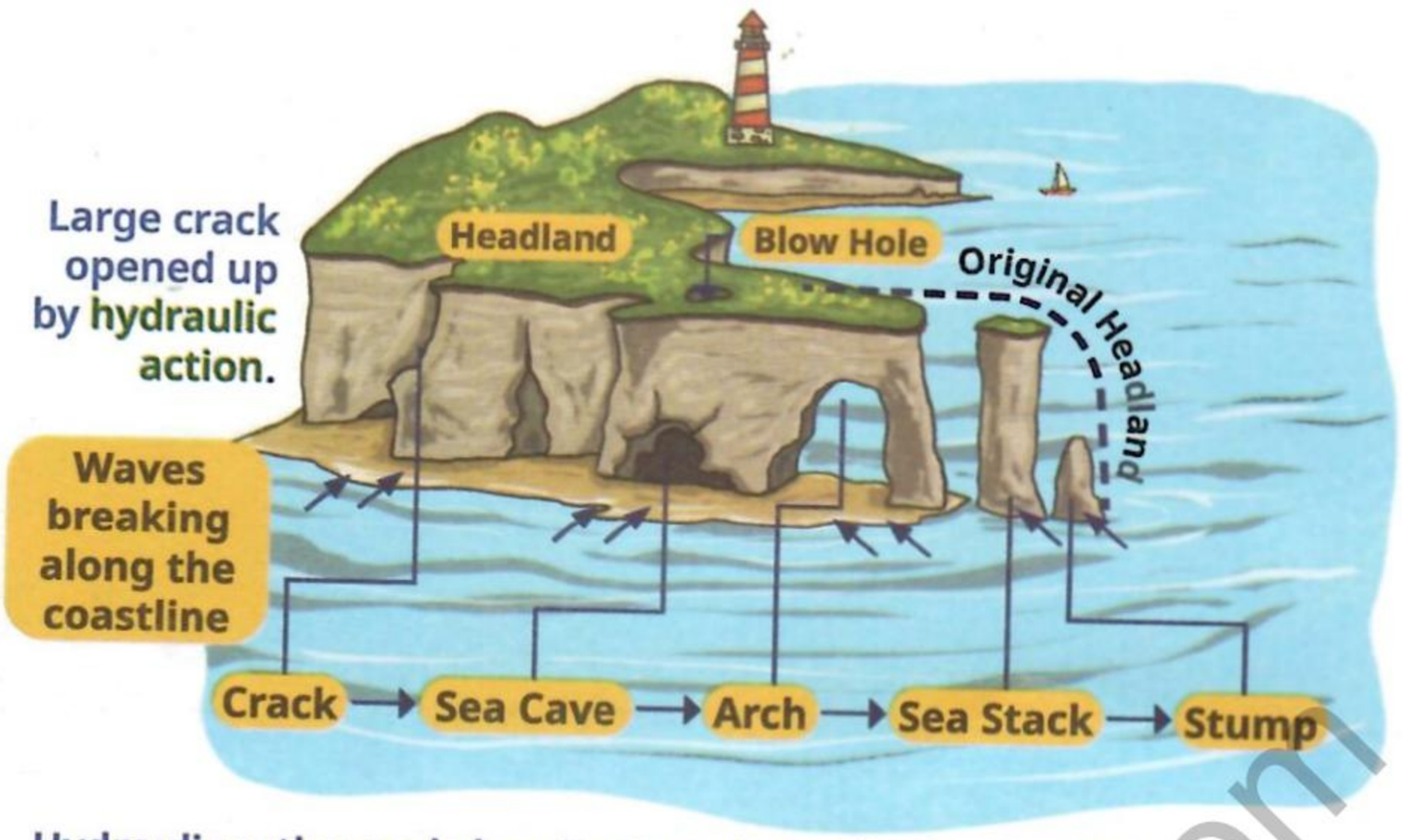

Fig. 1.4. Landform features formed by waves

LET'S EXPLORE

Find out the names of some famous sea beaches in our country and locate them on the map of India.

During storms, strong winds can generate huge waves that may cause significant damage along coastlines. The giant waves majorly caused by underwater earthquakes are called **tsunamis**, which can be extremely destructive. On reaching the shore, tsunamis may attain a maximum vertical height above sea level, known as the run-up height, of up to 30 metres, or even more in extreme cases.

Tides

Tides are the rhythmic rise and fall of ocean water that occur twice a day in most parts of the world. When the water rises to its highest level and covers much of the shore, it is called high tide (flood tide). When the water falls to its lowest level and moves away from the shore, it is called low tide (ebb tide). Observe Fig. 1.5 and note the change in water level during high and low tides. Do you notice the oval layer around the Earth indicating water level? Can you also see how the water level rises during high tide and falls during low tide?

Hydraulic action: The erosion of rock and soil caused by the force of moving water, which loosens and removes material.

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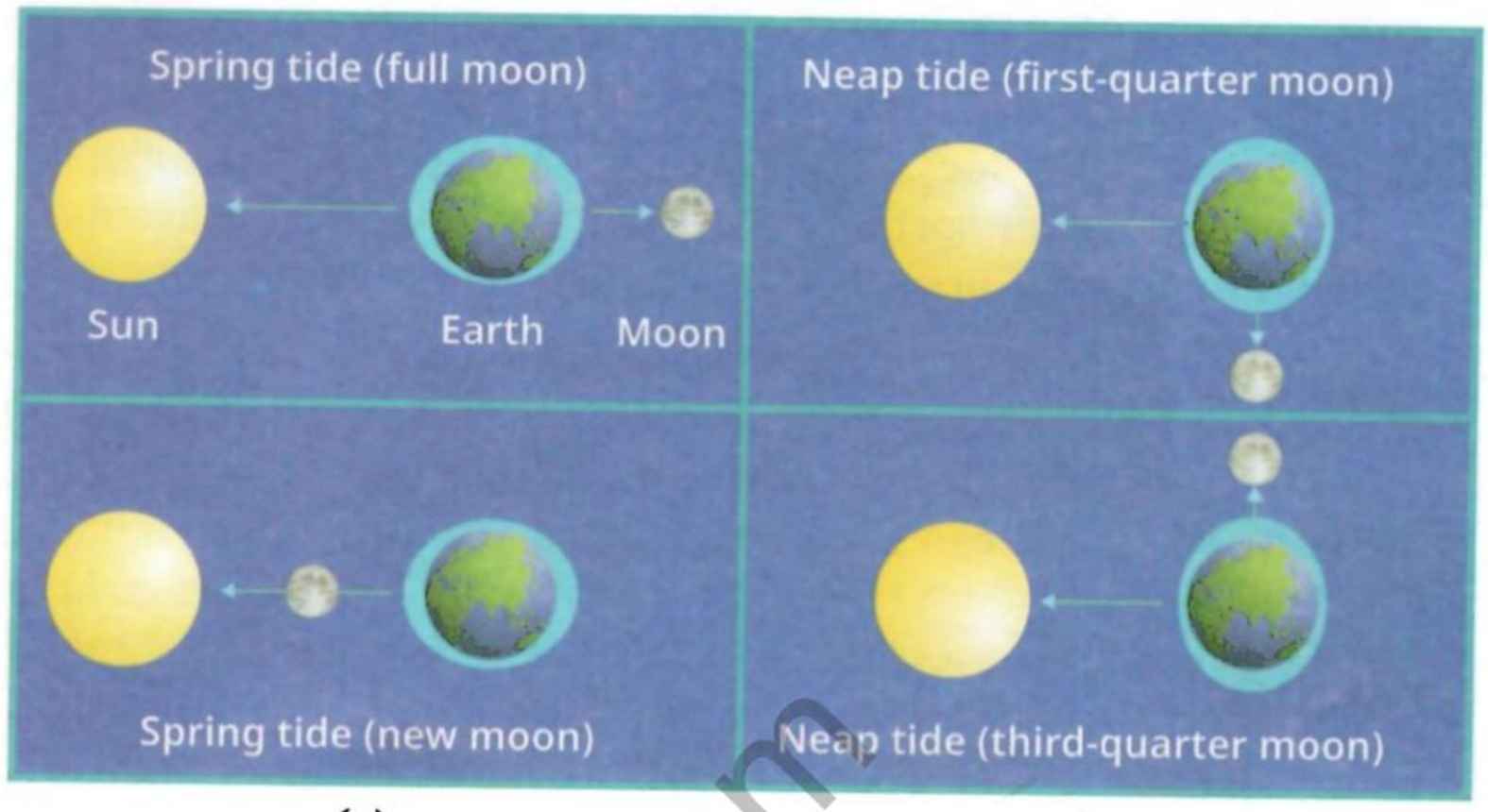

Fig. 1.5 (a) Spring tide (b) Neap tide (Schematic diagram of tidal forcing showing idealised bulges)

Spring Tides and Neap Tides

You may wonder what causes this regular rise and fall of ocean water. To understand this, let us explore how tides are formed.

- **Spring tides:** Observe Fig. 1.5(a), where the Sun, the Earth, and the Moon are aligned in a straight line. Spring tides occur during this alignment, which happens on full moon and new moon days. At this time, the combined gravitational pull of the Sun and the Moon causes high tides of unusual heights and low tides that are unusually lower. The figure also shows the bulging of water on opposite sides of the Earth due to this strong gravitational pull, leading to the formation of spring tides.
- **Neap tides:** When the Moon is in its first- and third-quarter phases, as shown in Fig. 1.5(b), the gravitational pull of the

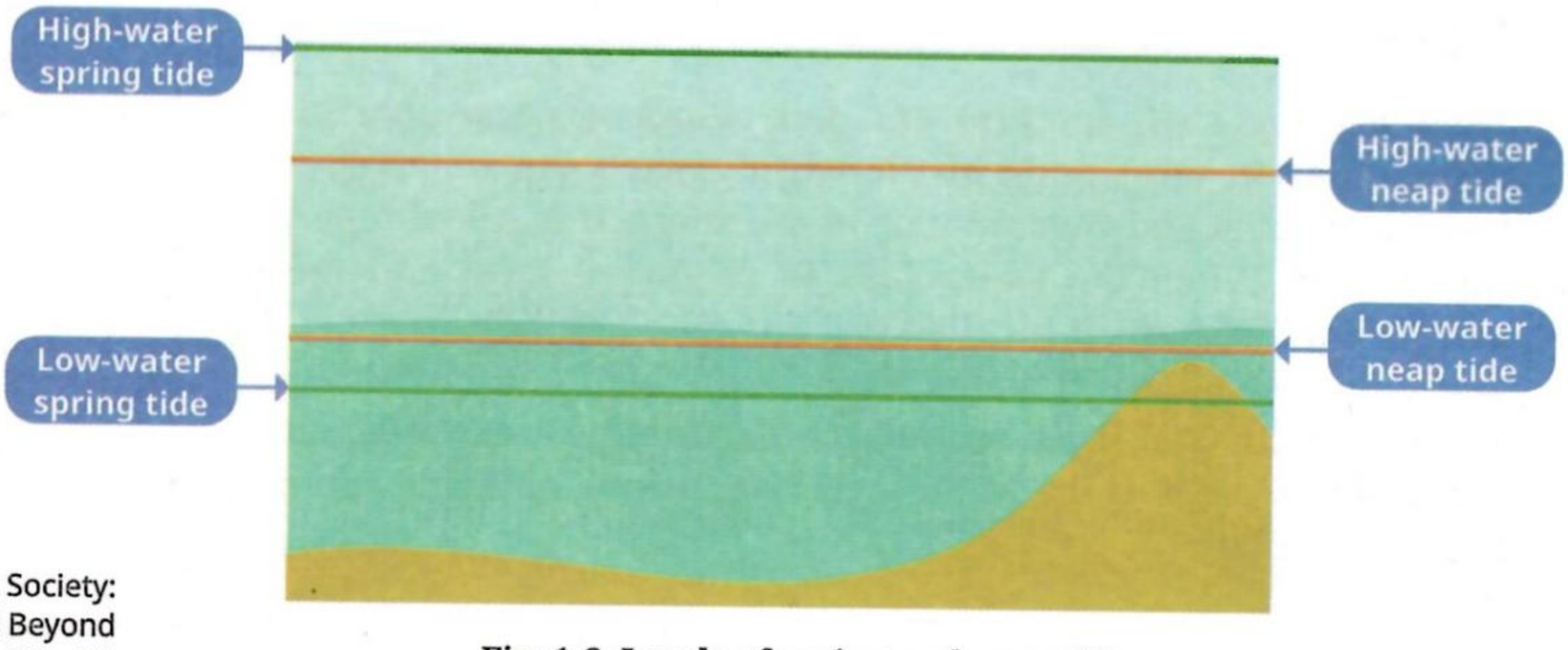

Fig. 1.6. Levels of spring and neap tides

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Sun and the Moon acts in different directions. As a result, the difference between high tide and low tide becomes smaller, producing neap tides. Fig. 1.6 shows the levels of high and low water during both spring and neap tides, along with the average low-water levels for each.

High tides are important for navigation because they raise the water level near the shore, allowing ships to enter harbours more easily. In ports and coastal areas, tidal information is essential, as tidal ranges in some ports can reach 4–5 metres. The Gulf of Khambhat has one of the highest tidal ranges, at about 12 metres. Before a ship berths at a port, it must check both the water depth and the tidal range to ensure safe docking, especially during low tide.

Some sea channels, such as the Singapore Strait, contain shallow areas. Hence, ships such as Very Large Crude Carriers (VLCCs) passing through these regions often wait for hours to transit only during high tide to ensure sufficient water depth and avoid running aground.

Tidal changes also generate strong currents because large volumes of water flow in and out of coastal areas. These tidal currents can reach speeds of 5–6 knots (about 9–11 km/h).

Fig. 1.7. Observe how high and low tides help in fishing activities

High tides also support fishing activities (Fig. 1.7). During high tide, many fish move closer to the shore in search of food and shelter, enabling fishing communities to obtain a better catch.

The rise and fall of tidal water are also used to generate electricity in some places (Fig. 1.8). India has a vast coastline of 11,098.81 kilometres, and although tidal energy is not yet widely used on a large scale, certain areas have been identified as suitable for its development. The Gulf of Kutch, the Gulf of Khambhat, and the Sundarbans are among the major tidal energy sites in India.

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Fig. 1.8. Tides at the Bhadkeshwar Mahadev Temple, Dwarka, Gujarat

The Bhadkeshwar Mahadev Temple in Dwarka is situated on a small hillock in the Arabian Sea. Visitors can reach the temple only during low tide, as the stone pathway leading to it is completely submerged during high tide.

Coriolis force: A force caused by Earth's rotation that deflects any free moving object from its straight path to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

Ocean Currents

Ocean currents are continuous streams of water that flow across the ocean surface in definite directions (Fig. 1.9). They are caused by factors, such as winds, the Earth's rotation (**Coriolis force**), differences in temperature and salinity, and the shape of continents and ocean basins. Ocean currents may be warm or cold depending on the direction from which the water flows. Currents flowing from warmer regions are called warm currents, while those flowing from colder regions are called cold currents. On maps, warm currents are usually shown in red and cold currents in blue.

THINK ABOUT IT

Observe the map carefully and note the movement patterns of cold and warm currents. Do you think these currents affect the temperature of nearby coastal regions?

The Gulf Stream, a warm current originating in the Gulf of Mexico in the Atlantic Ocean, flows towards Norway and keeps its waters relatively warm. As a result, sea ice does not form easily in the Norwegian region despite its high latitude (60°–70°). In contrast, the Labrador Current, a cold current flowing along the east coast of Canada, carries freezing water and supports the formation of sea ice and icebergs even at lower latitudes.

LET'S EXPLORE

Observe Fig. 1.9 and take a note on ocean currents along the coastline of India. What is the nature of ocean currents?

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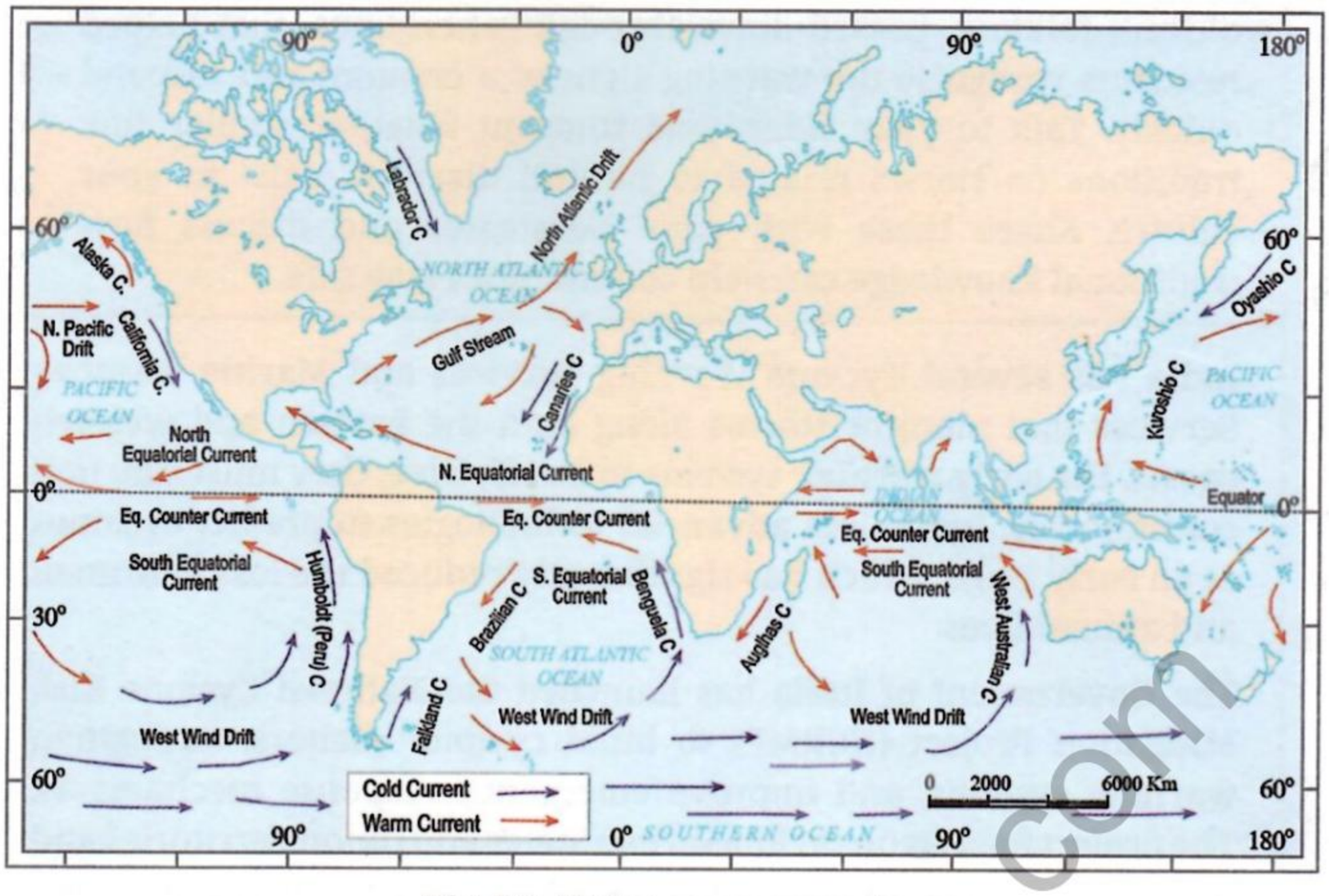

Fig. 1.9. Major ocean currents

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The regions where warm and cold currents meet are known as fishing grounds. These waters are rich in nutrients, supporting a large variety of fish and making them important fishing grounds.

Cyclones and Tsunamis—Early Warning Systems

Oceans and seas can give rise to sudden natural hazards such as cyclones, which are large rotating storms, and tsunamis, which are giant waves caused by earthquakes. To protect people from such dangers, scientists have developed Early Warning Systems that work like giant alarm systems for coastal areas. To reduce risks, people, communities, and governments must act in advance. Early Warning Systems help detect cyclones and tsunamis, predict disasters, and issue timely warnings so that people can evacuate and stay safe.

LET'S EXPLORE

Most deaths during the 2004 Indian Ocean tsunami occurred in Indonesia. However, the Island of Simeulue was largely spared from the disaster. Researchers suggest that this was partly because

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of local folklore, passed down through generations, that helped residents recognise the warning signs of a tsunami and respond quickly. Talk to your elders and find out whether similar folk traditions or stories related to natural disasters exist in your culture. Share these with your classmates and discuss how traditional knowledge can help communities stay safe.

India has several Cyclone Warning Services and Marine Weather Services that monitor storms along both the eastern and western coasts. For early warning systems to be effective, they must function continuously. India uses advanced technologies to predict cyclones at an early stage, which has significantly reduced the loss of human and animal lives.

The Government of India has launched the National Cyclone Risk Mitigation Project (NCRMP) to build cyclone shelters, strengthen warning systems, and improve emergency response mechanisms. The project focuses on cyclone-prone States and Union Territories and includes a Decision Support System (DSS) for disaster management authorities. The Indian National Centre for Ocean Information Services (INCOIS), in collaboration with the India Meteorological Department (IMD), has also established the Storm Surge Early Warning System (SSEWS) for the Indian coasts. Its main objective is to protect coastal communities by forecasting cyclone-induced storm surges and the extent of coastal inundation.

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The **GAGAN-Enabled Mariner's Instrument for Navigation and Information (GEMINI)** is a handheld device designed for fishermen that plays an important role in ensuring the safety of fishing communities at sea (Fig. 1.10). The device receives satellite information and transmits it to mobile phones through Bluetooth. Using its dedicated mobile application, fishermen can access

Fig. 1.10. Local fishermen at work in the Arabian Sea; Alappuzha, Keralam

important information in their local language, including alerts about high waves and strong winds, cyclone and tsunami warnings, potential fishing zone advisories, current speeds, **bathymetry** data, mapping, and risk management information.

Under the Pradhan Mantri Matsya Sampada Yojana (PMMSY), the government provides financial support for GEMINI devices to improve the safety of coastal communities while they are at sea (Fig. 1.11).

Bathymetry: The study and mapping of the underwater topography of water bodies, such as oceans, rivers, lakes, and streams.

How GEMINI Works

- The GEMINI device receives data from the GAGAN satellite and transfers it to fishermen's mobile phones through Bluetooth.
- The INCOIS mobile application decodes and displays this information in nine regional languages.
- At present, the communication channel can transmit information only up to 50 nautical miles into the sea.
- An upgraded version is expected to provide a two-way communication system on a pilot basis, allowing fishermen to send feedback as well.

Fig. 1.11. GEMINI

- This upgraded system will benefit fishermen travelling up to 300 nautical miles in search of better fishing opportunities.

Focus on Cyclones

Cyclones are caused by atmospheric disturbances over warm ocean waters.

Naming the cyclone: Cyclones in the North Indian Ocean are named sequentially from a list of 169 names submitted by 13 member countries, including India. This system aids clear communication and public awareness. Recent examples include Biparjoy (2023) and Remal (2024). Note: You may track cyclones and their affected areas in India on the School Bhuvan–NCERT Geoportal (Fig. 1.12).

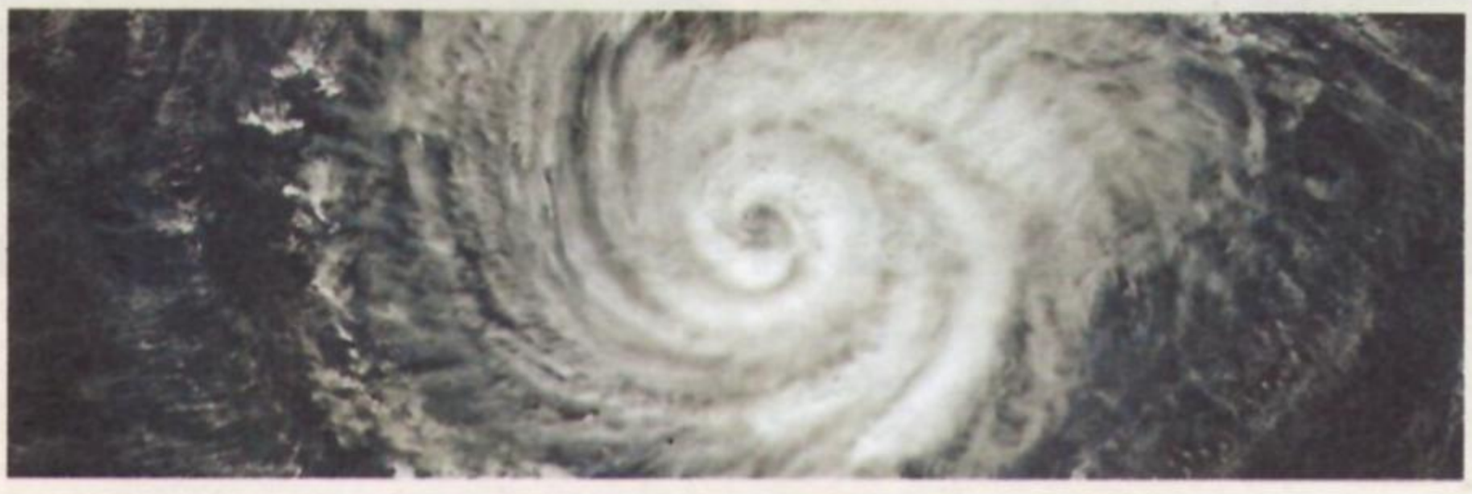

Fig. 1.12. Satellite image of a cyclone

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On 26 December 2004, the world witnessed unprecedented devastation caused by the Indian Ocean Tsunami. The disaster claimed around 2,30,000 lives and caused severe environmental damage across many countries bordering the Indian Ocean. India was among the worst-affected countries, as massive waves swept away people and structures along the coasts. Tamil Nadu, Puducherry, Andhra Pradesh, Keralam, and the Andaman & Nicobar Islands especially suffered extensive damage. The scale of the disaster led many countries to establish effective tsunami early warning systems and stronger disaster-management mechanisms. Under an initiative of the Indian Tsunami Early Warning Centre (ITEWC), the Government of India set up a real-time network of tsunami detection buoys (Fig. 1.14) designed to detect, measure, and monitor tsunamis.

The photograph in Fig. 1.13 shows some areas of Chennai affected by the Indian Ocean tsunami (2004).

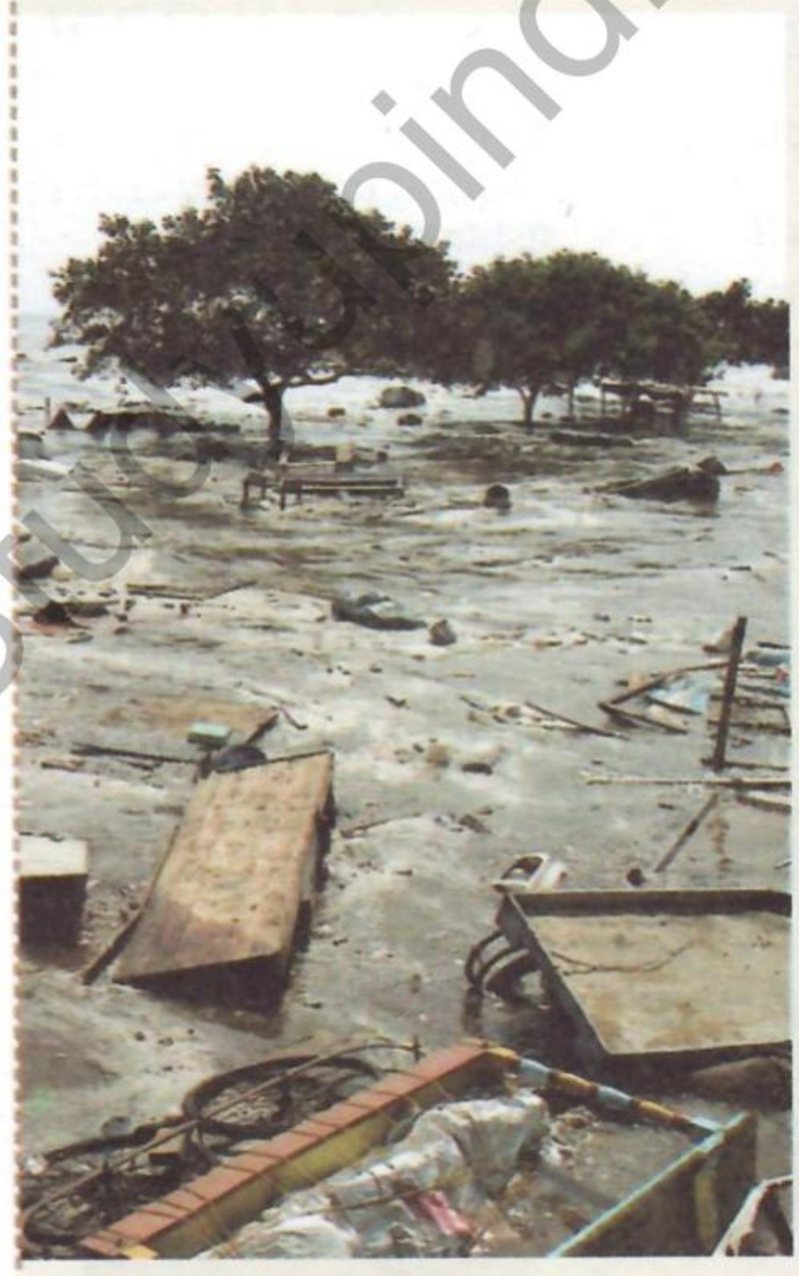
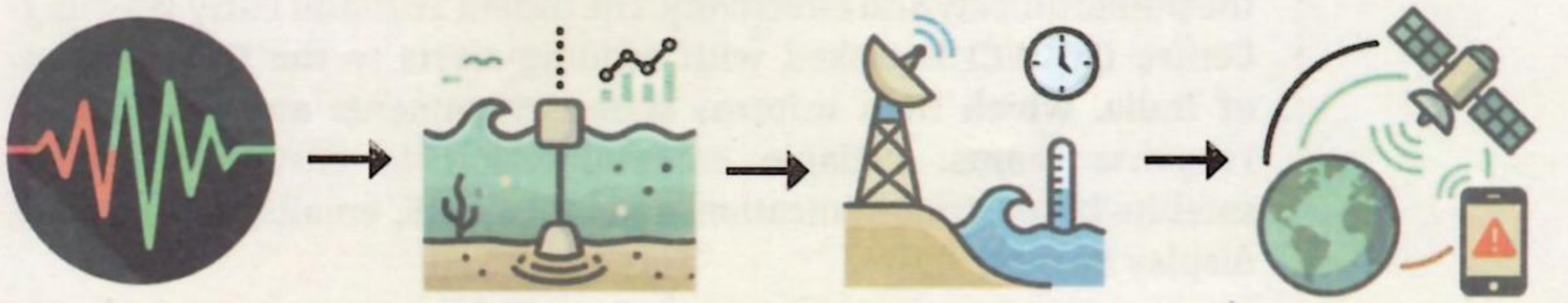


Fig. 1.13. Some of the tsunami-affected areas in Chennai (2004)

Fig. 1.14. Tsunami detection buoy

Study Point



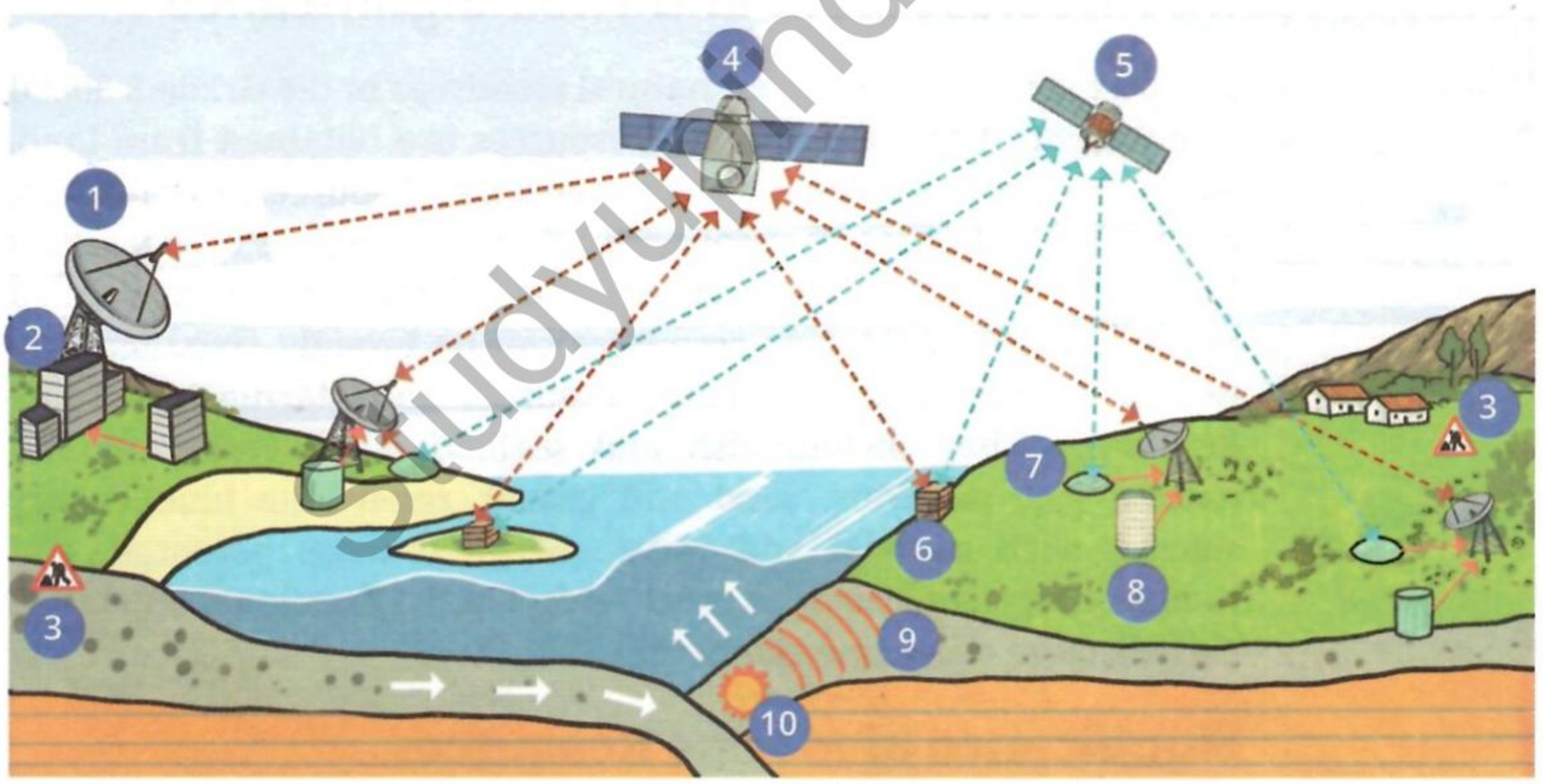
Detection
A network of seismic stations detects undersea earthquakes in real time.

Confirmation
Ocean Bottom Pressure Recorders (OBPRs) placed on the seafloor detect the passage of tsunami waves, confirming the threat.

Tracking
Coastal tide gauges and radars monitor changes in sea level to track the arrival and movement of tsunami waves.

Warning Dissemination
Alerts are transmitted via satellite to government agencies and are also sent directly to officials and the public through SMS and other communication channels.

Fig. 1.15. Functioning of the tsunami early warning system



1 Warning centre	2 Modelling/ simulation	3 Capacity development	4 Communication satellite	5 GPS satellite
6 Tide gauges station	7 GPS station	8 Seismometer station	9 Seismic waves	10 Earthquake

Fig. 1.16. The different elements of the tsunami early warning system

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For an early warning system to save lives, information must reach the public quickly and effectively. The Indian Tsunami Early Warning Centre (ITEWC) is tasked with sending alerts to the Government of India, which then informs state governments and emergency response teams. Reliable communication is ensured through satellite-based communication networks, SMS, emails, and coastal display boards.

Besides technology, public awareness and training programmes are equally important for an effective early warning system. Educating officials and coastal communities on how to interpret warnings helps ensure that when disasters such as cyclones are predicted, people know how to respond appropriately.

THINK ABOUT IT

If you receive a tsunami or cyclone alert on your phone right now, do you know the five steps your family should take to stay safe?

Marine Resources and Their Significance

Do you recall learning about natural resources in the Grade 8 Social Science Part 1 textbook? These resources are obtained from land, water, forests and animals, and other natural sources. Oceans, too, are vast storehouses of resources, and the materials and features found in oceans that have intrinsic or economic value are called marine resources. Marine resources can be broadly classified into two categories—Marine Living Resources and Marine Non-living Resources. They include fish and seafood, biodiversity, oil and natural gas, minerals, sand and gravel, renewable blue energy sources such as tidal and wave energy, tourism potential, and unique ecosystems such as coral reefs. Fig. 1.17 presents a detailed classification of marine resources.

Significance of Marine Resources

Marine resources are vital for food security, economic activities, and maintaining a healthy environment (Fig. 1.17). Countries with long coastlines, such as India, benefit greatly from the sea. Fishing, trade, and tourism provide livelihoods to millions of people, while oceans also support biodiversity and sustain numerous plant and animal species. However, marine resources must be used responsibly without disturbing marine ecosystems.

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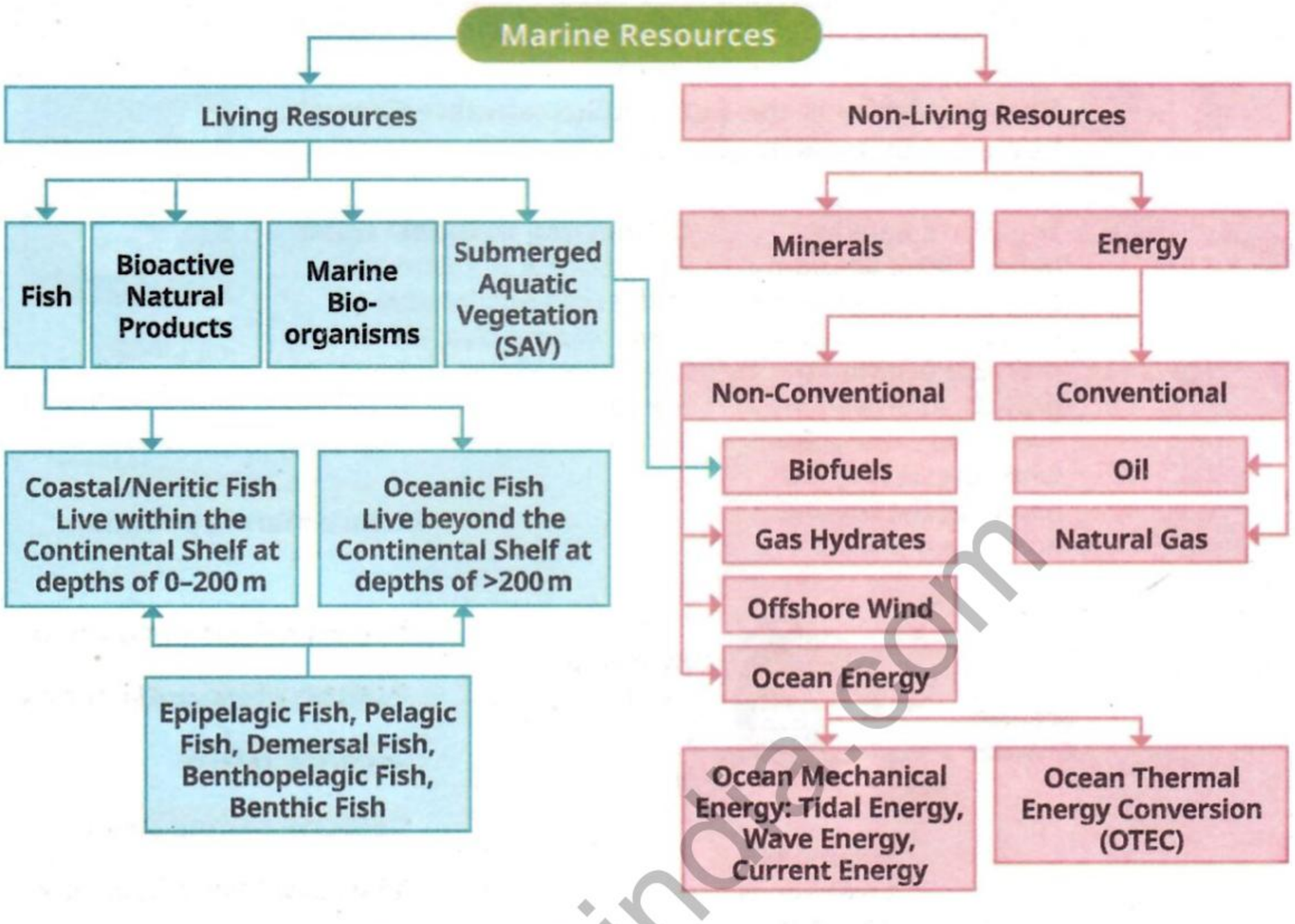

Fig. 1.17. Classification of marine resources

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The Ocean is a Vast Storehouse of Resources Vital for Life, Livelihoods, and Culture

Marine resources hold immense value for humanity and for coastal nations like India, they are the bedrock of local economies and traditions.

Food and Livelihood: Fish are a major source of protein, while fishing and aquaculture support important industries and livelihoods.

Minerals and Materials: Sand, gravel, and deep-sea minerals.

Energy: Offshore oil and gas reserves; immense potential for renewable 'blue energy' from tides and waves.

Trade and Tourism: 90% of global trade by volume is carried by sea; coastal tourism is a multi-billion dollar industry.

Cultural Significance:

The sacred *śhaṅkha* (conch) and *kapardikā* (cowrie shells), once used as currency, are regarded as divine treasures from the mythological *Samudra Manthana*, reflecting the close connection among the sea, prosperity, and culture.

Fig. 1.18. Significance of marine resources

DON'T MISS OUT

Blue Economy is the Future: Sustainable Growth Powered by Healthy Oceans

The World Bank defines **Blue Economy** as "the sustainable use of ocean resources to promote economic growth, improved livelihoods, and jobs, while preserving the health of the marine ecosystem".

Connecting to Global Goals: This concept is enshrined in the UN's Sustainable Development Goal (SDG) 14: Life Below Water. Its key aims are to:

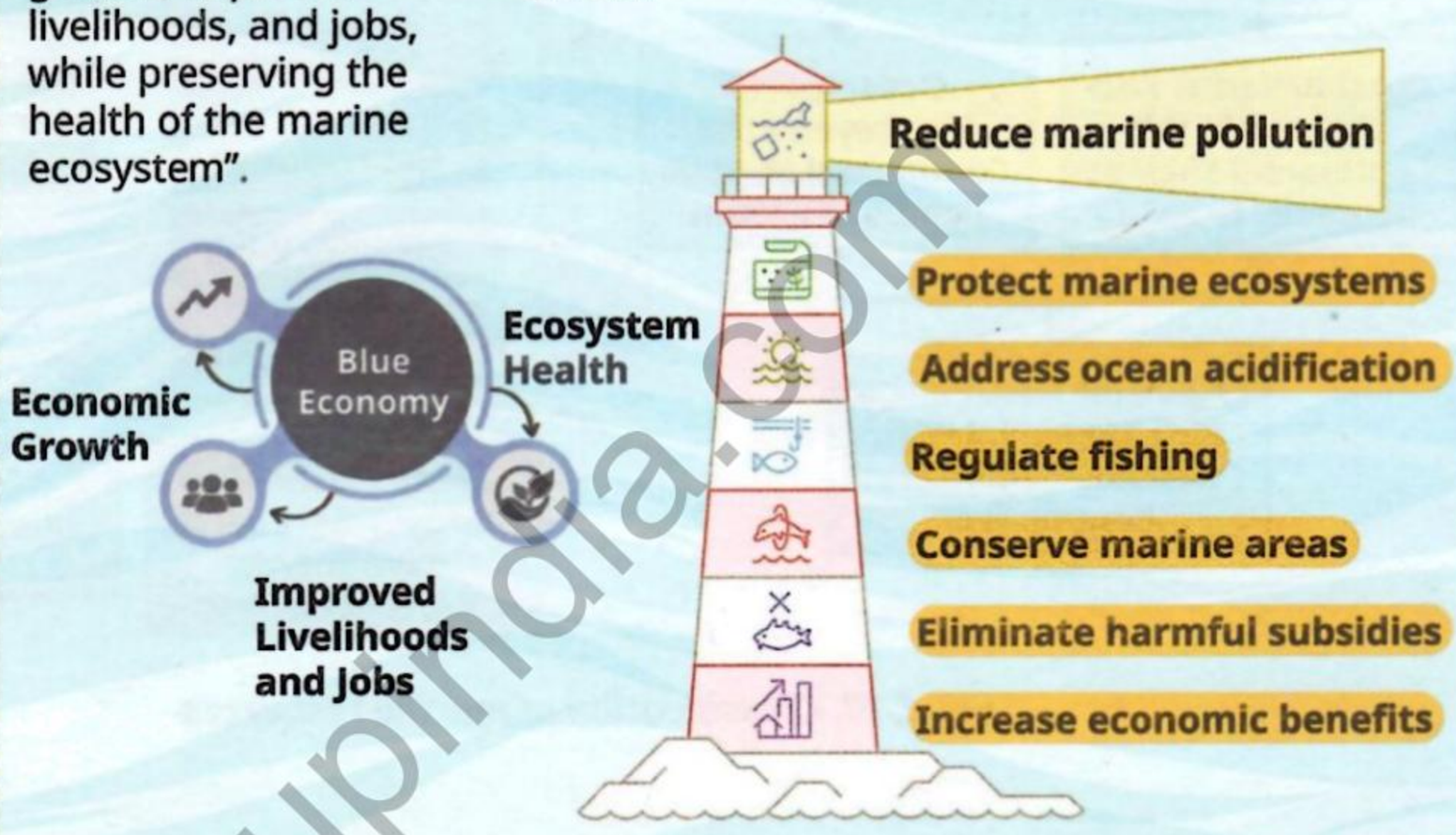

Fig. 1.19. The role of Blue Economy

Maritime Zones

Have you ever wondered who owns the oceans and how countries decide where they can fish or travel? To understand this, we need to learn about maritime zones—different areas of the ocean defined by international law.

The sea area closest to a country is called its territorial waters, over which the country has near-complete control, subject to the right of innocent passage for foreign ships. Just as on land, the government can make laws relating to customs, immigration, and pollution control in this zone.

Observe Fig. 1.20 and note the extent of the territorial sea in **nautical miles**. Beyond this lies the contiguous zone, extending from the baseline up to 24 nautical miles, where the country has limited control over customs, fiscal matters, and illegal immigration.

Nautical mile: A unit of measurement used in maritime and air navigation. It is defined as exactly 1,852 metres (about 6,076 feet). Based on the Earth's circumference, one nautical mile corresponds to one minute of latitude making it a convenient unit for charting and navigation across the world's seas.

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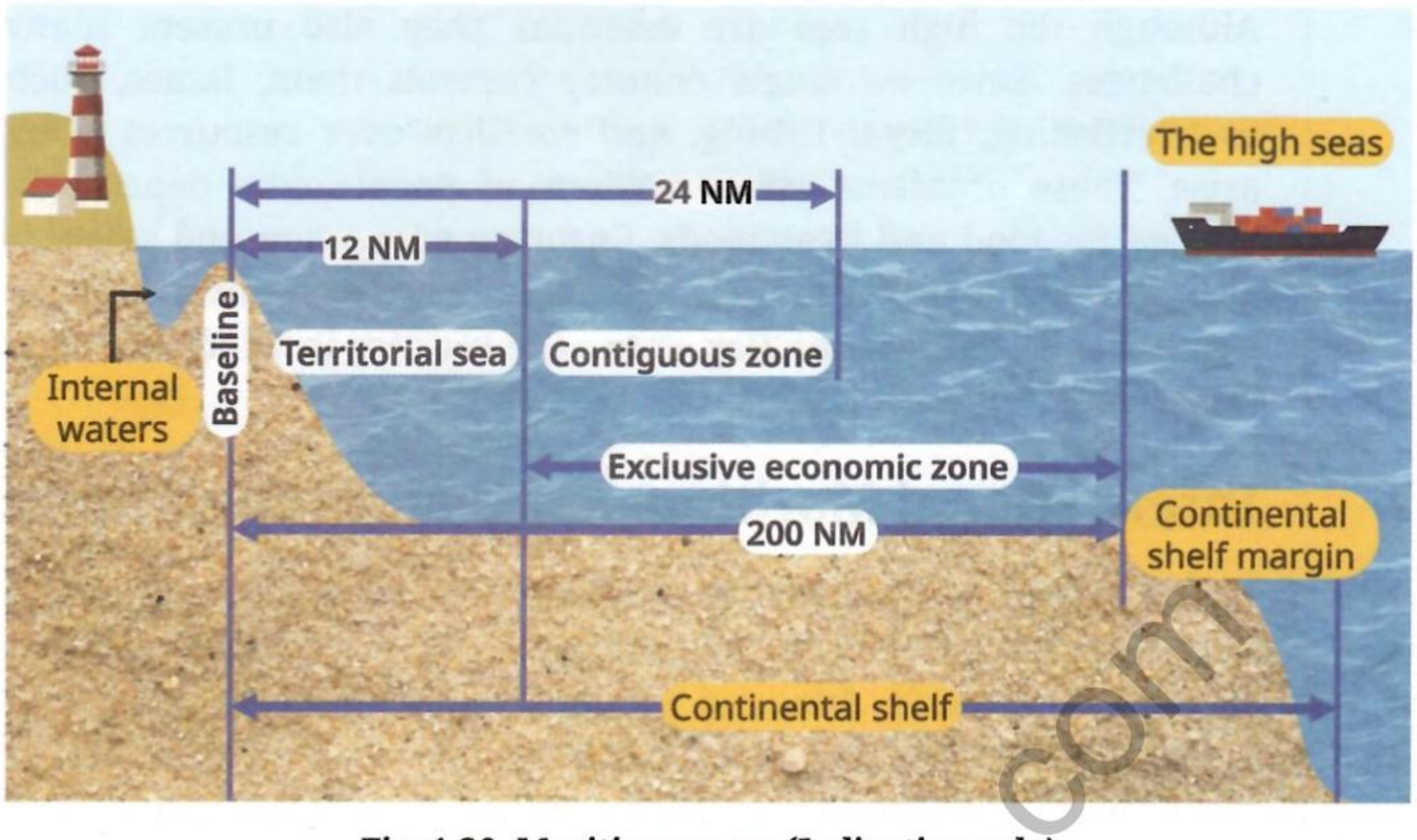

Fig. 1.20. Maritime zones (Indicative only)

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Innocent Passage is a principle that allows foreign ships to pass through the territorial waters of a country as long as they do not threaten its peace, good order, or security. Ships must move continuously without stopping, except in emergencies or cases of breakdowns. This right does not permit activities such as fishing, military operations, or other non-peaceful actions.

Beyond the contiguous zone, lie the Exclusive Economic Zones (EEZs), which extend from the baseline up to 200 nautical miles from the coast, and are defined under the United Nations Convention on the Law of the Sea (UNCLOS) by global agreement. In these zones, a country has special rights to use ocean resources, such as fish, oil, and gas, although ships from other countries may still pass through peacefully. It is important to note, however, that the country does not fully own or control the waters of the EEZ.

Beyond the EEZs of all countries lie the high seas, also known as international waters. Covering about 64 per cent of the world's oceans, these waters do not belong to any single country and are therefore considered part of the global commons. All countries enjoy equal rights on the high seas, including navigation, fishing, flying over them, laying undersea cables, building lawful structures such as artificial lands, and exploring for minerals.

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Although the high seas are essential, they also present many challenges. Since no single country controls them, issues, such as overfishing, illegal fishing, and conflicts over resources often arise. These problems affect millions of people who depend on the sea for food and livelihoods. Ensuring navigation and safety is also more difficult without clear enforcement mechanisms. On the high seas, the laws of the flag state—the country in which a ship is registered—apply.

International Maritime Rules and Regulations

In the earlier sections of this chapter, you learnt that oceans are vital for trade, travel, and communication. To ensure that activities at sea remain safe, fair, and peaceful, countries follow a set of international rules and agreements known as maritime law or maritime regulations. These laws regulate how ships operate and move; protect the marine environment; address issues such as piracy, terrorism, natural disasters, and accidents; safeguard trade routes; and help settle disputes between countries.

LET'S EXPLORE

Maritime disputes are conflicts or disagreements between countries over territory, resources, or rights in naval areas such as oceans, seas, or coastal regions. Can you find out some countries that have conflicts over their territorial sea waters?

Important Maritime Rules and Regulations

Just as there are traffic rules on land, ships must follow "rules of the road at sea" to avoid accidents. These rules, known as COLREGs (Convention on the International Regulations for Preventing Collisions at Sea), include maintaining a proper lookout, deciding which side to pass on, identifying the give-way and stand-on vessels, proceeding at a safe speed, and using lights, sound signals, shapes (for different types of vessels such as fishing boats), along with appropriate actions to prevent collisions. In addition, ships use international signals and radio communication to share information and prevent accidents, especially in crowded or hazardous waters.

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Ships must also meet safety standards and are prohibited from polluting the seas. The MARPOL (International Convention for the Prevention of Pollution from Ships) Convention is an important international agreement aimed at reducing pollution from ships.

Why are Maritime Rules Important?

International maritime rules, regulations, and agreements help keep the oceans safe and ensure the fair and sustainable use of ocean resources. These rules enable countries to cooperate in using marine resources without conflict while protecting marine life, ships, sailors, and coastal communities. Without such regulations, the oceans could become dangerous and chaotic, with increased pollution, accidents, collisions, and disputes over resources. By establishing common standards for all ships, regardless of the country they belong to, maritime rules help keep the oceans safe, clean, and open for trade and travel.

Before we move on...

- Oceans are vital for global trade, travel, communication, and climate regulation.
- Submarine relief refers to the physical features of the ocean, such as the continental shelf, ridges, trenches, and seamounts, which influence marine life, resource availability, and undersea exploration.
- Movements of ocean water—waves, tides, and currents—affect marine ecosystems, global climate, and human activities like fishing and navigation.
- Maritime zones include territorial waters, EEZs (Exclusive Economic Zones), and the high seas.
- Sustainable use of marine resources through cooperation, regulation, and conservation is essential to protect aquatic life and support future generations.
- Maritime rules and regulations exist to resolve disputes over marine boundaries.
- International laws, such as UNCLOS (United Nations Convention on the Law of the Sea), help define maritime zones and settle disputes among countries.

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Questions and activities

1. How do different features of the ocean floor affect human activities such as fishing, transport, and underwater exploration?
2. How do waves, tides, and ocean currents influence climate, marine life, and human activities around the world?
3. Describe the importance of marine resources.
4. Draw a poster that encourages people to protect marine resources. Include a slogan (like Save Our Seas or Fish for the Future) and explain your message in 2–3 sentences.
5. Using a map of India, mark at least five coastal states that might need tsunami warnings.
6. Work in a group and act out what would happen when a cyclone warning is issued. Highlight what role different people, like scientists, officials, and coastal villagers, would play.
7. Suppose a foreign ship is sailing through India's territorial sea. What principle allows the ship to pass through these waters, subject to certain conditions?
8. A country has an Exclusive Economic Zone (EEZ) extending up to 200 nautical miles from its baseline. What rights does the country have over the natural resources in this zone?
9. Create a model of an ocean with ship routes, protected zones, and safety measures. Based on the model, explain how it follows maritime laws and protects the ocean.

Notes

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Chapter 2

Life on Earth

The Big Questions

1. *How do physiography and climate shape life on Earth?*
2. *How are humans and other life forms interconnected?*
3. *What is the importance of preserving various life forms, and how can we contribute to their conservation?*

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Geography helps us understand how climate, landforms, and water shape life on Earth—and how humans adapt their lives to these conditions. From forests and grasslands to deserts and coastal regions, different environments support different forms of life. At the same time, human activities influence these natural systems in significant ways. This chapter explores the close relationships among physical features, climate, and living organisms, and encourages us to reflect on our responsibility towards preserving the diversity of life on Earth.

Imagine our planet as a giant patchwork quilt, where each patch has its own colours, weather, and living beings. These huge patches are called **biomes**. The word 'biome' is made up of the word *bio*, derived from the Greek word *bios*, meaning 'life', and *-ome*, a suffix used in biology to denote a group of a specific type of biological constituents. Biomes are large natural regions of the world where certain types of plants and animals live under similar climatic conditions. For example, icy tundras, hot deserts, green forests, and vast grasslands are all different types of biomes. In simple terms, a biome is like a broad environmental region, while an **ecosystem** is like a neighbourhood within it, where plants, animals, and non-living things interact closely. Each biome is unique, but together they form the living world of our planet, supporting diverse forms of life and helping maintain the balance of nature.

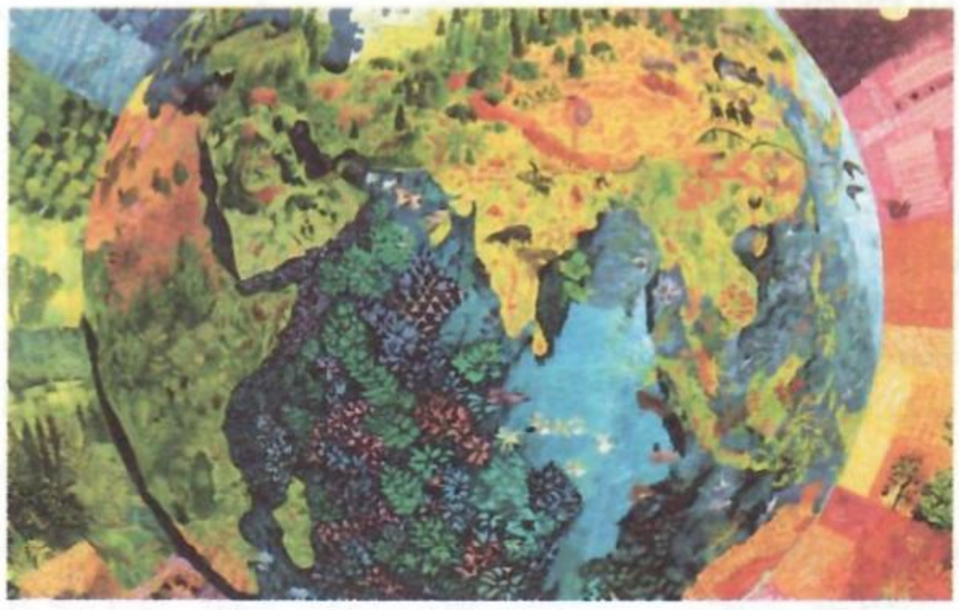

Fig. 2.1. Earth as a giant patchwork quilt

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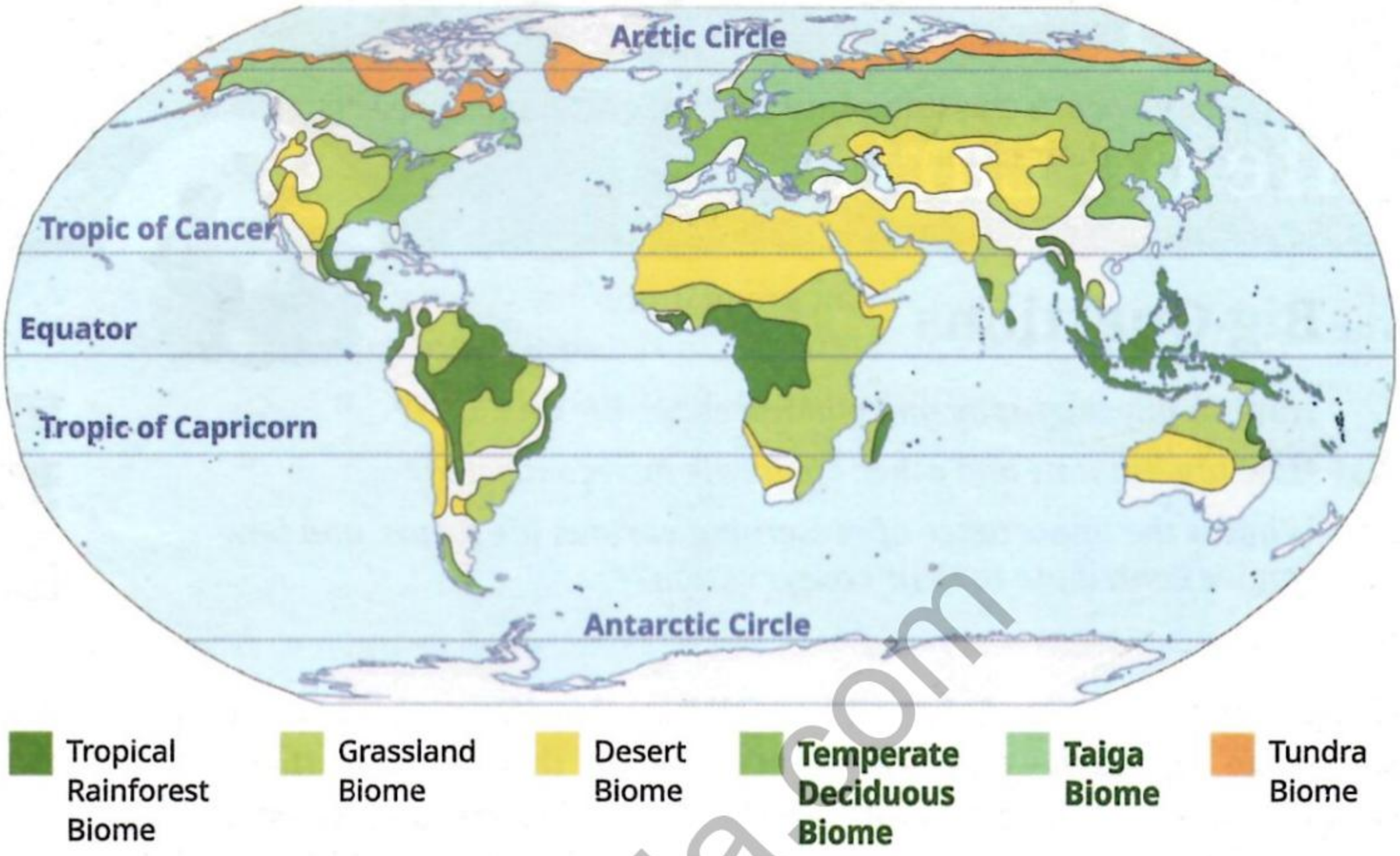

Fig. 2.2. World map showing distribution of biomes (map not to scale)

Temperate deciduous biome: This biome is characterised by forests of broad-leaved trees that shed their leaves seasonally, with moderate temperatures and distinct seasons.

Taiga biome: This biome is characterised by coniferous forests consisting mainly of pines, spruces, and larches.

LET'S MAP

Look at the world map in Fig. 2.2 and answer the following questions.

→ Which type of biome is found in smaller, separate regions, and why?

→ Which continent has the least number of biomes? What could be the reason for this?

→ Why are there areas that do not have any specific biomes?

Types of Biomes

Now that you have understood the meaning and uniqueness of biomes, let us identify the different types of biomes. Different **biomes** support diverse plant and animal life and can be categorised into **terrestrial** (land-based) and **aquatic biomes** (water-based).

There are several factors that determine which types of plants and animals can survive and thrive in a biome. These include temperature, precipitation patterns, dominant plant species, growth forms of vegetation, etc.

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Terrestrial Biomes

Terrestrial biomes are classified into forests, grasslands, deserts, and tundra, based on their climate.

Forest Biome

A forest biome is like nature's giant green umbrella—warm, wet, and bursting with life at every level, from the tallest trees to the damp forest floor. **Tropical rainforests** are mainly found near the Equator, between the Tropic of Cancer and the Tropic of Capricorn (23.5° N to 23.5° S). These regions remain hot and wet throughout the year, which supports the growth of dense forests. The largest tropical rainforests are located in the Amazon Basin in South America, the Congo Basin in Africa, and parts of Southeast Asia, such as Indonesia and Malaysia. Smaller rainforest regions are also found in Central America, as well as in India's Andaman & Nicobar Islands, and the Northeastern states. Together, these areas form the world's richest storehouse of biodiversity with rich genetic resources. They preserve unique genetic material, supporting conservation and scientific research, and are hence also known as a 'gene bank'. Forests also play an important role in maintaining oxygen levels and balancing the Earth's climate.

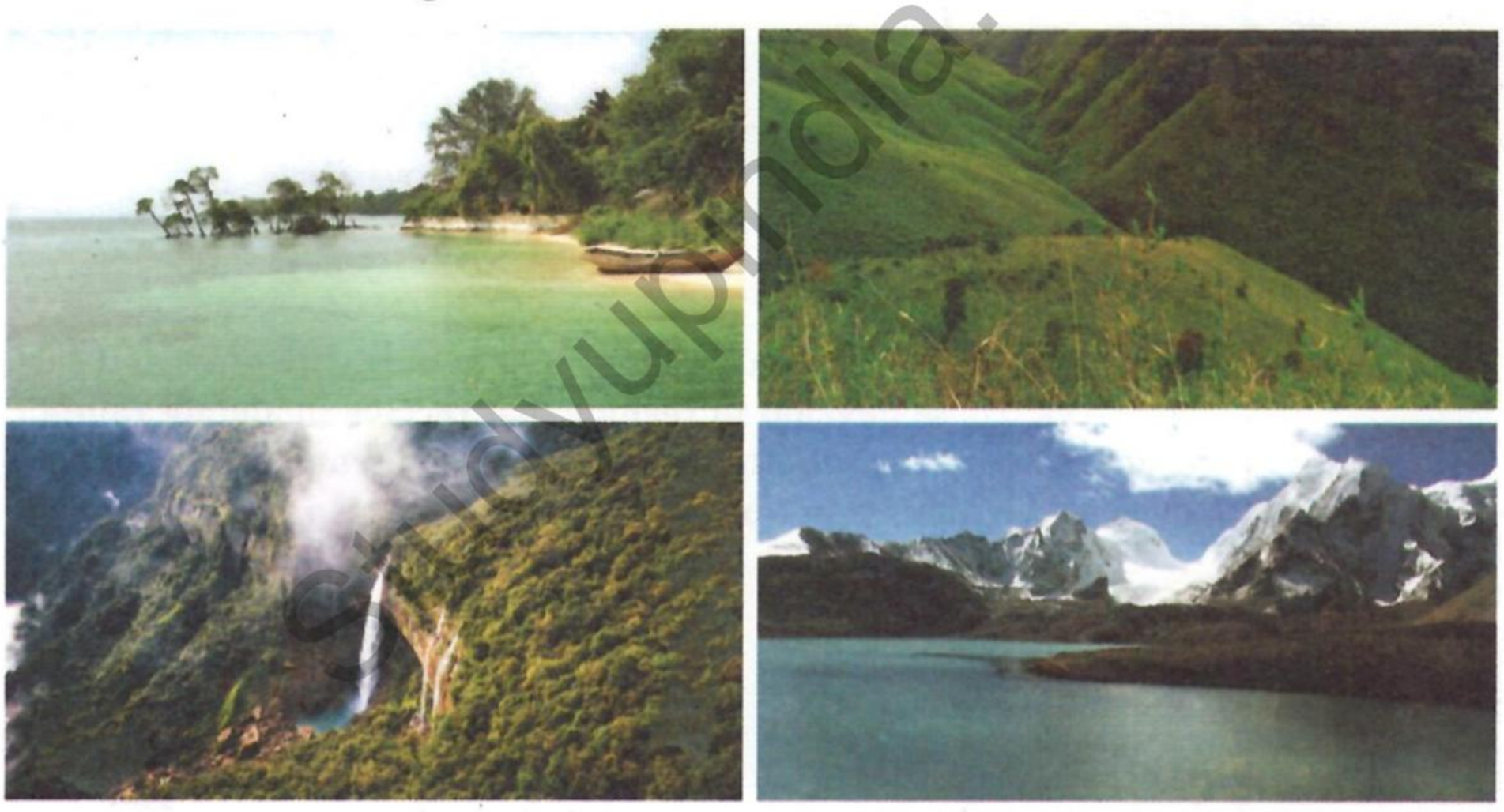

Fig. 2.3. (Left to right, top to bottom) A beach, Andaman Islands; Dzukou Valley, Nagaland; Nohkalikai Falls, Sohra; Gurudongmar Lake, Sikkim

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LET'S EXPLORE

Look at the layers of vegetation in a rainforest. What do the yellow shapes in the figure indicate?

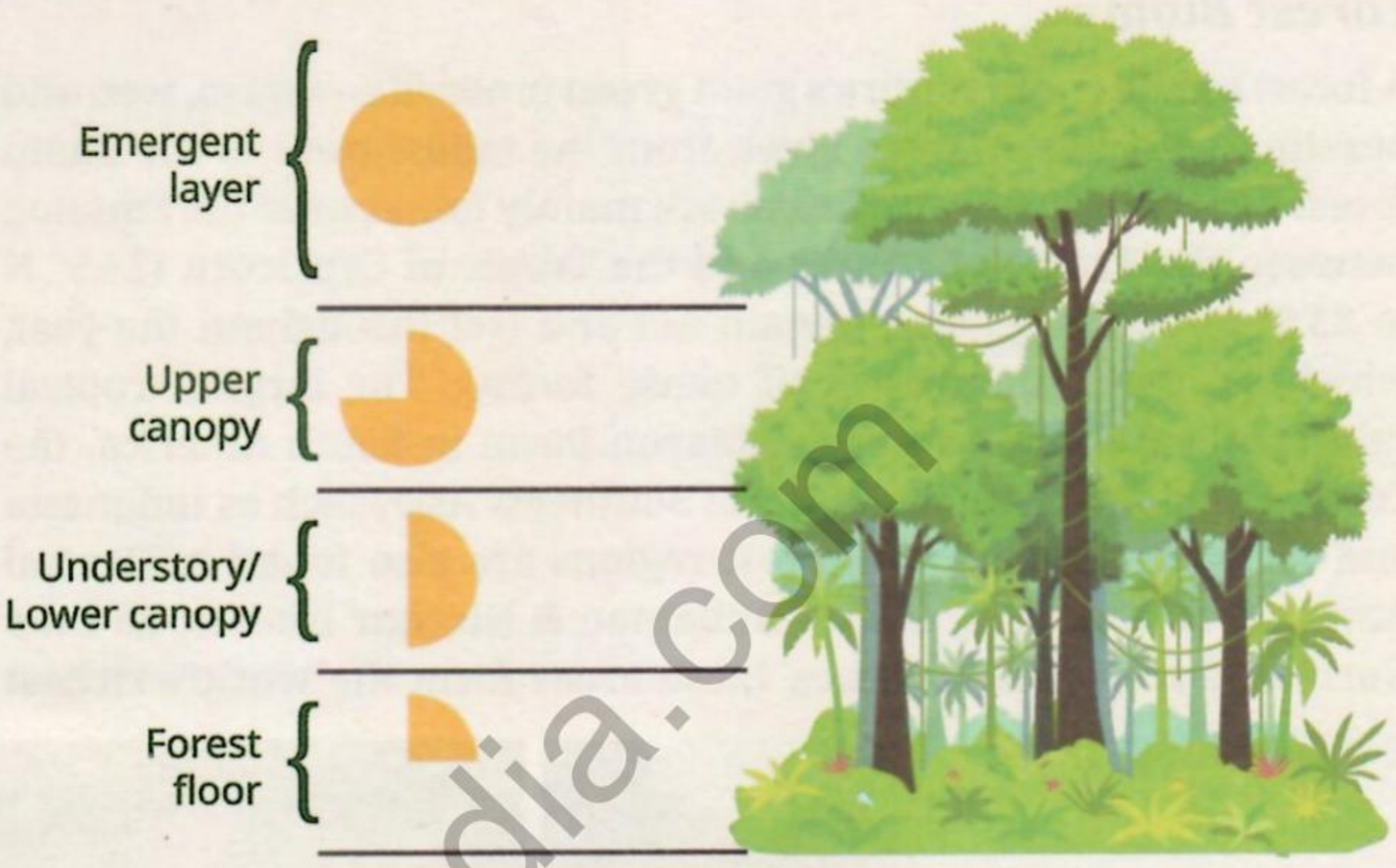

Fig. 2.4. Ideal structure of a rainforest biome

Grassland Biome

Grasslands are large open areas covered mainly with grass and very few trees. These develop in regions that receive moderate rainfall, which is not enough to support dense forests. Temperate grasslands (such as the prairies in North America and the steppes in Asia) have

Fig. 2.5. Typical grassland landscape

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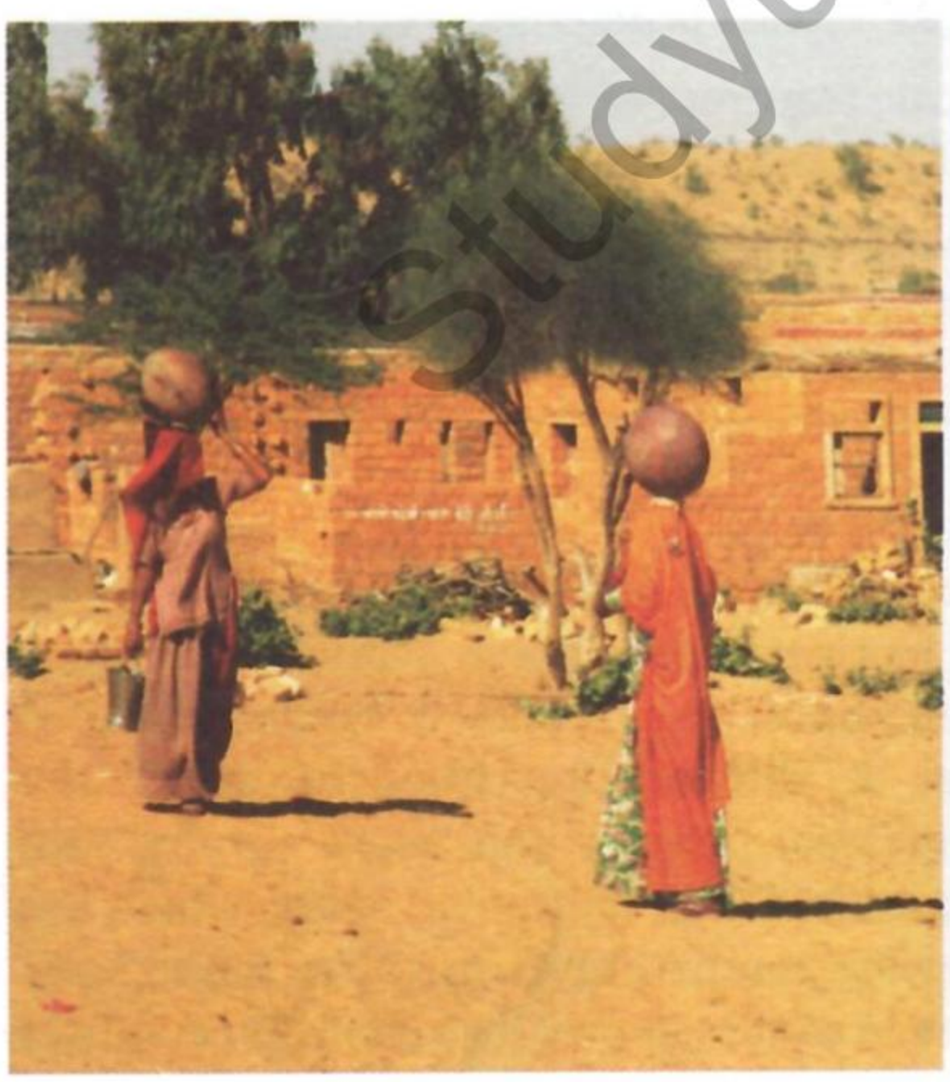
fertile soil and are often used for farming. Tropical grasslands, also known as Savannas (as in Africa), have scattered trees and are home to large animals such as lions, elephants, and giraffes. Grasslands support grazing animals and are important for human activities like agriculture and livestock rearing.

LET'S EXPLORE

Observe Fig. 2.5 and explore which of these animals you will find in the grasslands of India.

Desert Biome

Deserts are the driest biomes, receiving very little rainfall. They can be hot, like the Sahara in Africa, or cold, like the Gobi in Asia. The Rann of Kutch in Gujarat is one of the world's largest salt deserts, turning into a marshy wetland during the monsoon and a vast white salt desert during the dry season. The extreme conditions of a desert make it difficult for life to survive; however, plants like cacti and animals such as camels, lizards, and snakes are specially adapted to these conditions. They can store water and withstand heat and dryness. Deserts often have sandy soil and sparse vegetation. Despite their harshness, they are home to unique ecosystems, and many of them also provide valuable resources such as minerals and oil.



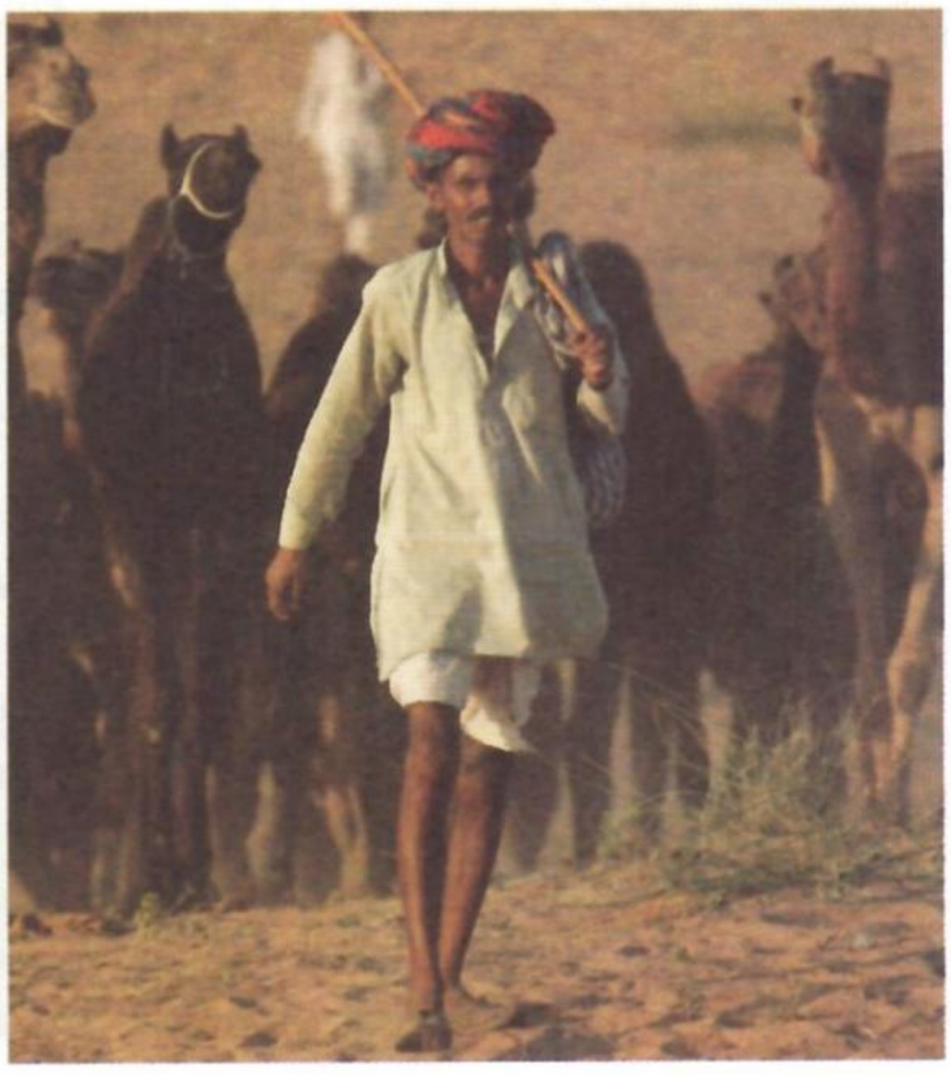


Fig. 2.6. Life in the Thar Desert

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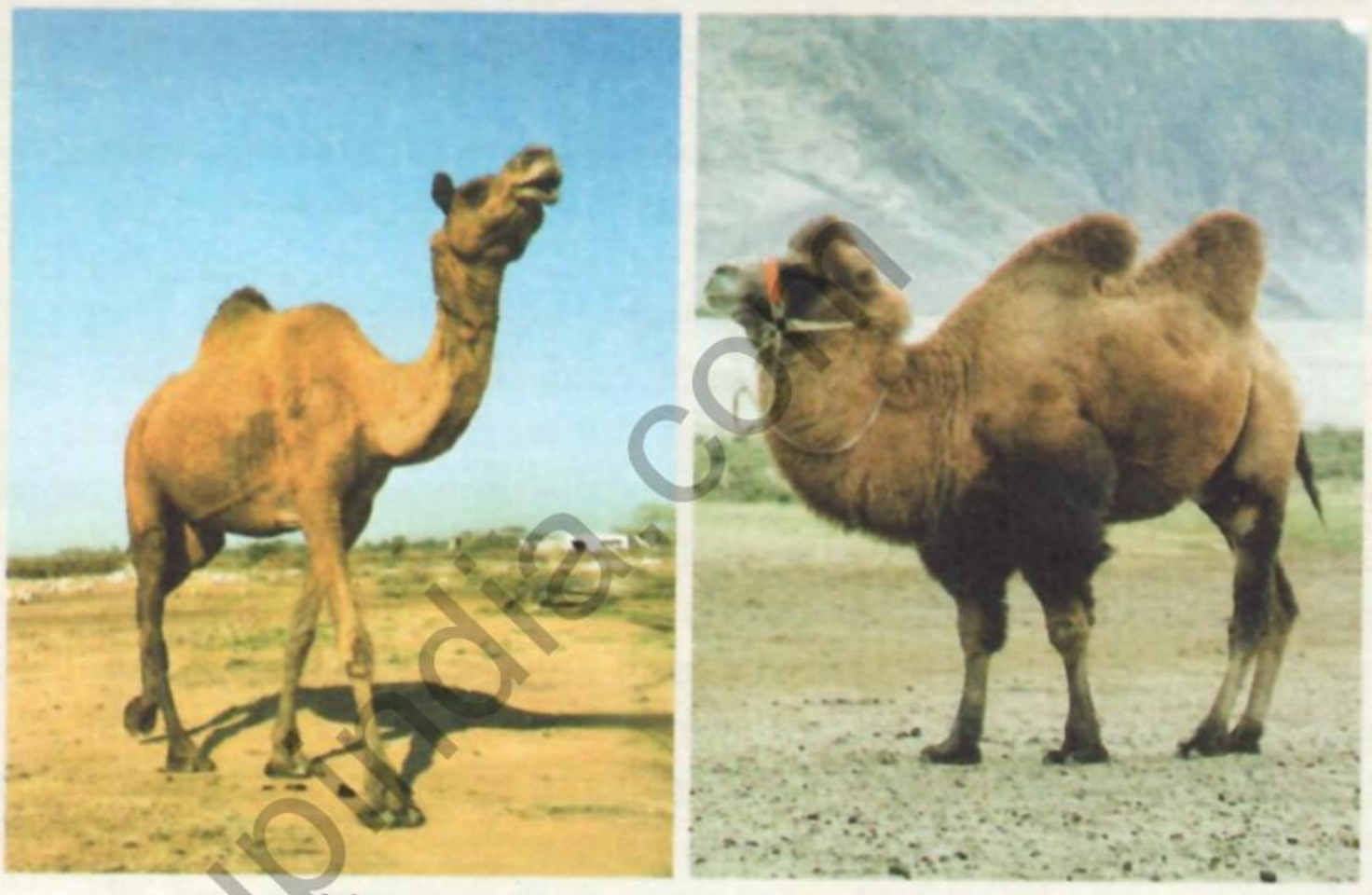
LET'S EXPLORE

→ What types of activities do you think people in desert regions generally undertake to sustain their households and livelihood?

→ With the help of your teacher, explore how do people address issues related to the availability of water.



(a) (b)

Fig. 2.7. Camels in deserts

→ Compare the physical features of the Rajasthan camel (Fig. 2.7a) and the Nubra Valley (Ladakh) camel (Fig. 2.7b).

→ Considering the climatic conditions, why do you think the Nubra Valley camel might have a different fur density than the Rajasthan camel?

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Tundra Biome

The tundra biome is the coldest of all and is found near the Arctic and Antarctic regions. It has very long, harsh winters and short summers. The soil is frozen almost year-round—a condition called permafrost—which makes it difficult for trees to grow. The vegetation is limited to mosses, lichens, and small shrubs. And animals, such as reindeer, arctic foxes, and polar bears, which are well adapted to survive freezing temperatures, are often found here. The tundra is fragile, and even small temperature changes can affect its ecosystem.

Fig. 2.8. Tundra (left) and the Himalaya (right)

THINK ABOUT IT

Tundra in polar regions has mosses, lichens, shrubs, etc., while the Himalaya, in the subtropics, is dominated by conifers. What could be the reason for this? What difference do you find in the animals found in these two regions? Which species are found only in the Himalaya? Discuss in class.

Aquatic Biomes

Aquatic biomes cover about three-fourths of the Earth's surface and are divided into freshwater and marine biomes (Fig.2.9). **Freshwater biomes** include rivers, lakes, ponds, and wetlands. These are crucial for agriculture, getting access to drinking water, and supporting species, such as fish, frogs, and insects. **Marine biomes** include oceans, seas, and coral reefs, which are rich in biodiversity and influence the Earth's climate by storing and distributing heat. They are also home to countless species, including whales, dolphins, fish, and plankton, and provide food and livelihood for millions of people.

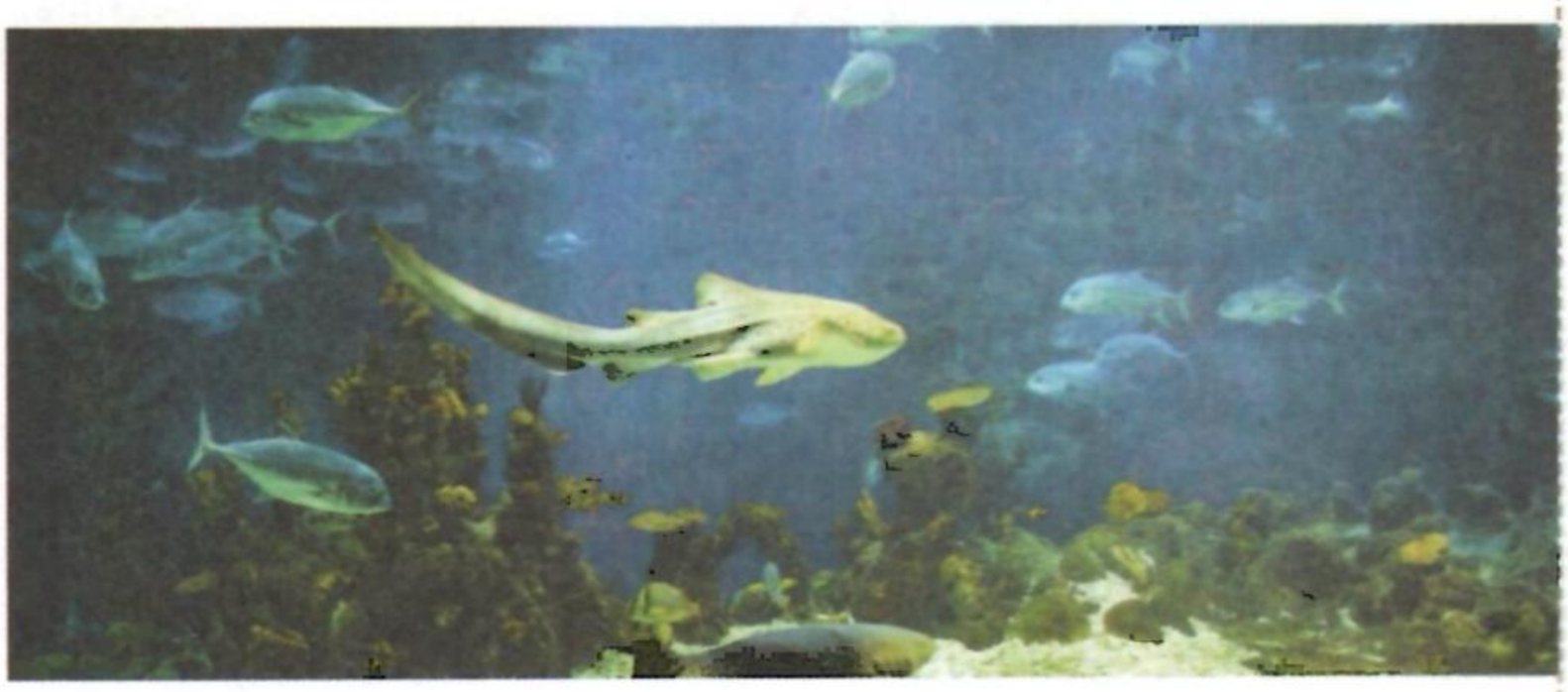

Fig. 2.9. The diversity of the aquatic biome

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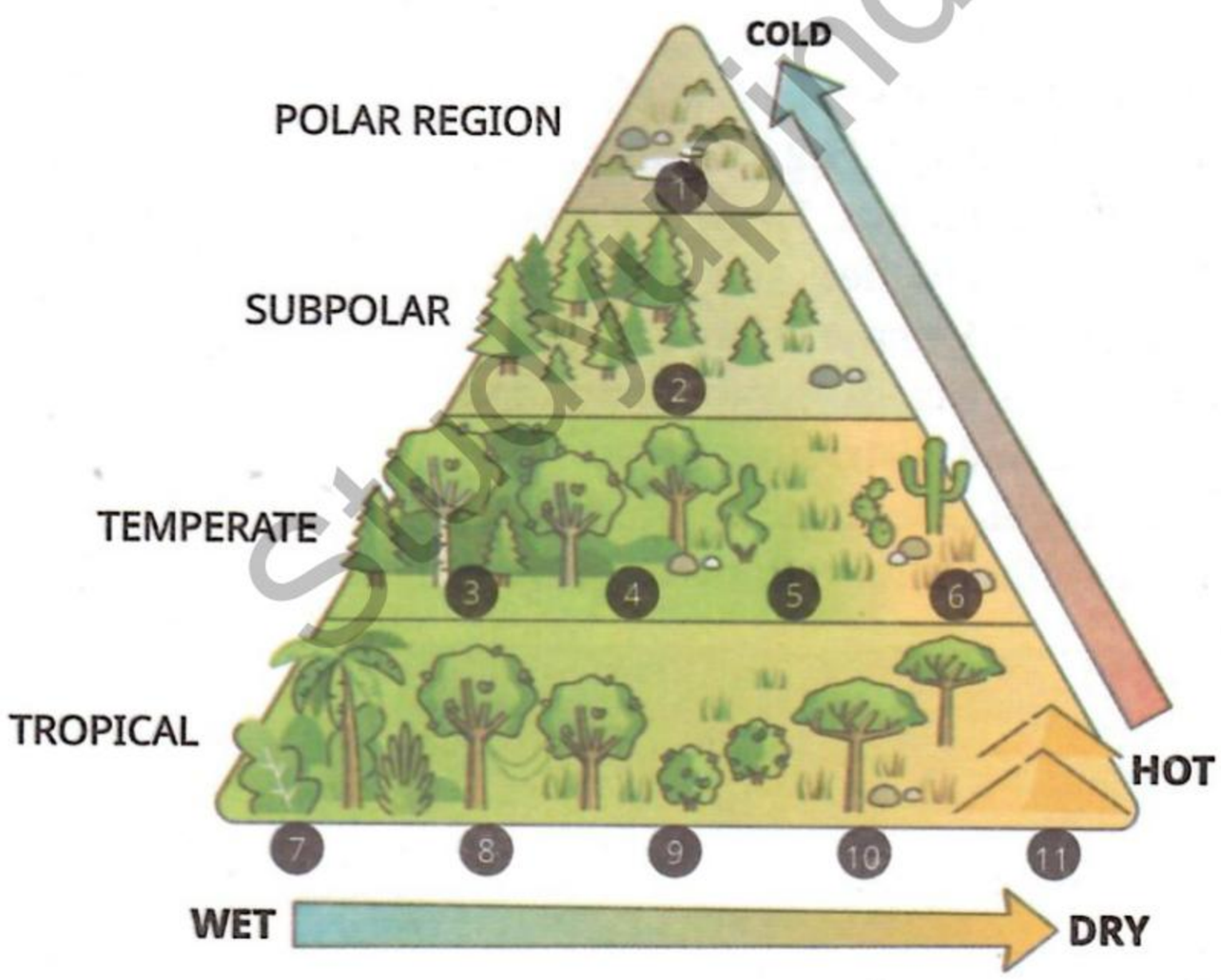
LET'S EXPLORE

Explore your local environment. This could be your village, city, town, or residential area. Observe how people, plants, and animals build structures and interact. Complete the table with your findings.

Component type	Examples observed	Interaction with humans
Trees/Vegetation/Plants		
Insects		
Animals		
Non-living things		

Biome Pyramid

A **biome pyramid** is a graphical representation that illustrates the relationship between different organisms within a biome, usually in terms of energy, number, or biomass. At the base of the pyramid



1 Tundra	**6** Desert
2 Coniferous Forest (Boreal Forest)	**7** Rainforest
3 Deciduous Forest	**8** Scrubland
4 Chaparral	**9** Tropical Seasonal Forest
5 Grassland	**10** Savanna
	11 Desert

Fig. 2.10. Biome pyramid

are the producers, such as plants and algae, which capture solar energy through photosynthesis. The next levels consist of herbivores (primary consumers), carnivores (secondary consumers), and top predators (tertiary consumers). As we move up the pyramid, the

Study Tripathi

number of organisms, biomass, and available energy decrease as energy is lost at each trophic level as heat during metabolic processes. Thus, a biome pyramid helps in understanding the structure of ecosystems, the flow of energy, and the balance between different living organisms in a biome.

LET'S MAP

Mark the following regions at appropriate levels on the Fig. 2.10: Andaman & Nicobar Islands, Keralam, Ladakh, Chhattisgarh, Odisha, Meghalaya, and Lakshadweep. Where will you place your home?

There is no arrow on the third side of the triangle. What would happen if we drew an arrow? Would it point upwards or downwards?

Interconnectedness between Biomes and Living Beings

Biomes are closely connected to living beings because they provide a natural environment where plants, animals, and humans can survive and interact. At the same time, living beings also influence biomes—animals help in pollination and seed dispersal, humans modify landscapes for farming or cities, and microorganisms recycle nutrients. In this way, biomes and living beings are interconnected systems that depend on one another to maintain ecological balance and biodiversity. We will discuss some of the aspects of this interconnectedness in the following section.

Maintaining Ecological Balance and Biodiversity

Biodiversity

Biodiversity is the variety of life on Earth, including plants, animals, microorganisms, and the ecosystems they inhabit (Fig. 2.11). It refers not only to the number of different species but also to the genetic differences within species and the variety of habitats and ecological processes that sustain life. In simple terms, biodiversity means the richness of life forms and the natural balance they create, which is essential for maintaining healthy ecosystems and supporting human survival.

Fig. 2.11. Biodiversity in wild animals

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The **geographical aspects of biodiversity conservation** focus on how location, climate, landforms, and ecosystems influence the protection and management of living organisms. Since biodiversity is unevenly distributed across the Earth, geography plays a vital role in conservation. Some key aspects of conservation are—

Hotspots of Biodiversity: Certain regions, such as the Himalaya, the Western Ghats, the North–East India, and the Andaman & Nicobar Islands, are rich in unique species found nowhere else. Identifying and protecting these hotspots is a priority in conservation.

Fig. 2.12. (Left to right, top to bottom) Endangered Nilgiri Tahr Wild Goat; Himalayan Snowcock Bird; Indian Spider, Western Ghats; Big Himalayan Yak; Mangrove Forest, Andaman Islands

Ecosystem-Based Conservation: Different biomes (forests, wetlands, deserts, coasts, and mountains) support distinct life forms. Conservation strategies must suit the geographical setting; for example, mangrove protection along coasts or afforestation in mountains.

Protected Areas: National parks, wildlife sanctuaries, biosphere reserves, and World Heritage Sites are chosen based on geography to preserve critical habitats and prevent habitat fragmentation.

Geographical Barriers and Corridors: Natural features like rivers, mountains, and valleys serve as barriers or corridors that influence animal movement and gene flow. Conservation planning uses this knowledge to create wildlife corridors.

Human–Environment Interaction: Geography also considers how local communities use land, water, and forests. Sustainable practices vary with geography, such as terrace farming in the hills or agroforestry in the plains.

In short, biodiversity conservation is an intrinsic part of geography because the survival of species depends on their location, habitat conditions, and human activity within those regions.

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LET'S MAP

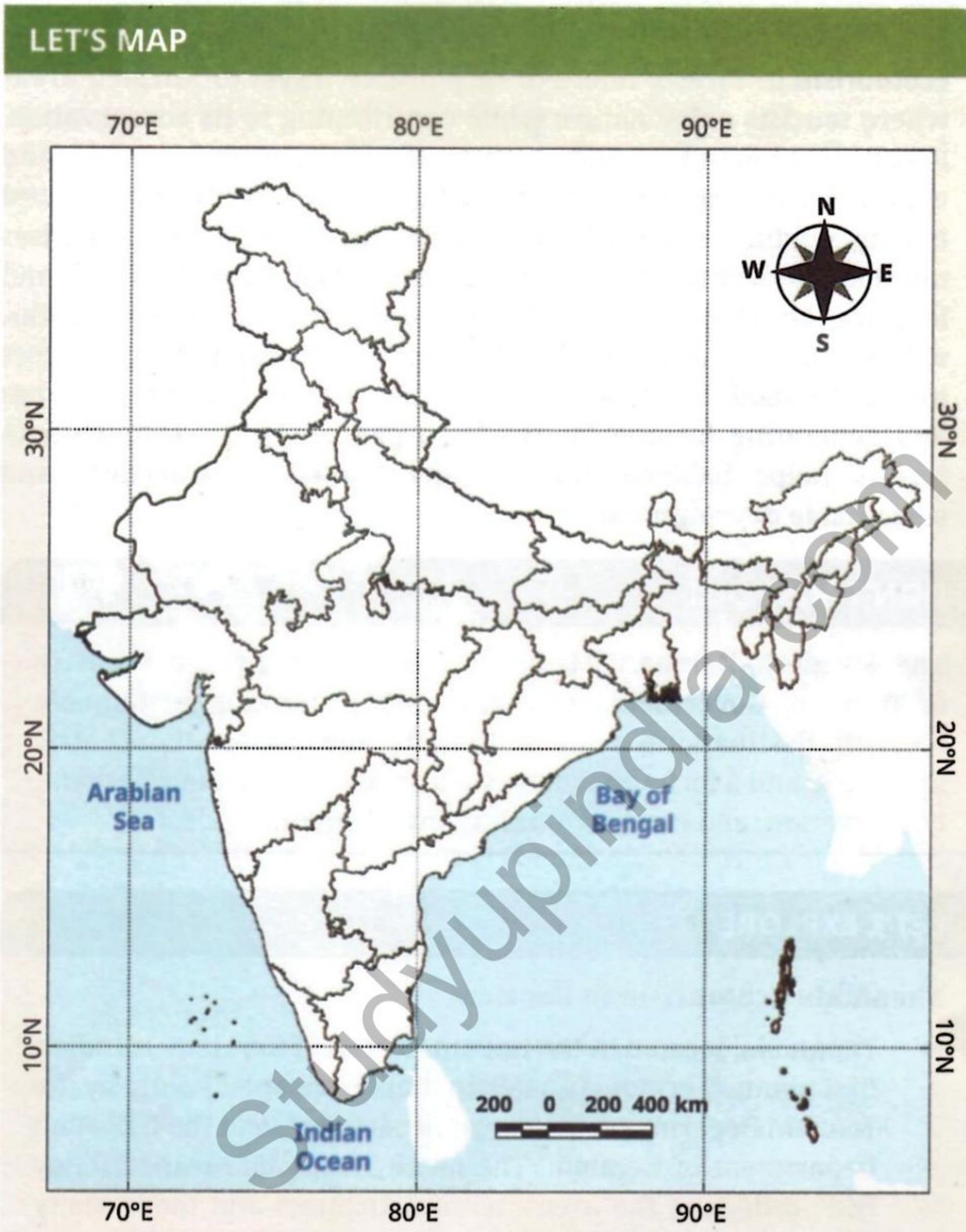

Fig. 2.13. Political Map of India

Mark and name the following regions on the map in Fig. 2.13. Also gather more information about these biodiversity regions from your teacher or other sources and discuss their features with your peers.

1. Namdapha National Park
2. Rann of Kutch
3. Gahirmatha beach
4. Gulf of Mannar
5. Pachmarhi
6. Achanakmar–Amarkantak

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Ecotourism: Travelling to nature-friendly places without harming the environment, and supporting local people.

Forest Ecotourism

Ecotourism in forests refers to responsible travel to forested areas where tourists enjoy nature while contributing to its conservation. Forests are home to a wide variety of plants, animals, rivers, and waterfalls, which attract people who want to experience natural beauty, wildlife, etc. Unlike mass tourism, ecotourism emphasises minimal environmental impact, respect for local communities, and learning about ecosystems. For example, visiting national parks, wildlife sanctuaries, or trekking through rainforests allows visitors to observe biodiversity while supporting conservation programmes and generating income for the local people. Thus, ecotourism in forests helps balance tourism, environmental protection, and sustainable development.

DON'T MISS OUT

The Swadesh Darshan 2.0 Scheme, launched by the Ministry of Tourism, Government of India, aims to develop sustainable tourism destinations with a tourist- and destination-centric approach, and a focus on factors such as sustainable development, conservation, and local community involvement.

LET'S EXPLORE

Thenmala Ecotourism in Keralam

→ Thenmala, located in the Kollam district of Keralam, is India's first planned ecotourism destination, developed jointly by the Tourism Department, the Forest Department, and the Irrigation Department of Keralam. The name *Thenmala* means 'Honey Hill', reflecting the area's natural richness and biodiversity within the Western Ghats—a UNESCO World Heritage Site. The project was launched to promote sustainable tourism while conserving forests and supporting local communities, by employing the inhabitants as guides, eco-guards, and service providers, ensuring that the tourism benefits are shared within the community. Thenmala's success lies in its balanced approach—combining recreation, environmental education, and conservation—making it a model for responsible forest-based ecotourism in India (Fig. 2.14).

→ Is there any such ecotourism site in your State/UT? Collect some information and discuss with your friends.

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→ What should be a tourist's behaviour while visiting any tourist site? What should one be mindful of?

Fig. 2.14. Exploring forest canopy in Thenmala

Forests and Livelihoods

People, particularly rural and tribal communities, rely heavily on forests for their livelihoods. Forests provide essential resources such as food, fuel, fodder, medicine, shelter, timber, bamboo, and firewood. Non-timber forest products, such as honey, fruits, nuts, resins, and medicinal plants, help families meet daily needs and earn income. Several communities including the Gonds and Bhils perform rituals to honour the forest deities before collecting resources, reflecting sustainable cultural practices and a deep respect for nature (Fig. 2.15). Forests also support livestock grazing, small forest-based industries, ecotourism, and employment, making them vital for the livelihood, culture, and survival of many.

Fig. 2.15. Rituals to honour the forest deities

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THINK ABOUT IT

Sacred groves are small forest patches protected by local communities for religious and cultural purposes. Across India, there are over 1,00,000 sacred groves, though many are not officially recorded. They are known by different names, such as *Devrai, Devarakadu, Kavu,* and *Sarna.*

Why are these groves so special? Despite being small, they are rich biodiversity hotspots, often preserving rare plants, medicinal species, and wildlife that may not survive elsewhere. Some groves are even linked to water sources and help to maintain streams and groundwater.

Timber and bamboo (Assam, Tripura, Keralam)	Wild fruits, honey, mushrooms, and roots (Jharkhand, Odisha)
Tendu leaves, mahua flowers, sal seeds, gums (Madhya Pradesh, Chhattisgarh)	Ecotourism (Uttarakhand, Assam, Keralam)
Herbs like ashwagandha, neem, and tulsi (Himalayan region, Western Ghats)	Handicrafts and cottage industries (Rajasthan, Keralam)
	You may add some examples of forest produces from your own state/UT in the map above.

Fig. 2.16. Forest produces from different Indian states (map not to scale)

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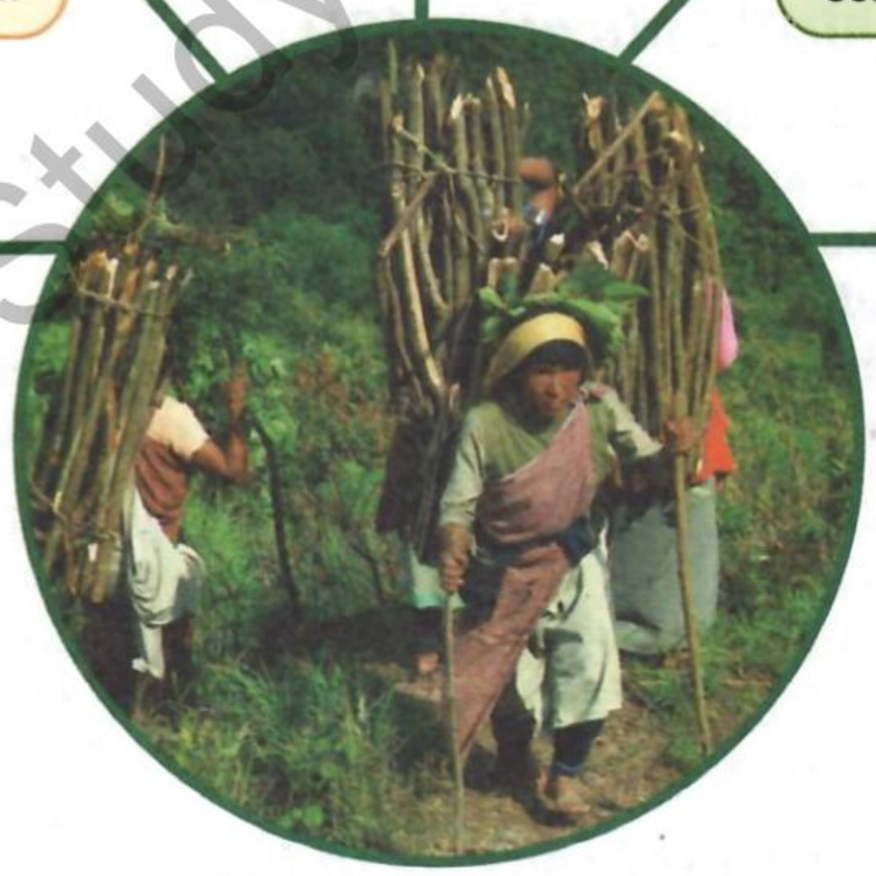
Forest Dwellers

Forest dwellers have a deep and multifaceted connection to forests as they rely on them for household needs, cultural practices, and livelihood activities. Through their roles and responsibilities, they are often deeply involved in caring for and using forest resources. Women, in particular, play a vital role in collecting and managing forest resources for household needs, sustaining traditional knowledge and practices, and contributing to the livelihoods. Some of these connections are:

Knowledge Holders: They possess rich traditional knowledge of using herbs, plants, and forest resources for food, medicine, and cultural practices. This knowledge is passed down through generations and helps in biodiversity conservation.

Livelihood and Income: They are actively engaged in gathering and processing non-timber forest products (NTFPs) such as tendu leaves, mahua flowers, honey, lac, and bamboo, which provide income for households.

Forest Protection: They have been at the forefront of forest protection movements, such as the Bishnoi Movement in Rajasthan led by Amrita Devi. This also influenced the Chipko Movement in Uttarakhand, where village women hugged trees to prevent deforestation. Hence, they often play a key role in community forest management and eco-restoration.



Household Needs: In many rural and tribal areas, forest dwellers collect firewood, fodder for cattle, wild fruits, roots, tubers, and medicinal plants from forests to sustain their families.

Cultural and Social Role: In many communities, forests are linked to several rituals, festivals, and beliefs making them guardians of ecological and cultural traditions.

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Fig. 2.17. Connections between forest dwellers and forests (Central image shows women carrying bundles of leaves and wood from a mountain forest in Sohra, Meghalaya)

LET'S EXPLORE

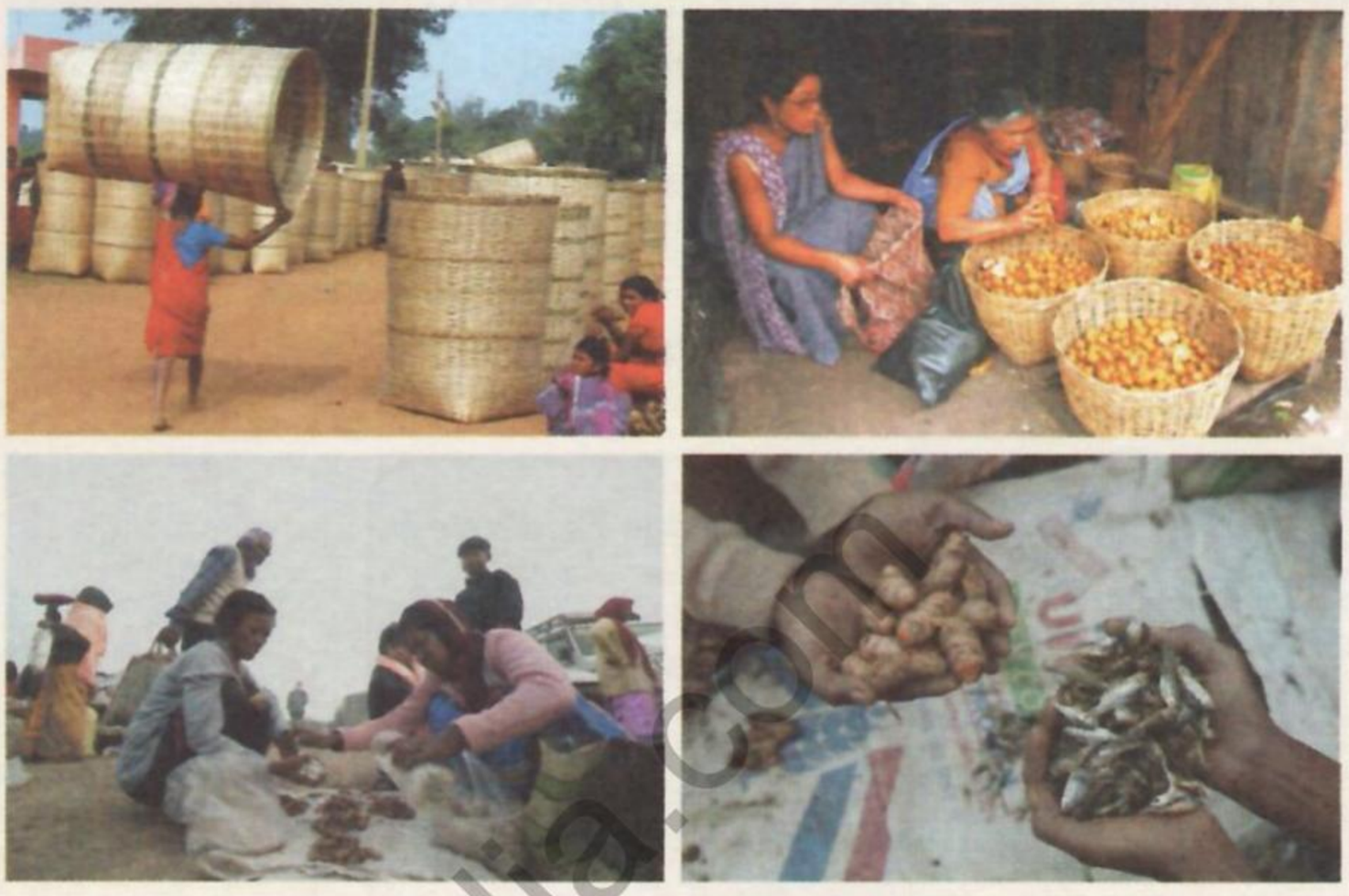

Fig. 2.18. Traditional tribal livelihoods

Try to identify and classify the items for sale in the images given in Fig. 2.18.

Also, make a list of the food and medicinal varieties available in your local market.

Geographical Indication (GI)

Did you know that some forest products are unique to specific regions and are recognised as Geographical Indications (GI)? A **Geographical Indication (GI)** tag is a type of intellectual property right that identifies a product as originating from a specific place, where a given quality, reputation, or other characteristics of the product are essentially linked to its geographic origin.

In simple terms, it is a label that indicates that a product comes from a particular region and derives its uniqueness from that place. Thus, GI tagging helps protect region-specific products, promote cultural heritage, and generate economic benefits for local communities through exclusive market recognition.

GI-tagged Forest Products in India

India has several forest-based products that have been granted Geographical Indication (GI) tags because they are unique to their region, linked to traditional knowledge, and depend on the forest ecosystem (Fig. 2.19).

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Fig. 2.19. Some GI-tagged forest products of India (map not to scale)

1. **Ramban Sulai Honey (Jammu and Kashmir):**
   It is highly nutritional with strong antioxidant and immuno-modulatory potential and has a distinct flavour, aroma, and taste.
2. **Tezpur Litchi (Assam):**
   - Grown in the forest-influenced soil and climate of Assam's Tezpur region.
   - It has a distinctive bright colour and high pulp content.
3. **Channapatna Toys (Karnataka):**
   The notable aspect of this forest-based craft is that they are made from *Wrightia tinctoria*, a forest tree locally known as 'Aale mara'.
4. **Kandhamal Haldi (Odisha):**
   - A variety of turmeric grown by tribal communities in the forest regions of Kandhamal.
   - It is known for its high curcumin content and medicinal value.
5. **Erode Manjal/Erode Turmeric (Tamil Nadu):**
   Cultivated in **forest-influenced soils**, it is used for cooking and ayurvedic medicine.
6. **Nilgiri Tea (Tamil Nadu, Keralam, Karnataka):**
   Though plantation-based, its quality depends on the forest ecosystem conditions of the Western Ghats.
7. **Tripura Queen Pineapple (Tripura):**
   With a forest-dependent cultivation, it is famous for its sweetness and aroma.
8. **Khasi Mandarin (Meghalaya):**
   A citrus fruit from the forest-rich hills of Meghalaya.

You may explore more GI-tagged forest products from other states through the School Bhuvan-NCERT Geoportal.

Forest-influenced soils:
Soil profiles shaped over long periods by forest vegetation, root systems, and organic litter accumulation.

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LET'S EXPLORE

→ Can you explain how GI tagging of the forest products mentioned above connects geography, cultural traditions, local governance, and rural economy?

→ Can you name some GI-tagged forest products from your state or union territory?

Challenges Faced by People Living in Forests

People living in forests often struggle to access essential facilities such as healthcare, education, and clean water. Women are exposed to fuelwood fumes and face, many health issues, making their daily lives difficult. Forest dwellers depend heavily on forest produce such as firewood, honey, bamboo, and medicinal plants for their livelihood. Restrictions on forest use, deforestation, and the commercial exploitation of resources often limit their access to these resources. This creates a conflict between conservation laws and their survival needs, leading to economic insecurity.

Forested hilly regions are prone to natural disasters such as floods and landslides. Floods can wash away homes, crops, and roads, while landslides can bury villages and cut off access to essential services. Forest fires can quickly spread through dense vegetation, destroying homes and resources. Human care and preventive measures can help prevent many forest fires.

Crops are often damaged by elephants, monkeys, or wild boars, and sometimes people face direct attacks. Such conflicts not only cause economic losses but also create fear and insecurity among forest communities.

Many forest dwellers also face discrimination and neglect from mainstream society. They are often engaged in selling forest products at relatively low prices. Therefore, creating opportunities for processing and marketing these products is essential to enhance their income. Forest communities also require assistance in accessing government schemes and programmes, as well as greater awareness of their legal rights.

THINK ABOUT IT

You may already be aware that incidents of forest fires have been reported from different parts of India and are a matter of serious concern. According to the Forest Survey of India (FSI), about 36

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per cent of the country's forest cover is estimated to be prone to forest fires, of which nearly 4 per cent is extremely prone. Climate change as well as human-induced factors are the main reasons for such occurrences, some of which are quite severe and cause extensive damage to human life and the environment (Fig. 2.20). *Note: You may explore various fire locations on the School Bhuvan–NCERT Geoportal.*

Fig. 2.20. A forest fire

LET'S EXPLORE

You may have heard about the Pradhan Mantri Ujjwala Yojana (PMUY). How are the benefits of this scheme impacting the health of women?

THINK ABOUT IT

- Which wild animals (like monkeys or deer) have you seen in your residential area?
- Identify the causes behind these animals' presence in human settlements.

Fig. 2.21. Forest bridge to avoid roadkill

Government Efforts to Support Forest Dwellers

To reduce the hardships of people living in forests, the Government of India has taken several steps. One of the most important measures is the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, commonly called the Forest Rights Act (FRA). The Act recognises the rights of eligible communities over the forest land and forest resources that they have traditionally depended upon. It allows them to collect minor forest produce such as bamboo, honey, and medicinal plants, and grants ownership of small patches of land cultivated by them.

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Along with the FRA, the Government runs schemes for tribal welfare, including education scholarships, healthcare programmes, and skill development initiatives. The Joint Forest Management (JFM) programme encourages local communities to participate in protecting forests while benefiting from forest produce. Other initiatives, such as the Panchayats (Extension to Scheduled Areas) Act (PESA), the Dharti Aaba Janjatiya Gram Utkarsh Abhiyan (DAJGUA), and the Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan (PM JANMAN), aim to promote the welfare of such communities.

Mendha Lekha: A Case Study in Community Forest Management

Mendha Lekha is a small tribal village located in the Gadchiroli district of Maharashtra, known for its effective community-based forest management and self-governance. The village, inhabited mainly by the Gond tribe, became the first in India to gain rights over its forest resources under the Forest Rights Act (2006). The people of Mendha Lekha follow the principle of *Mawa Nate Mawa Raj* (our village, our rule), ensuring that all decisions are made collectively through the Gram Sabha. They protect and sustainably utilise their 1,800 hectares of forest by regulating tree cutting, preventing forest fires, and promoting bamboo harvesting as a renewable resource. The revenue from the forest produce is used for community welfare activities such as education and infrastructure. This also upholds the fundamental rights of the people to participate in governance and control resources that sustain their livelihoods. Simultaneously, the villagers fulfil their fundamental duties by protecting biodiversity and preventing illegal exploitation. This model has not only strengthened local livelihoods but also demonstrates how traditional knowledge, rights, and duties, as well as democratic participation, can coexist to achieve sustainable forest conservation (Fig. 2.22).

Fig. 2.22. Community-led protection of forests through self-governance, traditional livelihood, and controlled grazing in Mendha Lekha

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Ecological Commons—Shared Resources

The term **ecological commons** refer to natural resources like forests, rivers, lakes, grazing lands, air, and biodiversity that are shared by a community and used collectively for its survival and well-being. Unlike private property, these resources are not owned by any single individual but belong to the community as a whole, and are often managed through traditional rules and practices. For centuries, people have depended on the ecological commons for food, water, fuel, fodder, and medicinal plants. However, with population growth, industrial use, and privatisation, many of these commons are shrinking or degrading. Protecting ecological commons is important not only for the livelihoods of rural and tribal communities but also for maintaining ecological balance, regulating climate, and conserving biodiversity.

A Case Study of Commons-based Fisheries Management in Bihar

After independence, The Indian Fisheries Act, 1897, remained the basis for the governance of inland fisheries. Historically, control over perennial water bodies and *diara* waterfronts in Bihar was held by large landowners who marginalised the fishing community, rendering them the most vulnerable group in an environment abundant with aquatic resources.

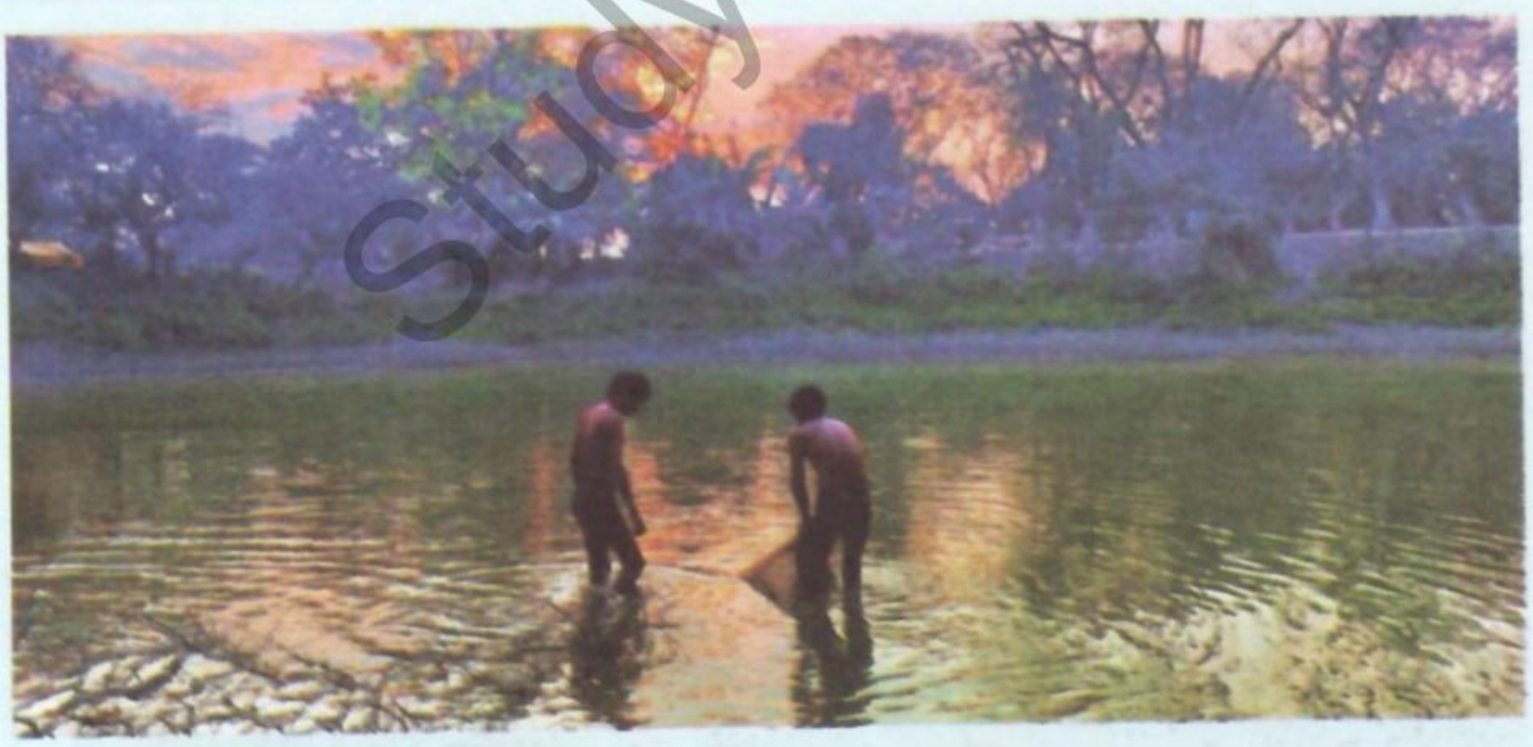

Fig. 2.23. Fishermen at work in a maun (lake) at dawn

The historical insights from Bihar's 19th century case study suggest that a commons-based management model is best suited to riverine regions. To promote community-led management, legal safeguards for public access, and cooperative resource stewardship as

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sustainable alternatives—along with the preservation of cultural heritage associated with riverine waterscapes—the Bihar Fish Jalkar Management Act, 2006, and its subsequent amendments, were introduced. Under this model, local communities develop context-specific solutions to balance the productivity of local resources with sustainable rates of exploitation. The success of this model is reflected in the persistent growth of fish production in the region.

Case Study of Water Tanks Built by the Gond Kings

In India, for centuries, regions with seasonal rainfall have tackled the issue of water demand and management as a collective concern and used common public resources for the same. In fact, between 800 and 1200 CE, the Chandella and Bundela kings built around 8,000 tanks in the Bundelkhand region. Similarly, *keres* were built in Karnataka, *eri* tanks in Tamil Nadu, and *chevuru* tanks in Andhra Pradesh and Odisha. Bihar had the famous *ahar pynes* to store water and the Gond kings funded huge *katas*—three-sided reservoirs—in the Gondwana region. According to official estimates, there are more than 15 lakh functional tanks in India at present.

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LET'S EXPLORE

Below is Fig. 2.24 from the School Bhuvan–NCERT Geoportal. Observe the image carefully and see if you can identify some of the tanks and mark the agricultural fields irrigated by them.

Fig. 2.24. Water tanks in Bhandara district, Maharashtra

Before we move on...

- Biomes are large areas on Earth with specific vegetation and animal life. Biomes are classified based on temperature, precipitation, flora, fauna, etc.
- Humans interact closely with different biomes and adapt their lifestyles to the conditions and resources available.
- Traditionally, different human cultures have developed various practices for conserving nature. Today, there is a legal framework to support conservation efforts.
- Occupations have been closely associated with forests, right from primitive hunting to modern ecotourism. The variety of human interactions with forests highlights their significance for various communities.
- With growing population, there is increasing pressure on forests, which leads to problems like human–wildlife conflict, depletion of resources, etc.

Questions and Activities

1. Why does India have diverse biomes? How does the local geography connect to the diversity?
2. Make a list of economically valuable native plants in your region and comment on their abundance.
3. Prepare a map of your state or Union Territory showing national parks and sanctuaries.
4. Study the impact of forest fires on people dependent on forests. Try to include the causes, losses, and preventive measures.
5. Why are forests often called the 'lungs of the Earth'?
6. If large forest areas continue to be destroyed, what do you think would be its long-term effect on the global climate?
7. Discuss some traditional forest conservation methods with your grandparents or an elderly person.
8. How do forest dwellers contribute to the protection and conservation of forests?
9. What is a forest fire? Prepare a plan for prevention and mitigation of forest fires.

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Notes

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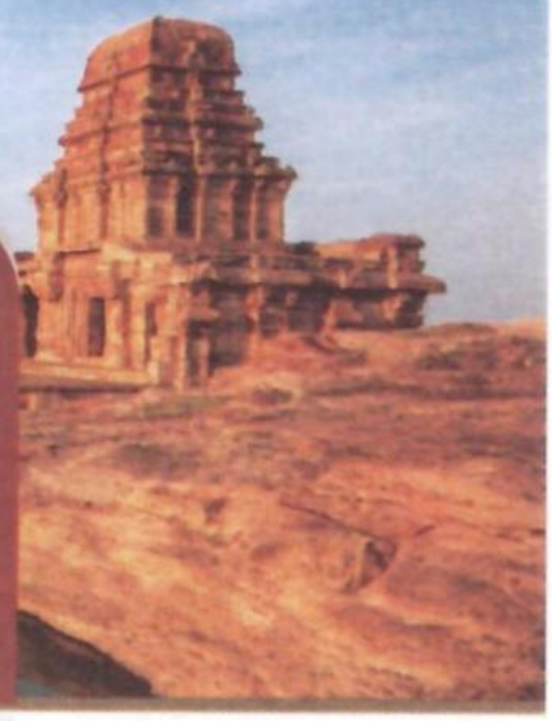
Chapter 3

Resistance and Resilience (1000–1700 CE)



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The Big Questions

1. *How did Indian rulers use resistance and alliances to safeguard their lands?*
2. *What factors contributed to the resilience of different kingdoms and communities during this period?*
3. *How did the Bhakti tradition foster a sense of solidarity?*
4. *How did the expansion of agriculture, trade, and commerce keep the economy sound?*

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The period between 1000 CE and 1700 CE witnessed profound political, social, and cultural changes. The Indian subcontinent experienced repeated invasions and wars accompanied by religious discrimination and social distress. Regional powers frequently resisted invaders who were making inroads into their territories. This resistance took both military and cultural forms. Nonetheless, this period manifested the resilience of the Indian civilisation, which safeguarded its core ideas and values. Indian society not only defended itself from external threats but also preserved its traditions. In this chapter, we will see how resistance and resilience shaped this period of Indian history.

Let us begin by understanding the meaning of resistance and resilience in the context of the period under study and how they are interconnected.

Resilience refers to the capacity to, survive and adapt in challenging situations, including times of political unrest and instability. It is also about how communities and groups across diverse regions respond to challenges and opportunities. This capacity to absorb and adapt to hardship while preserving inherited socio-cultural and spiritual traditions can also be understood as resilience. The idea of India's resilient spirit extends beyond political and military spheres.

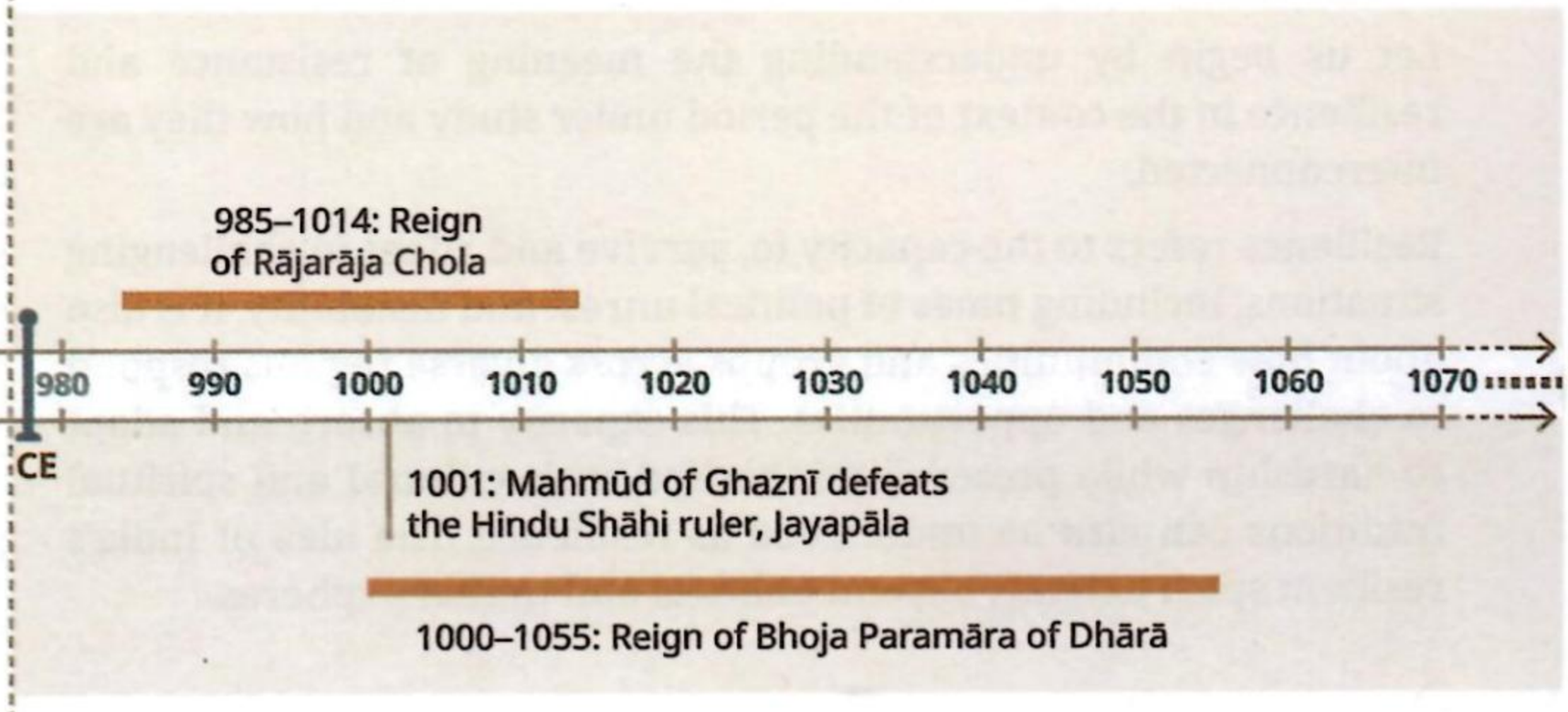
For instance, rulers and communities restored places of worship, modified customs to suit changing circumstances and, in many regions, established lasting patterns of coexistence across religious and cultural lines.

Resistance means actively opposing or challenging a force, authority, or threat in order to protect political power, territory, interests, or cultural and spiritual identity. Resistance may take the form of warfare, alliances, diplomacy, negotiation, or other political actions.

Resistance and resilience are, in fact, closely linked. Resilience is the backbone of resistance. While a political cause or leadership provides the condition for resistance, the ability to endure comes from resilience. Together, resistance and resilience help explain how Indian society sustained itself through different ages, particularly the period under study. This period is characterised by both conflict and cooperation, rather than being merely a story of invasion and resistance.

Safeguarding Sovereignty: Resistance, Alliances, and Confederacies

From the 7th century onwards, the Indian subcontinent witnessed successive phases of invasions driven by territorial expansion and religious motives. It may also be noted that the surplus agrarian produce of the fertile plains fed by perennial river systems contributed to the people's economic well-being and also became a major reason for the invasions. Against this backdrop, let us examine instances of resistance, as well as the strategic alliances and confederacies that emerged in these centuries at both the regional and pan-Indian level.



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From the mid-7th century, Arab forces launched invasions by sea and land. These early invaders were confronted and repulsed by Indian rulers across the Deccan, Malwa, Kashmir, and Rajasthan. Mahmūd, the Sultan of Ghaznī (in present-day Afghanistan and northeastern Iran), launched his first invasion in 1001 against Jayapāla of the Hindu Shāhi dynasty. During the 11th and 12th centuries, rulers of the Hindu Shāhi kingdom, and the Chandellas, Solankīs, and Chauhāns opposed and fought against the Turkish invaders. In 1178, Queen Nāikī Devi of the Chaulukya kingdom defeated Muhammad Ghurī at Kasahrada, near Mount Abu. In northern India, some powers, such as the Chauhāns, Gahaḍavālas, and Senas, however came under Turkic control towards the end of the 12th century, even as Gujarat, Malwa, and Bundelkhand under the Chandellas put up stiff resistance for about a hundred years thereafter. The Turks failed to conquer present-day Odisha, and parts of present-day Assam. Several states of Rajputana, in particular, remained centres of sustained resistance throughout the period under study. Thus, during the 11th and 12th centuries, northern and western India bore the brunt of Turkish invasions, while peninsular India remained free from Turkic control.

LET'S RECALL

You may recall learning about the invasions of Mahmūd of Ghaznī and Muhammad Ghūrī in the 11th and 12th centuries, covered in Grade 7 Social Science textbook. These invasions had seriously disrupted the political, social, and religious fabric of the Indian subcontinent.

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1178: Battle of Kasahrada: Chaulukya queen Nāikī Devī defeats Muhammad Ghūrī

1206: Delhi Sultanate established

1180 1190 1200 1210 1220 1230 1240 1250 1260 1270

1192: Second Battle of Tarain: Muhammad Ghūrī defeats Prithvīrāj Chauhān

1228: Ascendancy of the Ahoms

1262–1289: Reign of Kākatīya Queen Rūdramā Devī

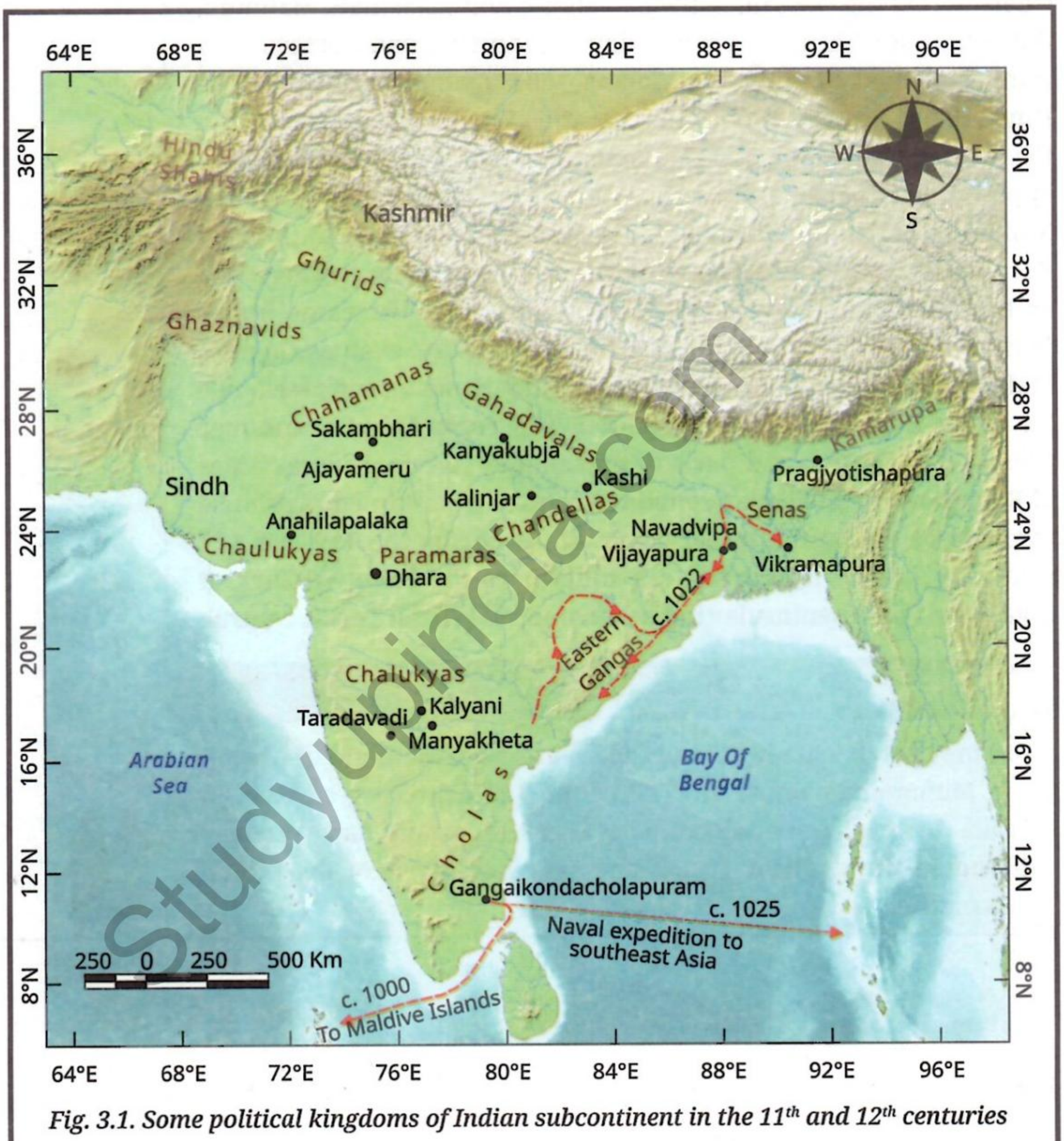

Fig. 3.1. Some political kingdoms of Indian subcontinent in the 11th and 12th centuries

Legend

- ● Major Centres
- ----→ Select Chola Campaigns (with dates)

**Names marked in red are dynasties*

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Rajput Dynasties: Confederacies and Resistance (1200–1500 CE)

The establishment of the Delhi Sultanate in 1206 was a turning point in Indian history. The new state introduced administrative practices that led to greater centralisation of authority. For example, the *iqta* system, under which territories were assigned to military commanders in lieu of salaries, became a mechanism for land revenue collection and a system that increased the burden on the peasantry. Furthermore, during their military campaigns for territorial expansion, the forces of the Sultanate demolished sites sacred to the local communities. This led to stiff resistance from several kingdoms and chieftains. Therefore, resistance throughout the Delhi Sultanate was fuelled by both political factors and the determination to preserve socio-cultural continuity. For instance, the Yādava dynasty of western Deccan (Devagiri), several Rajput dynasties of present-day Rajasthan, the Kākatīya rulers of eastern Deccan (Warangal), and the Hoysālas of present-day Karnataka repeatedly resisted the Sultans of Delhi. These states played an important role in checking the expansion of the Sultanate.

Over the years and under different Sultans, the Delhi Sultanate faced opposition and defiance from several states and regions including Mewar, Gwalior, Malwa, Awadh, Etawah, Ajmer, Jalore, Kalinjar, and Kannauj.

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Rajput–Mughal Dynamics: Politics and Complexities of Rule (1500–1700 CE)

In the early 1500s, the weakening and subsequent decline of the Delhi Sultanate under its last ruler, Ibrāhim Lodī, opened the door to two parallel developments. First, Zahiruddin Muhammad, also known as Bābur, from Central Asia launched an invasion on India. Second, Indian rulers began to assert their own authority. In 1526, Bābur defeated the Lodi forces in the First Battle of Panipat, bringing the Sultanate to an end. In 1527, Rāṇā Sangrām Singh I, popularly known as Rāṇā Sāngā, of Mewar led a Rajput confederacy against Bābur in the Battle of Khanua. The battle ended in favour of Bābur and is seen as an important turning point in Indian history.

Mansab: It means 'rank' or 'office', and its holders were known as mansabdars. Akbar introduced the mansabdari system, which combined military organisation with civil administration. It established a clear hierarchy among the nobility and helped strengthen imperial control.

As Mughal rule consolidated its power, relations between the Rajputs and the Mughals evolved over time, shaped by strategic considerations. Some Rajput kingdoms such as Mewar, maintained stiff resistance to Mughal authority, while others like the Kachhwāha Rajputs of Amber, negotiated cooperative relationships. The Mughals, for their part, had a strong incentive to collaborate with the Rajputs—they needed the Rajputs' military might, their loyalty as well as their administrative strength to consolidate and stabilise the Mughal Empire. Towards this end, the Mughals integrated some of the Rajputs into the empire through the grant of **mansabs** and other understandings.

Mewar's Resistance against Mughals

Akbar attacked the kingdom of Mewar and laid siege to the fort at Chittor in 1567–1568. The Mughal army paid a heavy price in the campaign. Under Rāṇā Udai Singh, the Rajputs put up a stiff resistance for more than five months before the fort was captured

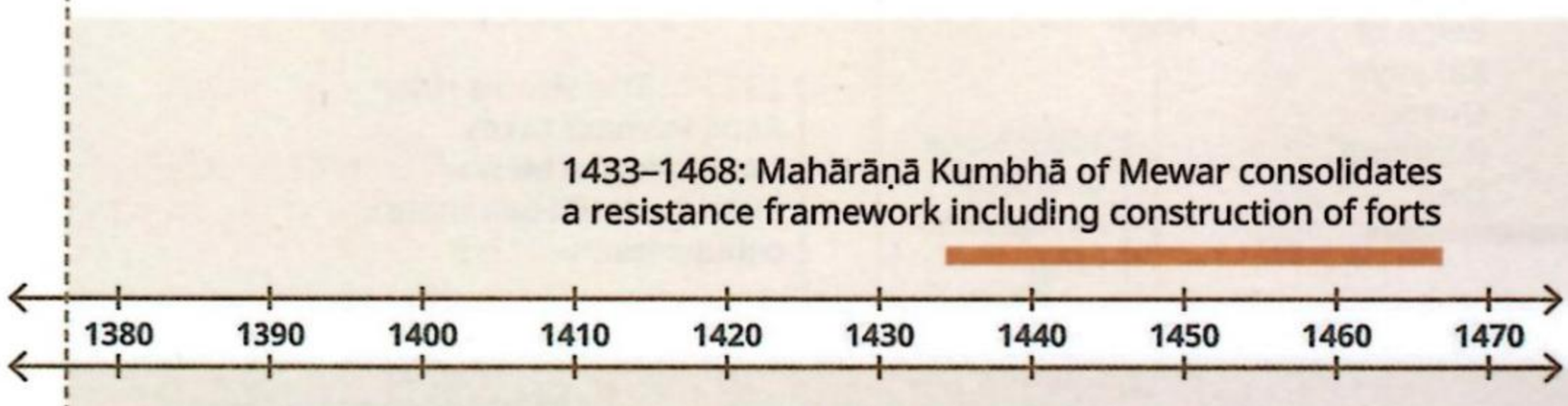

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Fig. 3.2. A painting depicting the Battle of Haldighati

by the Mughals. By this time, Akbar had already concluded several successful campaigns and the empire was at the peak of its power. In a strategic move, Rāṇā Udai Singh retreated to the Aravalli Hills along with a large number of his subjects.

In 1576, Akbar launched another attack on Mewar. This time, Mahārāṇā Pratāp, the son and successor of Rāṇā Udai Singh, fought against Mughal forces in the Battle of Haldighati. Continuing his fierce resistance, Mahārāṇā Pratāp modified his military strategy and adopted guerrilla warfare. He used the rugged Aravalli terrain to his advantage and conducted surprise attacks, raids, and ambushes on the Mughal forces and their networks. He continued to resist and eventually reclaimed most of his ancestral territories in the 1580s.

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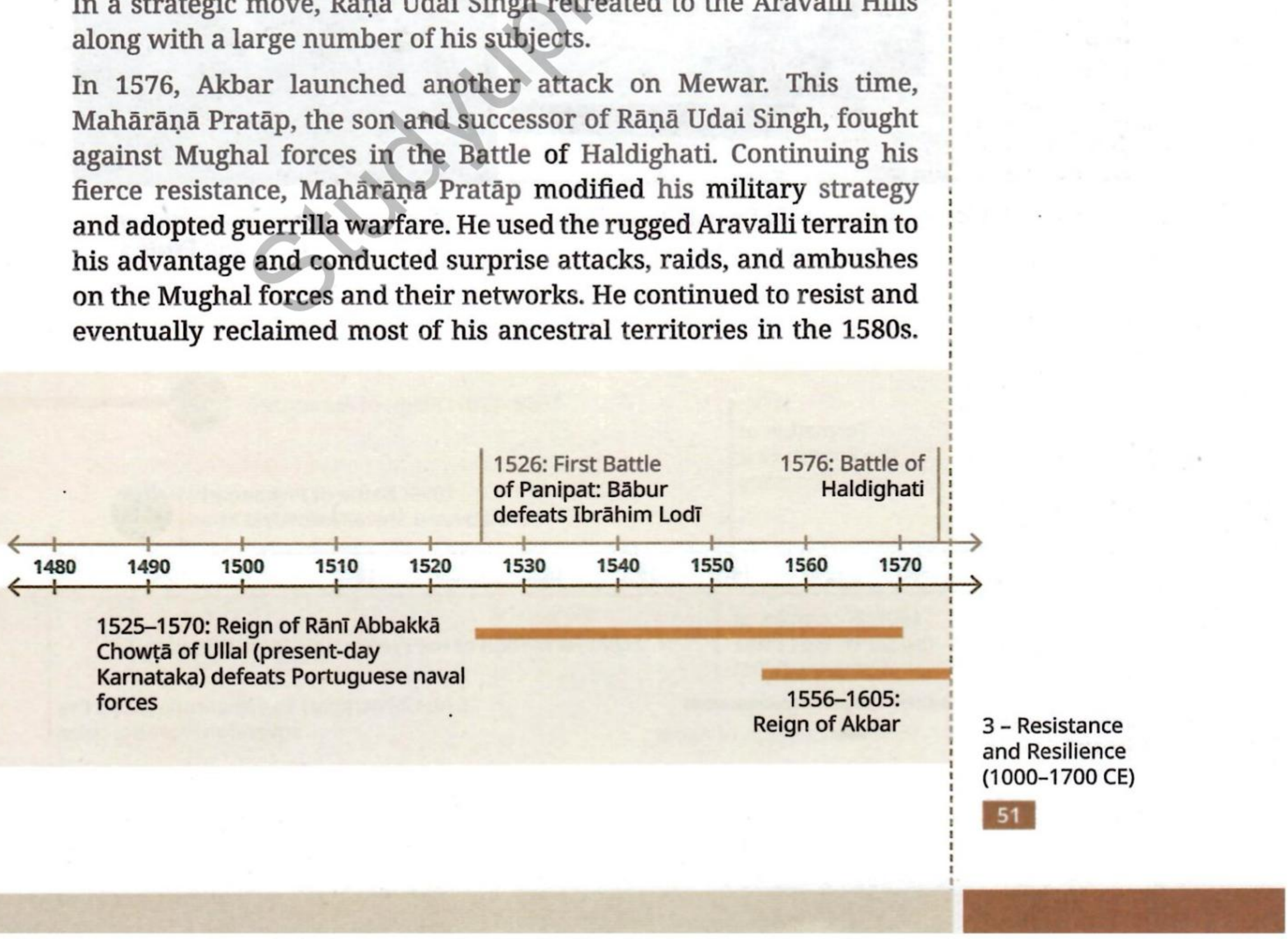

The Battle of Dewair fought near Udaipur in 1582 was significant as it resulted in the collapse of key Mughal military outposts in Mewar and helped Mahārāṇā Pratāp reclaim control over Mewar.

Mahārāṇā Pratāp's son and successor, Mahārāṇā Amar Singh, continued Mewar's tradition of military resistance against the Mughals. Jahāngīr made military advances towards Mewar in 1605, 1608, and 1609. But each attempt was countered by a strong and organised resistance from the Rajputs. Eventually, the Mughals were compelled to acknowledge Mewar's distinct political position.

Fig. 3.3. Victory memorial in Dewair, Rajasthan

Fig. 3.4. A 19th century miniature painting of Mahārāṇā Pratāp

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On one hand Mewar continued its defiance in the form of a strong, organised, and determined resistance, and on the other hand, the Mughal state began to recognise and appreciate Mewar's unique geo-political position with trade routes leading to the major port of Surat. Needing peace with the rulers of Mewar more than ever, the Mughals were eventually compelled to negotiate a treaty. Under the terms of the treaty, Mewar leadership was exempted from personal attendance at the Mughal court, thus clearly acknowledging Mewar's continued autonomy.

LET'S RECALL

Do you remember reading about the Battle of Haldighati in your Grade 8 textbook? You may recall that the Bhil community provided crucial, tactical, and logistical support to Mahārāṇā Pratāp in his resistance against the Mughals. Their extensive knowledge of the Aravalli terrain, expertise in guerrilla warfare, skills in archery, role as guides, and above all, their bravery strengthened Mewar's resistance. This enduring alliance was later commemorated in Mewar's state emblem (Coat of Arms), which depicts a Rajput warrior and a Bhil warrior with the local deity Eklingji (Shiva) shown at the centre.

Fig. 3.5. Mewar's Coat of Arms

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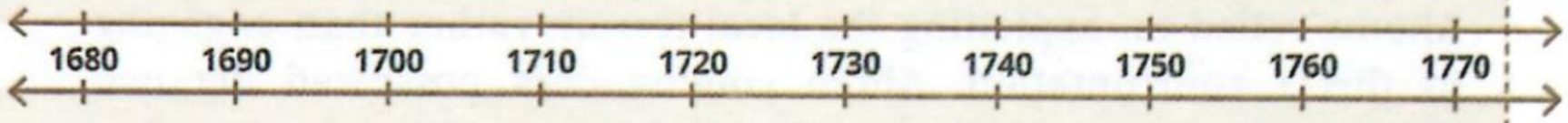
1658–1707: Reign of Aurangzeb



LET'S EXPLORE

After the death of Mahārāṇā Pratāp, renowned poet Dursā Arhā gave a vivid account of Mahārāṇā Pratāp in the following lines:

O Pratap, you kept your horses unbranded, your head unbowed, your fame untarnished. You were strong enough to carry on your work against heavy odds. You never participated in the Nauroz festival, nor did you mount guard for the imperial presence during Jharokha Darshan. You attained a very high place in this world.

Fig. 3.6. Source: Rawat Saraswat, Dursa Arha (New Delhi: Sahitya Akademi, 1983)

Identify any two specific actions mentioned in the text which show that Mahārāṇā Pratāp maintained his independence from the Mughal court.

DON'T MISS OUT

Other Rajput rulers like Kānhaḍa Deva Chauhān (Jalore), Rāṇā Hammīra Deva (Ranthambore), Rāṇā Kumbhā (Mewar), and Rāo Surjan Hāḍā (Bundi) put up a strong resistance at different points of time.

The Ahom Kingdom: Centuries of Endurance

The Ahom kingdom of Assam maintained its political independence for almost six centuries (1228–1826) and successfully resisted several invasions and attempts at annexation. The military strategy of the Ahoms relied on exploiting the local terrain rather than engaging in direct confrontation. Ahom commanders possessed detailed knowledge of Assam's forests, river systems, and monsoon cycles, which invading forces generally lacked. Despite repeated attempts by the Mughals to annex the Brahmaputra valley, the Ahoms successfully defended their territories against Mughal invasion.

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Fig. 3.7. A monument at Guwahati, Assam commemorating the Battle of Saraighat

The first Ahom–Mughal clash took place in 1615 near Samdhara. Mughal forces under Abu Bakr made early gains but were soon compelled to retreat. By 1616, the Ahoms had recovered the lost territory, setting a pattern that would recur over the next sixty years. In 1662, Mīr Jumlā captured the Ahom capital, Garhgaon. Subsequently, a treaty was imposed on King Jayadhvaj Singha. However, the king defied the Mughals and renounced the treaty within a year of the Mughal withdrawal, and the Ahoms retained their independence.

A decisive check on Mughal ambitions in the region came in 1671, when the Ahom commander Lachit Borphukan defeated a Mughal fleet at the Battle of Saraighat, securing the Brahmaputra as an effective territorial boundary. The Ahoms' successful resistance against the might of the Mughal empire is a remarkable story of bravery, pride, and endurance.

Ahom endurance rested on a combination of strategic alliance-building, guerrilla tactics, Assam's challenging geography, and the *Paik* system (about which you have read in Grade 8) of military-administrative organisation. The Ahom navy also played a key role, using light, fast boats, called *bāchārī*, built specifically for river warfare, which gave them an advantage over Mughal fleets.

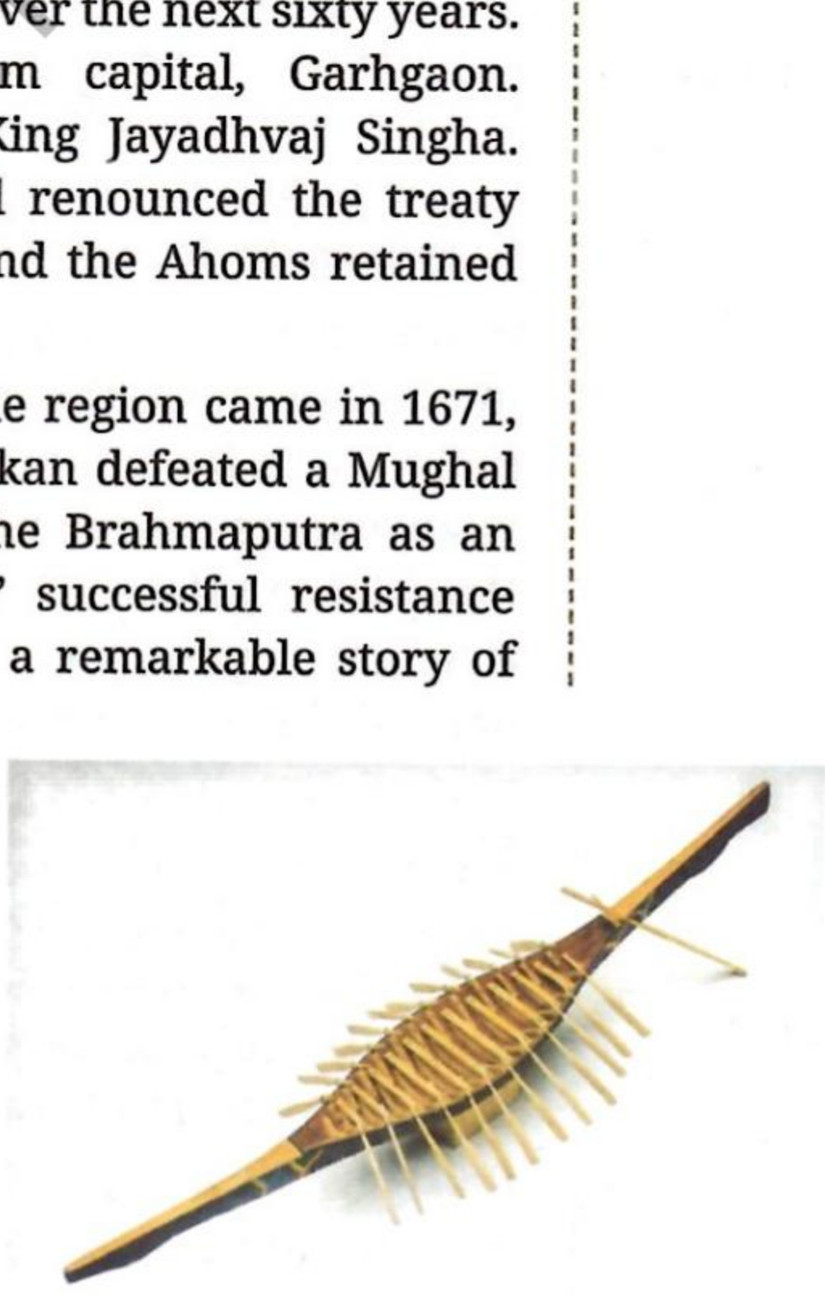

Fig. 3.8. Model of a bāchārī boat

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Fig. 3.9. An aerial view of Hampi, the capital of the Vijayanagara Empire

Vijayanagara Empire: Resilience against Expansion

Resistance to foreign invasions was not confined to northern India. In the Deccan and beyond, the expanding campaigns of the Delhi Sultanate steadily weakened the older kingdoms of the south. Malik Kāfūr, Alauddin Khiljī's general, attacked major cities and temples, including those at Warangal, Belur, and Madurai. Later, Muhammad bin Tughlaq's rise to power extended the Sultanate's influence even farther south, ultimately contributing to the decline of the Pandya Kingdom and the eventual emergence of a new sultanate territory known as the Madurai Sultanate.

It was against this backdrop that the Vijayanagara Empire rose in 1336 to check the expansion of the Sultanate into peninsular India. Harihara and Bukkā established the independent kingdom of Vijayanagara—meaning 'City of Victory'—located near present-day Hampi, Karnataka. The region's geography, including a formidable river and rugged, rocky terrain, provided Vijayanagara with a strong natural defence and a secure base for military operations. They made determined efforts to defend and expand their state's frontiers. The empire gradually expanded by annexing territory south of the Krishna and Tungabhadra rivers and became one of the richest and most influential states, encouraging the flourishing of literature in Kannada, Telugu, Tamil, and Sanskrit. Vijayanagara's capital city became a centre of trade, culture, and political power in southern India.

During Bukkā's reign, Vijayanagara made major territorial gains, including the subjugation of the Madurai Sultanate and the annexation of Arcot. In 1342, the ageing Hoysāla king Vīra Ballāla III

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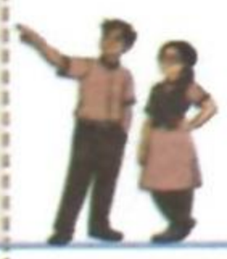
fought the Madurai Sultanate, resulting in the Battle of Kannanur, which was a pivotal episode in this struggle. However, the Sultan launched a surprise counter-attack and executed the Hoysāla ruler. For almost four decades, Vijayanagara wrestled with the Sultans of Madurai. Later, the campaign against Madurai Sultanate was led by Kumāra Kampana, the son of Bukkā. He killed the Sultan and brought the Sultanate of Madurai to an end. Kumāra Kampana also reinstated the divine images in the Rajasimheshwara Temple at Kanchipuram and the Sri Ranganathaswamy Temple at Srirangam.

LET'S RECALL

In Grade 8, Part I, you have read about how the Vijayanagara Empire reached its peak in the 16th century, under its celebrated ruler, Kṛiṣhṇadevarāya, who expanded and secured the empire's dominance over the Deccan. Under his rule, the empire achieved both military power and cultural glory. Kṛiṣhṇadevarāya provided grants to many temples, including Tirupati in Andhra Pradesh and the Vitthala temple in his own capital Vijayanagara, which had many grand temples, palaces and other buildings. After winning many wars against his neighbours, Kṛiṣhṇadevarāya died of illness in 1529.

Decades later, the Vijayanagara army was defeated at the Battle of Talikota in 1565 by an alliance of the Deccan sultanates. In the aftermath, the city of Vijayanagara was sacked and largely abandoned, while the empire gradually fragmented. Several of its territories came under the rule of smaller Nāyaka kingdoms.

DON'T MISS OUT

By the end of the 16th century, the Vijayanagara Empire had split into smaller kingdoms ruled by the Nāyakas. They also continued the tradition of resistance—not always through military action, but sometimes through bold acts of defiance aimed at Mughal authority. An interesting account is of Muthu Vīrappa Nāyak III of Madurai. Local tradition holds that the Mughal emperor would dispatch a royal slipper to rulers as a symbol of imperial supremacy and domination. Recipient rulers were expected to receive the slipper ceremonially at the borders of their territories and pay tribute. When the slipper arrived at the borders of Madurai, Muthu Vīrappa Nāyak refused to observe this protocol. In an act of defiance against

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this ritual of subordination, he refused to meet the procession at the border, compelling the Mughal emissary to carry the token right up to the palace hall. Even then, Muthu Vīrappa Nāyak refused to bow before the slipper. Instead, he forced the Mughal representatives to place it on the floor, put his own foot inside it, and mockingly asked why the Badshah had not sent a matching pair. He then had the messengers expelled from his kingdom.

Fig. 3.10. A painting of Chhatrapati Śhivājī with his personal guards

Marathas: The Quest for Swarājya

The Marathas were a formidable force in the Deccan particularly during the 17th and 18th centuries. The establishment of an independent Maratha state in 1674 is closely associated with the rise of Chhatrapati Śhivājī. The Marathas had been involved in the region's political affairs for centuries. Their knowledge of the Deccan's complex geography and landscape, along with their bravery and martial abilities, made them a formidable force in both administrative and military affairs.

LET'S RECALL

You have already learnt about the foundation of Maratha power and the rise of Chhatrapati Śhivājī in your Grade 8 textbook, Part I. Think back to his struggle and the ways in which he built and strengthened his kingdom. What do you remember about the strategies and institutions that helped the Marathas establish their power?

Over time, the Marathas emerged as a prominent political and military force in India. In the Maratha state, territorial control and sovereignty were closely linked to the idea of *swarājya*, or self-rule. Chhatrapati Śhivājī is regarded as one of the earliest proponents of

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the spirit of *Swarājya*—the idea of an independent, self-governing territory free from external control.

As you have read in your Grade 8 textbook, the Marathas fought many battles against the Mughals and other powers. The prolonged warfare exhausted the Mughal Empire economically and politically, to the point that Aurangzeb spent more than two decades in the Deccan in an attempt to control the region and subdue Maratha resistance. As political boundaries shifted constantly, the Marathas relied on strategic alliances to extend their control over diverse territories. Through these alliances, they built a vast network of tributary relationships that extended their influence far beyond the areas under their direct rule. The key to the success of the Marathas was their bravery and military skills.

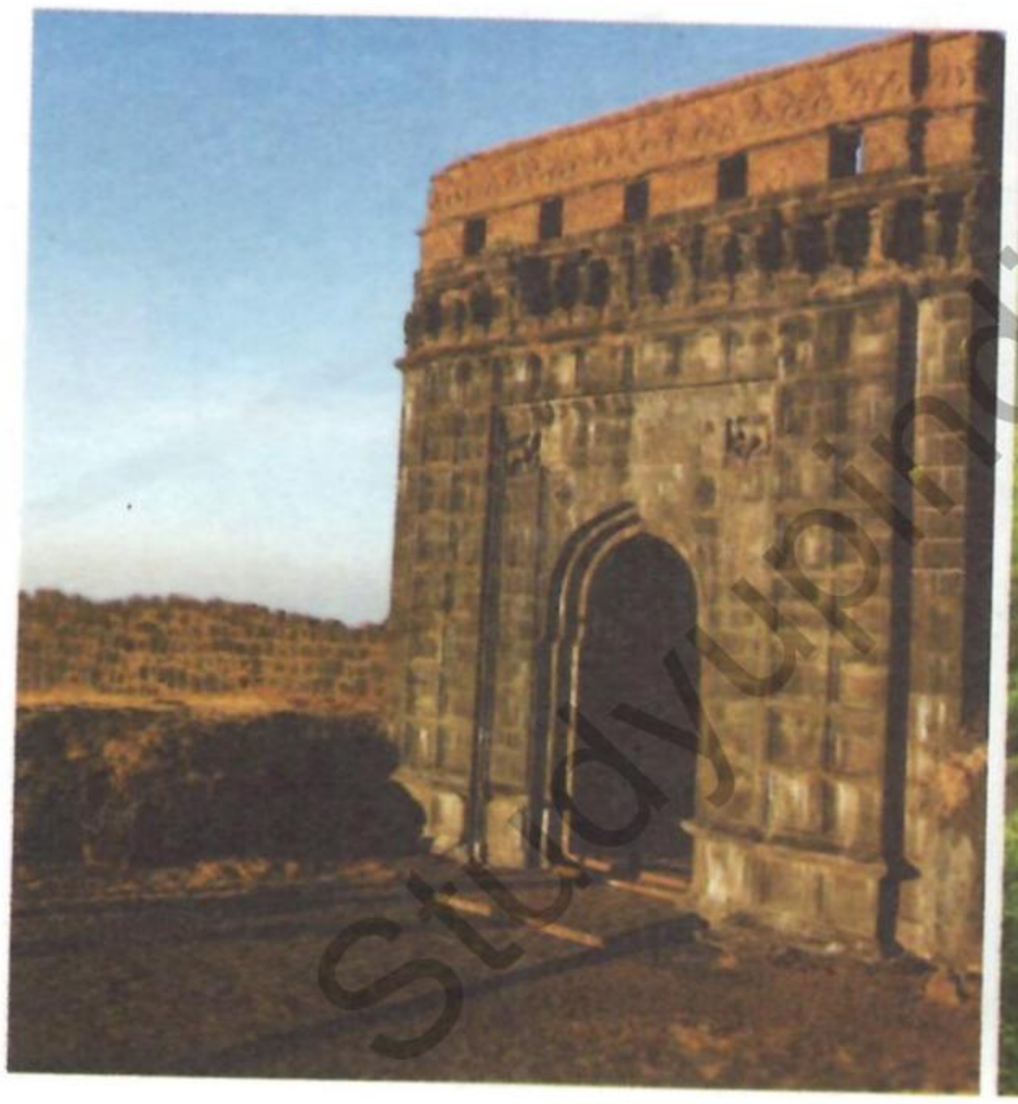

Fig. 3.11. Raigad fort

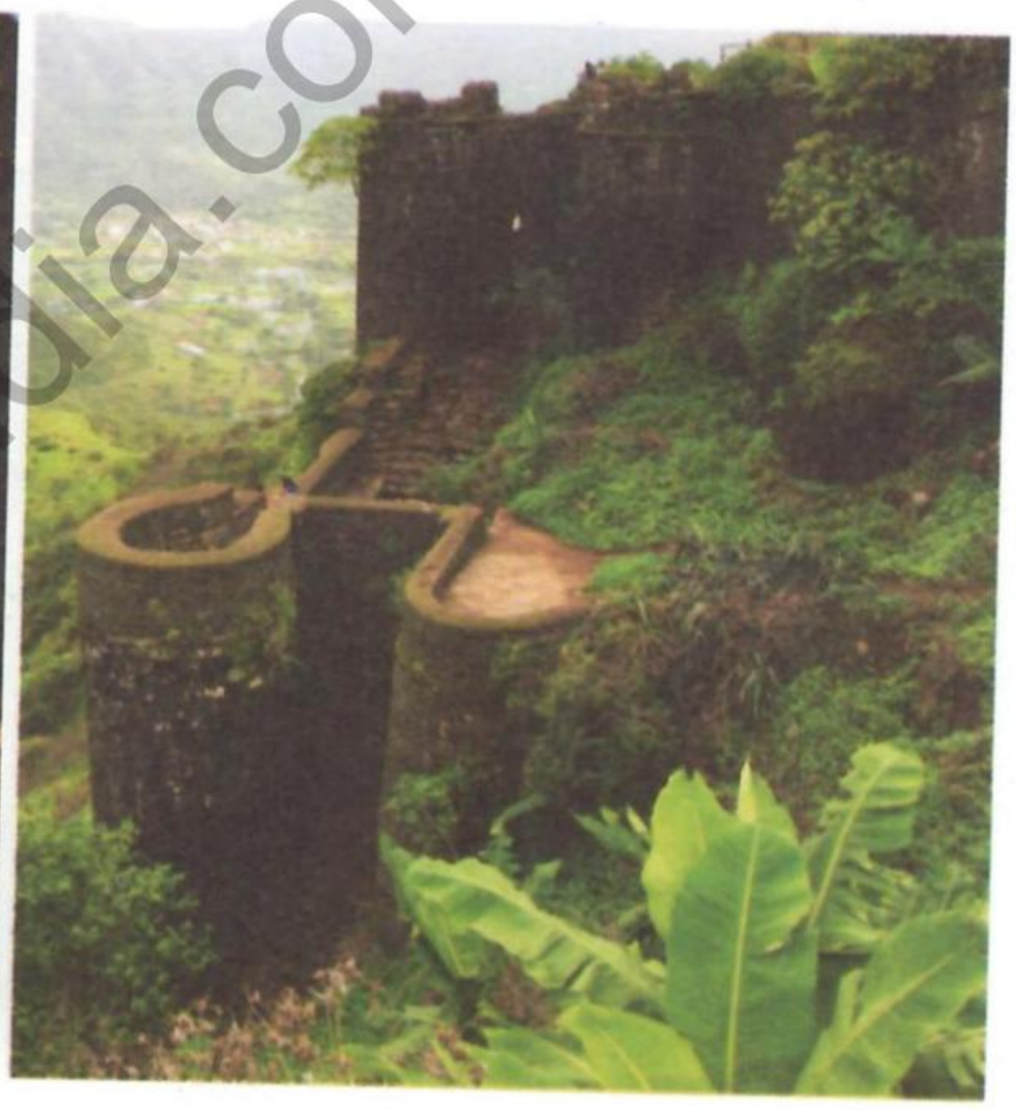

Fig. 3.12. Sinhagad fort

The Marathas strategically built forts such as Raigad and Sinhagad which played a crucial role in resistance against the Mughals. The Maratha territories were characterised by hills and deep valleys. Maratha forts, therefore, functioned as strategic nuclei for guerrilla warfare. This network of forts enabled the Marathas to launch swift attacks from different directions and provided safe retreats. Different types of forts were constructed to serve distinct purposes, and while some functioned as advanced outposts, those on the frontiers served

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as fallback capitals to avoid total defeat. Many formed part of the wider defensive network, and most were designed to withstand prolonged sieges. Forts also served as centres of administration and judicial authority. Strong defences, water sources, and armouries provided resilience and self-sufficiency, enabling the Marathas to sustain prolonged resistance during sieges.

DON'T MISS OUT

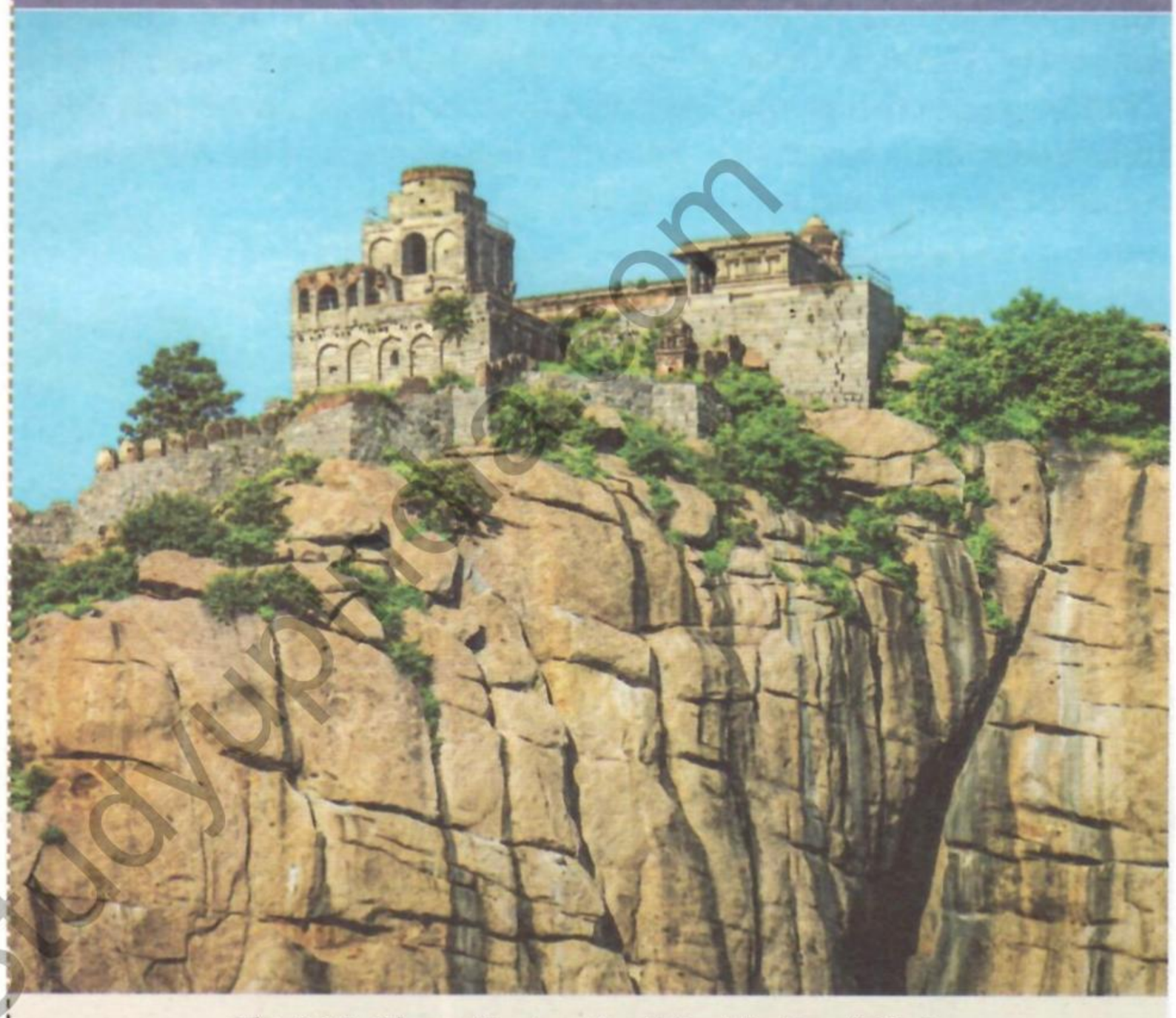

Fig. 3.13. Gingee Fort or Senji Fort in Tamil Nadu

The Fort of Senji, or Gingee, is situated in the Villupuram district of Tamil Nadu. It is considered a marvel of military and civil engineering and military architecture. One of its gateways is said to have been designed so that a small contingent of soldiers could defend it against an army of ten thousand. The fort came under the control of several powers, including the Vijayanagara Nayakas, Bijapuri Sultans, Mughals, Marathas, French, and English. Its changing control also highlights the fort's strategic importance and centrality to the political affairs of the region.

In 2025, the Fort was declared a UNESCO World Heritage Site during the 47th session of the World Heritage Committee in Paris.

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Fig. 3.14. Some political dynasties in the Indian subcontinent in the 17th century

The Sikhs: A Story of Spiritual and Political Resilience

A major spiritual movement was founded by Guru Nānak in the late 15th century which later came to be known as Sikhism. It emphasised devotion to one God, social equality, and the rejection of rituals. Under Guru Nānak and his successors, the Sikh community expanded and flourished across the subcontinent, particularly in the Punjab, Sindh, and Gangetic plains. During this period, Punjab was largely under the Lodī Sultans and later under the Mughals. It was a fertile, densely populated region, with a large peasant population.

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Fig. 3.15. A 19th century image of Sri Harmandir Sahib (Golden Temple), Amritsar

Due to the growing socio-cultural influence of the Sikh Gurus, the Mughal state increasingly intervened in their affairs from the beginning of the 17th century. As the Guru's influence was increasing, Jahāngīr ordered the execution of Guru Arjan Dev (the fifth Sikh guru) on the pretext of the Guru's alleged support to Jahāngīr's rebellious son, Khusrau Mirza. The martyrdom of Guru Arjan Dev marked a significant turning point in the history of the Sikhs and contributed to a gradual shift in their organisational and political orientation. Later, Guru Hargobind (the sixth Sikh Guru) was detained by Jahāngīr in Gwalior Fort. During this period, the Sikhs visited Gwalior in large numbers and paid their respects to the Guru from outside the fortress. In 1675, Aurangzeb ordered the execution of Guru Tegh Bahādur, the ninth Sikh Guru. The martyrdom proved a turning point in Indian history. The brutality of the Mughal state against the Sikh Gurus particularly Guru Arjan Dev, Guru Tegh Bahādur, and the tenth Guru, Guru Gobind Singh has few comparisons in history. The valour and sacrifice of the Gurus for the cause of the faith became the defining features of Sikhism. Beyond political motivations, the act was also deeply rooted in the rigid religious and discriminatory practices of the Mughals. The martyrdom of the Sikh Gurus to uphold their faith and principles firmed the conviction of the Sikhs that persecution had to be met with firm resistance and resilience. Such persecution and political pressure could only be countered through collective organisation, and armed resistance as and when required. Many armed conflicts took place between the Sikhs and the Mughals. Guru Hargobind established the *Akāl Takht*, a seat of temporal authority, to administer justice and address matters of governance. Scholars suggest that the establishment of the *Akāl Takht* was intended, in part, to counter Mughal political dominance.

The concept of *mīrī-pīrī* became vital in this context, representing the Guru's authority over both spiritual (*pīrī*) and worldly (*mīrī*) affairs. In Amritsar, the Harmandir Sahib (Golden Temple) represented spiritual authority (*pīrī*), while temporal power (*mīrī*) was represented by the *Akāl Takht*. In 1699, Guru Gobind Singh established and developed a

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Sikh order known as the Khalsa. The Khalsa adopted a distinct martial code in response to the changing socio-political environment. Through this process, spiritual identity became closely intertwined with the pursuit of political identity.

Due to sustained Mughal hostility, including the execution of Guru Gobind Singh's sons, the Sikhs organised themselves into armed groups known as *misals* to defend themselves and the territories through military and administrative structures (Eventually, these *misals* became a network of confederacies that united the Sikhs against the Mughals.). In this process, many people from peasant communities became soldiers and fought against Mughals in Punjab.

Jat Resistance

During the 17th century, the Jats, a farming community from Ganga–Yamuna Doab were burdened with heavy taxes and exploitation by Mughals. These economic hardships combined with Aurangzeb's religious policies led to growing resentment amongst the Jats. In 1669, under the leadership of Gokul Jat, they rebelled against the Mughal Empire. Mughal forces struggled for many months to suppress the uprising.

However, the Jats continued their resistance. In 1686, they took advantage of Aurangzeb's prolonged military campaigns in the Deccan and rebelled once again under the leadership of the *zamindars* of Sinsani and Sogar. The Mughals sent several armies to suppress the rebellion, leading to a long and fierce struggle in which both sides suffered heavy losses. By 1691, the Mughals had gained the upper hand. Yet, the Jat resistance continued.

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DON'T MISS OUT

Niccolao Manucci, an Italian traveller, wrote an extensive account about India during his journey (17th century) including descriptions of Jat resistance against the Mughals.

Bhakti Tradition: Affirmation of the Spirit of Resilience

India underwent a profound transformation in its religious and spiritual landscape between the 11th and 17th centuries. Bhakti—a tradition of personal devotion to the divine—was expressed through devotional

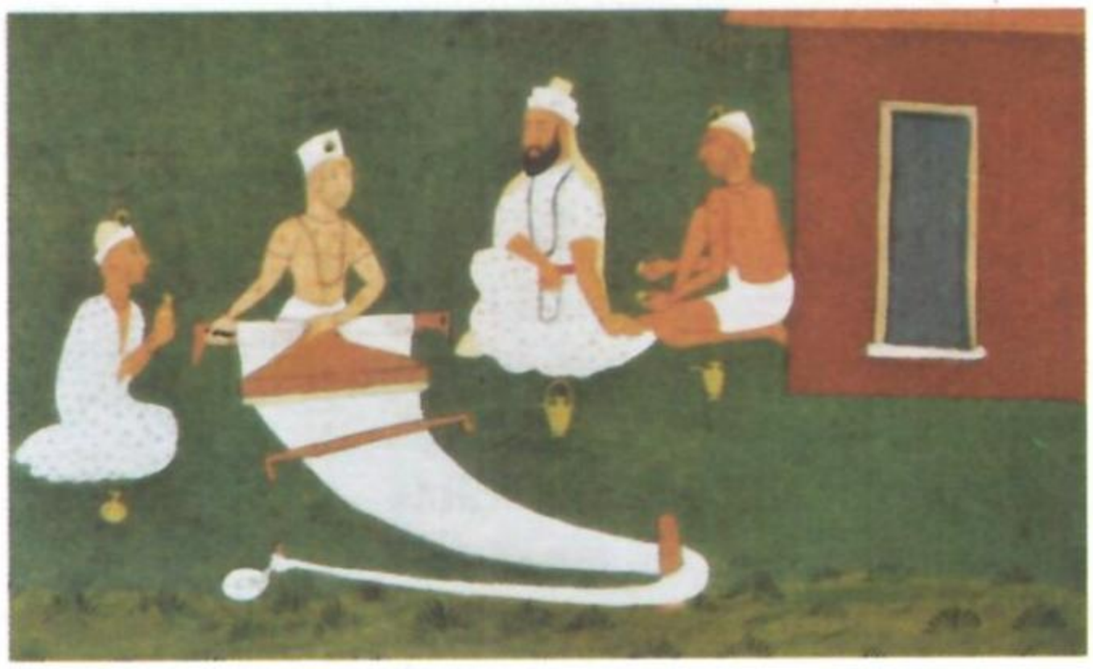

Fig. 3.16. Saint Kabīr with Nāmdeva, Rāidās, and Pipājī, Jaipur, early 19th century

hymns composed in local languages. From the 7th century onwards, the rise and expansion of Bhakti in peninsular India shaped social, religious, and political life. Bhakti tradition emphasised inclusivity, socio-religious equality, simplicity, and connecting with people's everyday life. For instance, Eknāth composed in Marathi and propagated a devotional path that was accessible to all without dependence on institutions.

LET'S EXPLORE

Explore the contributions of the Nayanar saints (devotees of Śhiva) like Karaikal Ammaiyar and the Alvar saints (devotees of Viṣhṇu) like Andal to the spread of the Bhakti tradition in peninsular India. Discuss their teachings and devotional practices and the ways in which they connected with people through their hymns.

DON'T MISS OUT

Bhakti saints sang divine lores in different languages, but the unifying factor was devotion to the ultimate reality.

From the 14th century onwards, a strong wave of Bhakti spread across northern India. Rāmānanda, a prominent saint of this period, helped bridge the Bhakti traditions of southern and northern India. The tradition developed into two broad streams—*Nirguṇa bhakti*, devoted to a formless divine, and *Saguṇa bhakti*, centred on a divine with attributes. One of the remarkable features of the Bhakti tradition was the diverse social and cultural background of its saints.

The Bhakti tradition not only provided a spiritual refuge, but also played a role in preserving and transmitting core cultural traditions. When the construction of temples faced restrictions or existing temples were being destroyed, devotees could turn to non-material forms of worship, such as *nām kīrtan* (the recitation of the divine Name). This made devotional practice less dependent on

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physical structures that could be destroyed. Bhakti flourished through local songs and community participation (*satsang*), making it accessible to wider sections of society and more difficult to suppress. For instance, Sankardev composed the *Kīrtana Ghosha,* a highly respected work that remains central to present-day Vaishnava traditions.

Bhakti saints such as Chaitanya Mahāprabhu and Guru Nānak used community singing to bridge social divides. Community singing, or *sankīrtan*, drew large gatherings and fostered a shared sense of community and collective emotion among devotees. In some instances, it also became an act of defiance. For example, Chaitanya Mahāprabhu led public *sankīrtan* in Bengal to protest against restrictions imposed by a Qazi on devotional practices. In Rajasthan, religious orders such as the Dādu Panth and Niranjani Sampradāya provided a form of stability. Some of them even maintained standing armies of warrior ascetics to protect trade routes, facilitate the movement of saints, and provide educational and medical services.

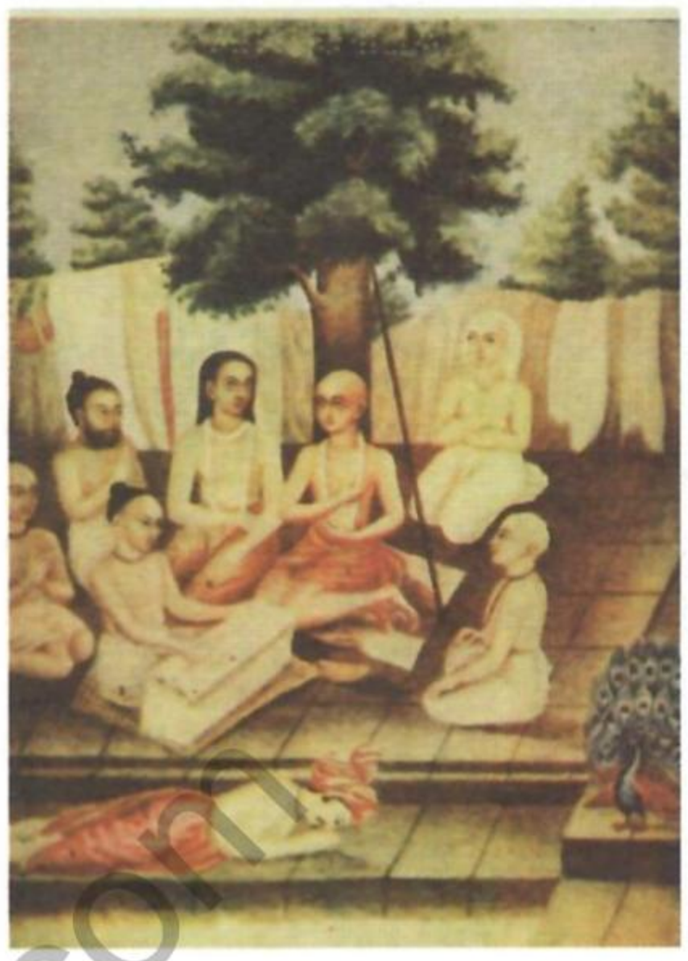

Fig. 3.17. King Pratap Rudra of Odisha bowing to Chaitanya Mahāprabhu

DON'T MISS OUT

The widespread popularity of Indian literature also strengthened social resilience over time. The two great epics, *Rāmāyaṇa* and *Mahābhārata*, were translated into many regional languages. Tulsidas, a well-known Bhakti saint from northern India, composed the *Rāmcharitmānas* in Awadhi, a widely spoken variety of Hindi in the Awadh region of present-day Uttar Pradesh. By doing so, he brought the story of Rāma into households across the region. The work played a transformative role in the Bhakti tradition by making the spiritual knowledge and teachings of the *Rāmāyaṇa* accessible to wider sections of society.

The Bhakti tradition's emphasis on equality and inclusion gradually encouraged greater participation of women. Many rose to prominence as saints, coming from diverse socio-economic backgrounds. They expressed their spirituality through devotional hymns composed in their own languages. The use of local languages strengthened shared identities and fostered a sense of solidarity.

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Fig. 3.18. A stamp commemorating Atukuri Molla

Atukuri Molla, a Telugu poet, composed the *Rāmāyaṇa* in Telugu, now known as the *Molla Rāmāyaṇam*. Her father was a potter in Gopavaram, a village in Badvel Mandal, Andhra Pradesh. She also recited the *Rāmāyaṇa* at the court of Kṛiṣhṇadevarāya.

Some Bhakti saints left their homes and devoted themselves entirely to the divine. Akkā Mahādevi (from present-day Karnataka) and Lāl Ded (from present-day Kashmir), both Shaivites, renounced worldly comforts and attachments, living as ascetics. In contrast, Bahinābāi, a saint from Maharashtra, fulfilled her familial duties while following the path of devotion to the deity Vithoba.

Northern and Central Region

- Lāl Ded
- Guru Nānak
- Rāmānanda
- Kabīr
- Mīrābāī
- Rāidāsa
- Vallabhāchārya
- Sūrdās
- Tulsīdās

Eastern Region

- Jayadeva
- Chaitanya Mahāprabhu
- Krittibāsa
- Mādhava Kandalī
- Sankardev

Southern Region

- Basavannā
- Mādhavāchārya
- Annamāchārya
- Pothanā
- Poonthanam Namboothiri

Western Region

- Ekanātha
- Jñānadeva
- Muktābāī
- Nāmadeva
- Janābāī
- Chokhamelā
- Narasimha Mehtā
- Bhalana
- Pīpājī
- Tukārām

Fig. 3.19. Some prominent Bhakti Saints from 12th–17th centuries

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LET'S MAP

On a map of India, locate and label the present-day states associated with the saints mentioned in the above infographic. Also, find and mark a saint from your own state who is not on the list. Are you familiar with any songs, poems, or verses composed by these saints? Share with the class and explain what it means or what you like about it.

THINK ABOUT IT

Vakh (*vachan* or saying) of Lāl Ded

To learn the scriptures is easy, to live them, hard.
The search for the Real is no simple matter.
Deep in my looking, the last words vanished.
Joyous and silent, the waking that met me there.

Fig. 3.20. Translation of a vakh composed by Lāl Ded

Lāl Ded says that it is 'easy' to learn scriptures but 'hard' to live them. What is the difference between knowing a rule and actually following it? Can you give an example from your daily life?

During this period, many devotees from diverse backgrounds contributed to the Bhakti traditions. Syed Ibrāhim Khān, popularly known as Raskhān, from present-day Uttar Pradesh, wrote in Braj and is known for his compositions like *Prem Vāṭikā* and *Sujān Raskhān*. Abdul Rahīm Khān-i-Khāna, popularly known as Rahīm, a Mughal military commander, wrote poetry in Braj to express his devotion to Kṛishṇa. Bibi Lachimar (of Turkish origin) devoted herself to the Bhakti of Viṣhṇu.

DON'T MISS OUT

Worship of goddesses was a long-standing tradition in India, and during the period under study, kings continued to invoke goddesses such as Kālī, Durgā, Bhavānī, and other forms of Shakti as protectors in times of conflict. Annamarājā, ruler of Bastar kingdom, is believed to have received his sword from Mānikyadevī (Danteśhvarī), while Kumāra Kampana was believed to have used a goddess-given sword against the Sultanate forces at Madurai. Similarly, Śhivājī is also believed to have received a sword from the goddess Tuljā Bhavānī to resist the Mughals.

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Fig. 3.21. A Mughal painting depicting Akbar worshipping the Sun

Sulh-i-Kul

During the later part of Akbar's reign, the philosophy of *Sulh-i-kul* was adopted as a state policy. It represented 'peace with all', 'universal toleration', or 'absolute peace'. This policy had political implications, as it appears to have been intended to secure the support of political powers other than Muslims. Over time, Akbar also began to distance himself from certain orthodox elements and adopted practices that included reverence for the Sun. The Mughal state adopted a relatively inclusive approach during the later part of his reign.

Within the Indian religious, spiritual, and philosophical landscape, the Bhakti tradition emerged as a transformative force. It spread across the Indian subcontinent, taking distinct forms in different regions and social contexts. It played a crucial role in shaping and preserving India's spiritual landscape and social fabric.

DON'T MISS OUT

Sufism began to expand in the subcontinent with the arrival of the Turks. It was a mystical dimension of Islam that generally adopted a comparatively liberal outlook towards the world. It emerged in West Asia as a form of spiritual response to the worldly orientation of some *Ulema* (Islamic theologians and jurists) and the orthodoxy of *Shariat* (Muslim jurisprudence and law). Sufism spread through various orders, or *silsilahs*. During the Delhi Sultanate, the Chishti and Suhrawardi orders became prominent, while under the Mughals, the Qaddiriyya and Naqshbandiya orders gained importance. Sufi *khanqahs* (hospices) and shrines developed as centres of popular religious life and also played a role in social and political affairs. Sufis also contributed to the spread of Islam. In India, Sufis adopted many local practices to connect with the wider population, including using Indian languages such as Hindavi and Bengali. Some Sufis also incorporated yogic practices, such as *prāṇāyāma*, known in Persian as *habs-i-dam*.

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Temples: The Resilience of Sacred Sites

Temples functioned as central pillars of the region's socio-economic and political life. These religious institutions evolved into major economic hubs that managed extensive land grants, provided employment to hundreds of specialised workers, and served as financial institutions for local traders. Rulers used temple patronage to strengthen their political legitimacy. Beyond their spiritual and religious functions, temples facilitated social integration and served as centres of education and culture, where artistic traditions such as Bharatanatyam and many other art forms were preserved. They also exercised quasi-judicial authority, collecting local taxes and administering legal penalties.

Temples also served as sources of legitimacy for political authority. The deities of these temples were regarded as protectors of the state and its people. In some cases, the deity was regarded as the ruler, while the actual ruler governed as the deity's deputy. For example, the Gajapati rulers of Odisha called themselves the *Rauta* (deputy) of Jagannath. Similarly, the Kachhwāhā rulers of Jaipur governed on behalf of Govindadevajī, while the Rajput rulers of Kota regarded themselves as representatives of Rāma.

Temples in India also functioned as significant economic institutions. They provided employment to a wide range of groups, including architects, artisans, sculptors, priests, musicians, and administrative staff. As substantial landholders, temples received donations, endowments, and land grants, often given as acts of

Fig. 3.22. Sri Ranganathaswamy Temple, Srirangam

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Fig. 3.23. Brihadeeswara Temple, Thanjavur

religious merit. The Brihadeeswara (Rajarajeshvara) temple at Thanjavur exemplifies this, having received extensive donations of gold, ornaments, and vessels. This wealth enabled temples to support irrigation works and agrarian development, reinforcing their role in economy and the society. Temples also extended credit to cultivators, village assemblies, and merchant groups. This convergence of economic and religious activity contributed to the emergence of temple towns such as Thanjavur, Kanchipuram, and Puri, as prominent urban centres.

Between the 7th and 13th centuries, temple construction proliferated across the subcontinent under the patronage of various dynasties. However, from the 11th century onwards, we begin to see many instances of temple destruction, particularly during the Ghaznavid and Ghurid invasions. From the 12th century onwards, many temples and *maṭhs* were not only plundered or destroyed, but mosques were also sometimes constructed on the same sites.

While there were instances of the destruction of sacred sites, local rulers, merchant guilds, and religious communities actively maintained and restored temples during this period, allowing sacred centres to continue functioning. Local rulers who negotiated alliances with the central authority were generally allowed to build major structures. A royal temple built by Vīra Someśhvara at Kannanur, an earlier Hoysāla regional capital, was converted into a mosque under

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Fig. 3.24. Kashi Vishwanath Temple, Varanasi

mosque under the Sultanate rule before being restored as a temple by Kumāra Kampana. Further examples of temple construction and restoration include the reconstruction of the Kashi Vishwanath Temple by Toḍar Mal, revenue officer of Akbar, and Rājā Mān Singh, chief of the Rajput Kachhwāha family (the temple was later demolished during the reign of Aurangzeb. The present temple was built by Ahilyābāī Holkar adjacent to the original site in 1780). The latter also patronised the construction of the Govind Devji Temple at Vrindavan, one of the largest temples built in northern India during this period. The Bundela king, Vīr Singh, patronised the construction of the new Keshav Dev Temple at Mathura.

From the 16th century onwards, a shift can also be seen in the design and location of Indian temples. Temples built during this period were often situated in less visible locations, along narrow lanes, with entrances that did not open onto main streets. Many lacked the conventional appearance of temples, and instead, resembled ordinary houses. Detailed exteriors became less common, and tall *shikharas* were frequently avoided. This pattern is captured in the term 'havelī temple', used by some historians to describe temples constructed in the style of *havelī* residences.

It reflects how communities adapted temple architecture to avoid destruction. Such examples can be found in Delhi, Varanasi, and Rajasthan, including the Jagat Shiromani Temple.

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The Jagat Shiromani Temple, also known as Mīrābāī Temple, was built by Queen Kanakwati, wife of Mān Singh I, in Amer. The temple is dedicated to Kṛiṣhṇa and Mīrābāī.

Fig. 3.25. Jagat Shiromani Temple, Amer

DON'T MISS OUT

Jagat Singh challenged the Mughal authorities through his architectural projects. In addition to fortifying Chittor, he built a large Jagdish temple near his palace in Udaipur. Scholars suggest that the temple was a religious and political response to the Jama Masjid constructed by Shah Jahan in Delhi.

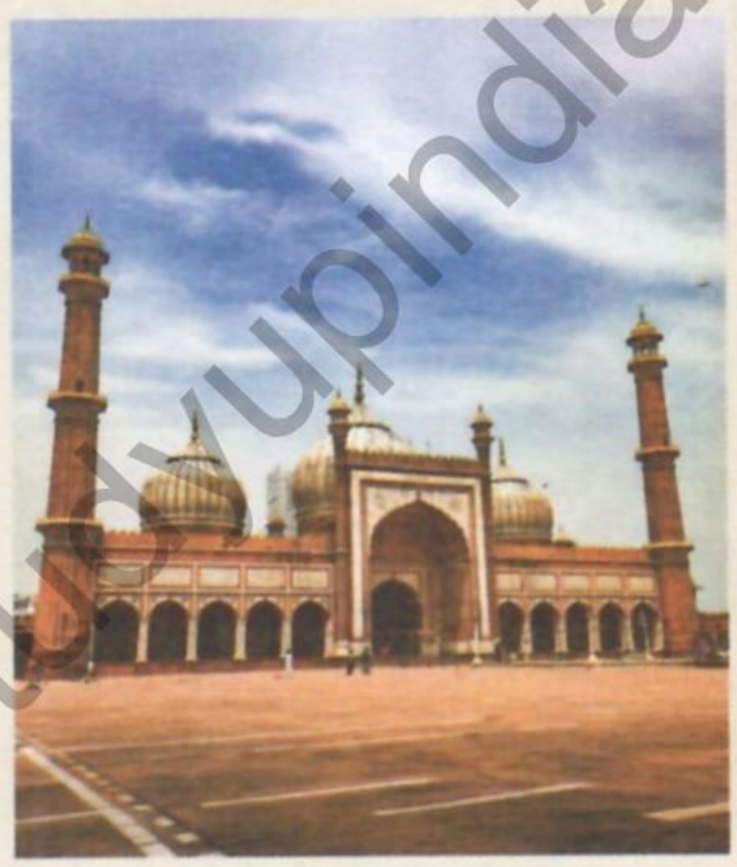

Fig. 3.26. Jama Masjid, Delhi

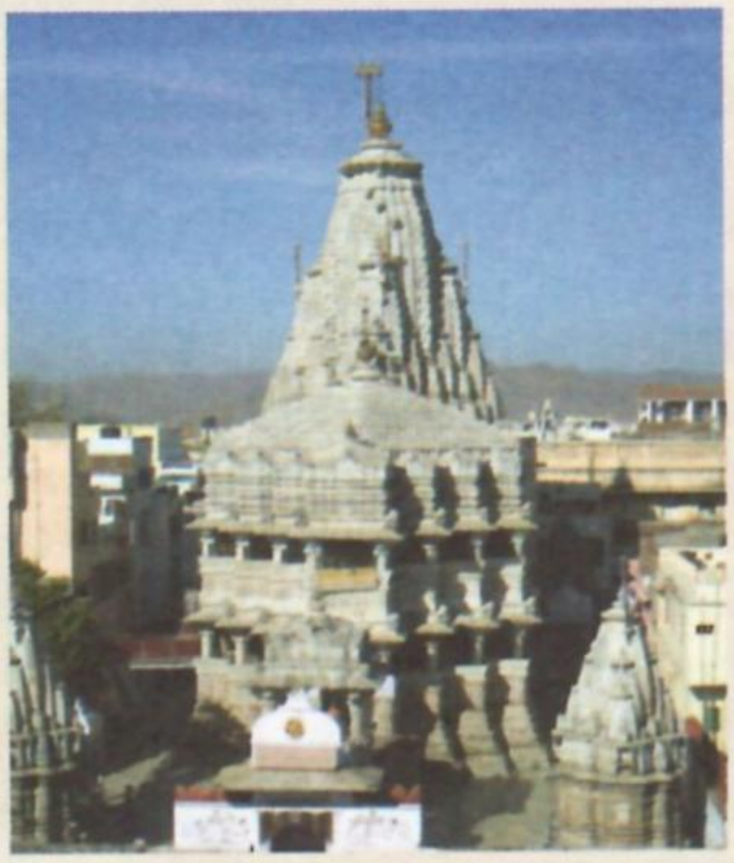

Fig. 3.27. Jagdish temple, Udaipur

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Indian Economy: Agriculture, Trading Networks, and Merchant Communities

The expansion of the agrarian economy and the gradual proliferation of centres of artisanal and craft production were significant developments of the early second millennium CE.

The main source of state income was land revenue, and the expanding frontiers of empires resulted in an increase in the area under cultivation. Over time, the collection of land revenue became increasingly strict and exploitative at various levels of administration. For example, during the reigns of Alauddin Khiljī and Mohammad bin Tughlaq, the land tax was raised to as much as 50 per cent of total agricultural production, in addition to other taxes. Moreover, the Sultans of Delhi, and later the Mughals had limited role in maritime trade. Their concerns regarding coastal areas and port cities were largely confined to political and military matters and, to some extent, the collection of customs duties. As a result, coastal areas and their commerce remained largely in the hands of regional power holders and local communities.

Irrigation technologies such as rope pulleys, spinning wheel, and Persian wheel were widely used by local populations, resulting in enhanced agricultural and economic output. In the Vijayanagara Empire, agricultural production varied greatly in both form and scale. Consequently, irrigation systems ranged from extensive canal networks to small rain-fed patches of land. A defining feature of the Vijayanagara economy was its investment in small-scale irrigation.

DON'T MISS OUT

During this era *hunḍī* was widely used as a financial instrument and can be compared in some respects to modern bank cheques. It functioned as a bill of exchange, facilitating large-scale commercial transactions without the need to carry large amounts of currency. It was also a trusted means of secured long-distance remittance. *Hunḍīs* were operated through networks of money-changers known as *sarrāfs* and their agents, called *gumasthas*.

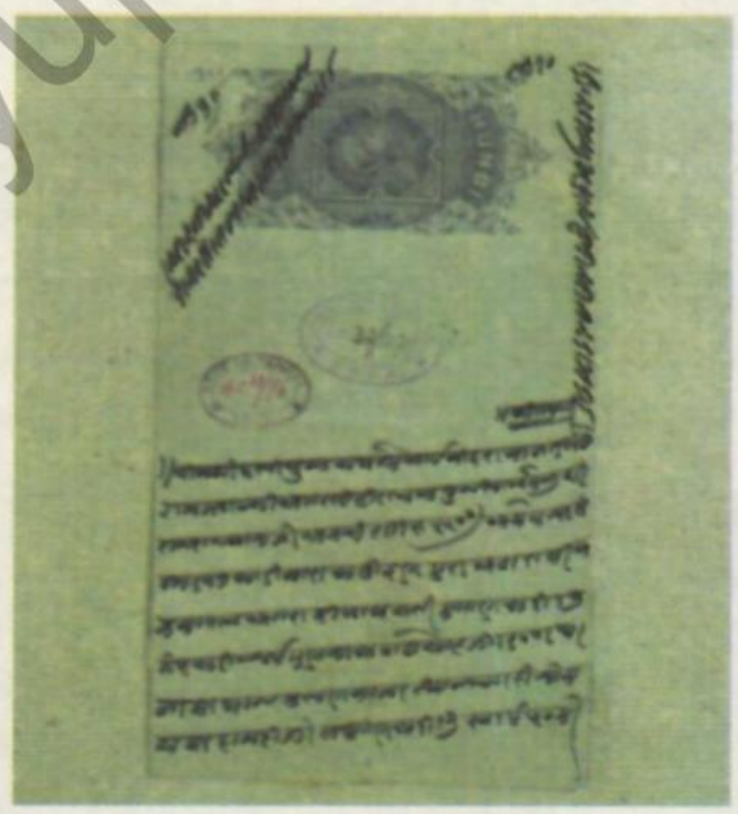

Fig. 3.28. A hunḍī stamped by the Bank of Bengal, Agra, with a denomination of one rupee and eight annas

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THINK ABOUT IT

What do you think made *hunḍī* a safer and more convenient choice than physically transporting money across long distances?

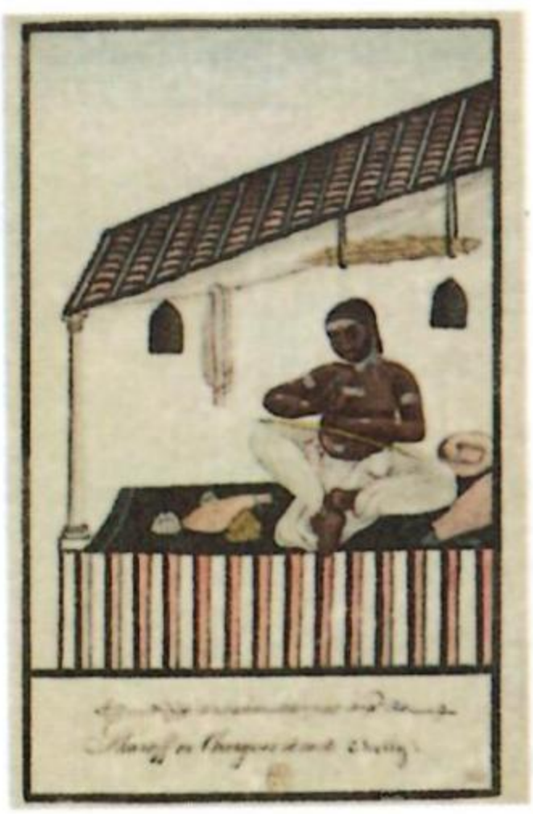

Fig. 3.29. A painting of a sarrāf or money changer

In addition to agricultural expansion, the period witnessed significant growth in trade and commerce. India's inland regions were connected to trade routes in Central Asia, facilitating a substantial flow of goods and artisanal knowledge, much of it carried by large groups of traders known as caravans. Inland trade networks expanded alongside the growth of urban centres. Local trading communities such as the *banjārās*, along with revenue-collecting officials, played a crucial role in connecting rural regions with urban centres.

From the 13th century onwards, seaborne trade grew significantly, with trading networks linking western India to the Persian Gulf and the Arabian Sea. Gradually, India became a major centre of the Indian Ocean trading network, with the eastern Mediterranean and China forming its vital axes until the arrival of Europeans in the subcontinent.

DON'T MISS OUT

Kṛiṣhṇadevarāya, the Vijayanagara ruler, greatly emphasised in his *Amukthamālyadā* the importance of oceanic trade, maintenance of ports, and fair treatment of foreign traders. He writes,

> *Manage your ports well, and let commerce increase in horses, elephants, and gems, pearls, and sandal-paste. When drought, sickness, and calamity make foreigners seek refuge in your lands, shelter them in keeping with their station. Give out gardens, yards, and mines to those whom you wish to favour...*

Fig. 3.30. A translated excerpt from the Amukthamālyadā

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